

Axion–Photon Conversion in Astrophysical Environments

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July 14, 2026

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Abstract

These notes collect a self-contained development of axion and axion-like-particle (ALP) phenomenology with an emphasis on axion–photon conversion in astrophysical settings. Starting from the strong- CP problem and the Peccei–Quinn mechanism, we introduce the QCD axion and ALPs, their characteristic parameters, their two-photon coupling, and their role as dark-matter candidates. We then develop axion electrodynamics from the θ -term, derive the coupled axion–photon equations of motion, and study resonant and non-resonant conversion in magnetised plasmas. Building on this machinery we examine axion–photon coupling in the Galaxy, conversion signatures near neutron stars and fast-radio-burst magnetars, astrophysical analogues of light-shining-through-walls bounds, and the misalignment mechanism. Throughout, intermediate steps are shown explicitly so that the notes serve as a detailed companion to the standard literature.

Acknowledgements

I am grateful to Prof. Subhabrata Majumdar (TIFR, Mumbai), Prof. Shiraz Minwalla (TIFR, Mumbai), and Dr. Abhijit Gadde (TIFR, Mumbai) for insightful discussions, guidance, and support.

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Chapter 1

Introduction

1.1 QCD axion and ALPs

The axion originated as the resolution of the strong CP problem: why is the CP symmetry conserved in the quantum chromodynamics (QCD)? R.Peccei and H.Quinn tackled this problem by postulating a new $U(1)_{PQ}$ global symmetry (PQ symmetry) [10, 11]. Spontaneous breaking of this symmetry $U(1)_{PQ}$ gives rise to a pseudo-NambuGoldstone boson, so-called QCD axion. The symmetry breaking energy scale f_a is called decay constant.

It is known that the string theory has the potential to provide many pseudoscalar particles in the low-energy four dimensional limit, so-called axionlike particles (ALPs). They are not necessarily the resolution for the strong CP problem.

QCD axion and ALPs share the basic feature. They arise as a massless particle with a shift symmetry,

$$a \rightarrow a + \text{const.} \quad (2.1.1)$$

where a is the axion. Due to the non-perturbative quantum effects (e.g., instanton), the axion obtain the periodic potential¹,

$$V(a) = \Lambda^4 \left[1 - \cos \left(\frac{a}{f_a} \right) \right] \quad (2.1.2)$$

where Λ is the energy scale which the non-perturbative effects turn on. Along with this potential, the shift symmetry becomes the discrete shift symmetry,

$$a \rightarrow a + 2\pi f_a \quad (2.1.3)$$

Provided that there are only small displacements from the potential minimum, $a \ll f_a$, we can expand the potential Eq. (2.1.2)

$$V(a) \simeq \Lambda^4 \left[1 - 1 + \frac{1}{2} \left(\frac{a}{f_a} \right)^2 \right] = \frac{1}{2} \left(\frac{\Lambda^2}{f_a} \right)^2 a^2 \quad (2.1.4)$$

Thus, axion obtains a mass:

$$m_a \sim \frac{\Lambda^2}{f_a} \quad (2.1.5)$$

1.2 Parameters which Characterize Axions

The physical properties of the QCD axion are essentially determined by the decay constant f_a . The mass of axion induced by QCD instanton is calculated in Refs. [12,48,49]. Here, we quote the result,

⁰¹ This potential expression is not unique but a qualitative one. In order to get a concrete form of $V(a)$, we need detailed knowledge about the non-perturbative effect.

$$m_{a,\text{QCD}} \simeq 6 \times 10^{-10} \text{eV} \left(\frac{10^{16} \text{GeV}}{f_a} \right) \quad (2.2.1)$$

In the case of QCD axion, there is a relation between mass $m_{a,\text{QCD}}$ and decay constant f_a . In the original model, the electroweak scale $f_a \simeq 250 \text{GeV}$ was assumed [44,50], but this parameter was ruled out immediately by experiments. Accordingly, nowadays, it is considered that f_a is much larger. However, the QCD axion cannot be lighter than 10^{-13}eV from the point of view that the decay constant f_a will never exceed the Planck scale no matter how big it is. Light QCD axions with mass range $6 \times 10^{-13} \text{eV} < m_{a,\text{QCD}} < 2 \times 10^{-11} \text{eV}$ are excluded by black hole superradiance [51,52]. There are upper bounds on the mass $m_{a,\text{QCD}}$ from stellar evolution, e.g., globular clusters [45,53,54] and the supernova SN1987A [45,55-57]. To sum up, we assume roughly

$$10^{-11} \text{eV} \lesssim m_{a,\text{QCD}} \lesssim 10^{-2} \text{eV} \quad (2.2.2)$$

as a mass range for QCD axion. At the same time, we assume

$$10^9 \text{GeV} \lesssim f_a \lesssim 10^{17} \text{GeV} \quad (2.2.3)$$

Since the interaction strengths are inversely proportional to f_a , axion has extremely weak interactions with other particles, so it is called the invisible axion. In the case of QCD axion, the energy scale Λ is given by $\Lambda^2 \sim \Lambda_{\text{QCD}}^{3/2} \sqrt{m_u}$ with $\Lambda_{\text{QCD}} \simeq 200 \text{MeV}$ and up quark mass m_u [44].

On the other hand, in the case of ALPs, there is no such relationship as Eq. (2.2.1), and its mass $m_{a,\text{ALP}}$ is an independent parameter by itself [14,15,44]. The energy scale Λ can be written as follows:

$$\Lambda = \mu e^{-S/4} \quad (2.2.4)$$

where μ is an energy scale which non-perturbative effect appears, and S is an instanton action. In general, the effective potential of ALPs is given by more than one nonperturbative effect. It is known that in many model f_a and S are related by

$$f_a \sim \frac{M_{\text{pl}}}{S} \quad (2.2.5)$$

where $M_{\text{pl}} = 2.4 \times 10^{18} \text{GeV}$ is the Planck scale. It is also known that depending on the model, f_a can be significantly lower than Eq. (2.2.5), but f_a never higher than Eq. (2.2.5). Thus Eq. (2.2.5) can be considered as an upper limit. String theory has the potential to provide hundreds of possible values for S . The values of f_a do not change so much between one axion and the other because f_a is inversely proportional to S . However, the value axion mass $m_{a,\text{ALP}}$ drastically changes with S , because it depends exponentially. ALPs have wide distributed masses $m_{a,\text{ALP}}$ on a log scale.

1.3 Interaction with Photons

Axions interact with photons:

$$\mathcal{L}_{a\gamma\gamma} \equiv -\frac{1}{4} g_{a\gamma\gamma} a F_{\mu\nu} \tilde{F}_{\mu\nu} = -g_{a\gamma\gamma} a \vec{E} \cdot \vec{B} \quad (2.3.1)$$

where $g_{a\gamma\gamma}$ is a coupling constant, $F_{\mu\nu}$ is the field strength of the electromagnetic field and $\tilde{F}_{\mu\nu}$ is its dual. $\vec{E}$ and $\vec{B}$ are electric and magnetic fields, respectively. Although axions also interact with other standard model particles, we dedicate to the two photon coupling. This is because the main subject of this thesis is related to this interaction. Moreover, it is also used as a common strategy for direct axion detection. Laboratory experiments restrict area on parameter plane $(m_a, g_{a\gamma\gamma})$.

Unlike ALPs, the area on the parameter plane $(m_a, g_{a\gamma\gamma})$ which can be occupied by QCD axion is limited due to the proportional relationship between m_a and $g_{a\gamma\gamma}$, Eq. (2.2.1). The width of the QCD axion area comes from the difference of phenomenological models,

$$g_{a\gamma\gamma} = -g_\gamma \frac{\alpha_{\text{EM}}}{\pi} \frac{1}{f_a} \quad (2.3.2)$$

where $\alpha_{\text{EM}} = 1/137$ is the fine structure constant, and model dependent dimensionless coupling constant g_γ . Kim-Shifman-Vainshtein-Zakharov (KSVZ) model [58, 59] $g_\gamma \simeq -0.97$ and Dine-Fischler-Srednicki-Zhitnitsky (DFSZ) model [60, 61] $g_\gamma \simeq 0.36$ are frequently referred, but $g_{a\gamma\gamma}$ can take various values for fixed f_a [62].

1.4 Axions as a Dark Matter Candidate

Axion is in the limelight as a dark matter candidate. In this section, we summarize the points which show that axion has adequate features as a dark matter candidate. Please see, e.g., Refs. [63-65] for more details on dark matter in general, not just about axions.

1.4.1 Dark Matter in General

The existence of non-luminous matter began to be pointed out by Zwicky in 1933 [66]. After decades, from the observations of rotation curves, it was suggested that there is mass also in dark places where there are few stars [67, 68]. Nowadays, with the addition of various observational data other than the above, we share the awareness for the existence of such "dark" components, which is called dark matter. To reveal the identity of these components is placed as a central problem in modern cosmology.

Observations in the cosmological scales, such as cosmic microwave background (CMB) anisotropy and the large scale structure (LSS), tell us properties which dark matter should have; that is, dark matter should be pressureless and act nonrelativistic by the time $z \sim 10^7$ [65]. This feature is expressed by the word, cold. According to the Planck 2018 analysis of CMB observations, the proportion of cold dark matter in the energy density of the present Universe is $\Omega_{\text{CDM}} \sim 0.26$ [69]. In addition, by definition, the dark matter should have feeble interaction with standard model particles such as photons, and its lifetime should be long enough. Please see [63] for details on the conditions which dark matter candidates must meet.

1.4.2 Axions in the Expanding Universe

Let us consider an axion with mass term Eqs. (2.1.4)-(2.1.5). In the early Universe, the axion is constant due to the Hubble friction. In accord with the Universe expands, the Hubble parameter H decreases. When it eventually becomes comparable to the mass of axion $H \sim m_a$, the axion starts to roll down toward the minimum of the potential and begins to oscillate. This coherent oscillation behaves like cold dark matter in the expanding Universe. The axion interaction with other components is so feeble that this oscillation does not dissipate until today and form cold dark matter energy density. This production mechanism of the cold axion is known as the misalignment mechanism. In the above, it was implicitly assumed that there is an axion during inflation. If the symmetry breaking occurred after inflation, then there would be other contributions to the relic energy density.

This scenario was firstly pointed out in Refs. [70-72], and the details about axion evolution can be found in, e.g., Refs [44, 45].

1.4.3 Population of Cold QCD Axion

The present density parameter of the cold QCD axion Ω_a is estimated in, e.g., Refs. [45, 46, 73],

$$\Omega_a \equiv \rho_a \frac{8\pi G}{3H_0^2} \quad (2.4.1)$$

where the axion energy density ρ_a is divided by the critical density $\rho_{\text{crit}} \equiv 8\pi G/3H_0^2$, and G is gravitational constant, and H_0 is Hubble constant. We quote the result:

$$\Omega_a \sim 0.3\Xi \left(\frac{f_a}{10^{12}\text{GeV}} \right)^{\frac{7}{6}} \quad (2.4.2)$$

where Ξ represents uncertainties due to a lack of knowledge about the early Universe. The factor Ξ is affected by whether or not the axion has already existed during inflation, and its value can change approximately by a factor of 10. The condition in Eq. (2.4.2) corresponds to $m_{a,\text{QCD}} \sim 10^{-5}\text{eV}$ because of the relationship between f_a and $m_{a,\text{QCD}}$, Eq. (2.2.1). Although there is a possibility that dark matter is composed of several species,

QCD axion, whose mass is heavier than 10^{-3}eV , cannot be the main component of dark matter. Therefore, if QCD axion is the true identity of dark matter, it is considered that its mass is within the range of $10^{-6}\text{eV} \lesssim m_{a,\text{QCD}} \lesssim 10^{-3}\text{eV}$, and experiments focusing on this range are also being conducted,

1.4.4 Decay Rate of Axions

The axion decay to two photons within a lifetime [44,45],

$$\tau_a = \frac{64\pi}{g_{a\gamma\gamma}^2 m_a^3} \quad (2.4.3)$$

due to the interaction with photons.

In the case of QCD axion, we obtain

$$\tau_{a,\text{QCD}} \simeq \frac{2.79 \times 10^{17}}{g_\gamma^2} \text{sec} \left(\frac{20\text{eV}}{m_{a,\text{QCD}}} \right)^5 \quad (2.4.4)$$

by using the relation Eq. (2.2.1). Given the age of the Universe is $t_0 \sim 1.38 \times 10^{17}\text{sec}$, axion with a mass $m_{a,\text{QCD}} < 20\text{eV}$ is stable enough. As mentioned above, there is a stronger upper limit to mass from stellar evolution

ALPs have leeway in its mass;

$$\tau_{a,\text{ALP}} \simeq 1.38 \times 10^{17} \text{sec} \left(\frac{10^{-10}\text{GeV}^{-1}}{g_{a\gamma\gamma}} \right)^2 \left(\frac{460\text{eV}}{m_{a,\text{ALP}}} \right)^3 \quad (2.4.5)$$

since coupling $g_{a\gamma\gamma}$ and mass $m_{a,\text{ALP}}$ are not related.

1.4.5 Fuzzy Dark Matter

The ALP whose mass is ultralight $\sim 10^{-22}\text{eV}$ is known as a fuzzy dark matter [74], and this is an alternative of the cold dark matter scenario. The following is a brief background of why such a substitute model is proposed.

The point is that the property "cold" is drawn from the large scale observations. In other words, it is necessary to carefully test whether the cold dark matter paradigm is also valid in the less than galaxy scale $\lesssim 10\text{kpc}$. In fact, it is pointed out that naive simulations assuming cold dark matter can not reproduce some galactic scale observations. This is called the small scale problem. Specifically, core-cusp problem, missing satellites problem, and too-big-to-fail problem can be mentioned. All these problems indicate that the amount of structure on a small scale should be suppressed compared to the cold dark matter predictions.

As a solution to the above, the ultralight axion dark matter which make small scale structure "fuzzy" has been proposed. As far as cosmological scale is concerned, the ultralight axion is indistinguishable from the cold dark matter behavior. However, it has a scale below which the structure is erased. Hence, the ultralight axion can successfully compensate for only the shortcomings of the cold dark matter while keeping the success of the cold dark matter intact.

As another approach to the small scale problem, there are simulations which suggest that structure suppression on a small scale can be interpreted as baryonic effects associated with astrophysical processes such as star formations and supernova explosions.

In any case, the small scale problem is a matter of debate and has not been settled and

continues to require careful consideration.

1.4.6 Local Dark Matter Density

It is considered that galaxies are surrounded by the so-called dark matter halo, which has spread beyond the edge of the visible radius. A standard assumption of the halo is that it goes through gravitational collapse, and now it is in virial equilibrium.

The local dark matter density means an averaged value over a few hundred parsecs around the Sun. Two approaches are often used to measure this local density, namely local measures and global measures. The former are so-called tracers and use the kinematics of the stars near the Sun. The latter extrapolates from the rotation curve of the galaxy. In the Reference [75], there is a summary of measurements. We adopt a typical value $0.3\text{GeV}/\text{cm}^3$ as a dark matter density near the solar system

Error bars of global measures often small, but they strongly depend on the shape of the dark matter halo. By contrast, local measures have large errors. Putting both observations together, we expect that the shape of the halo would be revealed. Just as the data of the Hipparcos satellite and the Sloan Digital Sky Survey improved local measure's error bars, our understanding of both the Milky Way and the nature of dark matter will be deepened by the Gaia satellite observations.

The upper bound of the local KSVZ axion density with mass range $(2.3 - 3.4)\mu\text{eV}$ is given in Ref. [76]; $\rho_{\text{KSVZ}} \lesssim 0.45\text{GeV}/\text{cm}^3$. There is also an attempt to impose a constraint on the local density of ultralight axion dark matter by using the pulsar timing array, e.g., Ref. [77].

1.5 Search for Axions

Indeed, the detection of axions is challenging due to its feeble interaction, but it might be possible. Over a wide parameter space, including ALPs, various ingenious detection methods towards the first detection of axions have been devised, and some of them are already up and running. In this section, we will take up only the representative examples and explain them briefly. For details on each experiment, please refer to the original papers of each group.

1.6 Solar Axion

Due to the axion photon coupling, the Sun is known as a powerful natural axion source. Inside of the Sun, axions are produced from photons in the presence of electromagnetic fields which are produced by the ionized plasma. The energy of the solar axion is comparable to the core temperature of the Sun. The axion flux on the Earth was calculated in Ref. [45]. Solar axion is detected as an X-ray by applying the strong static magnetic field in the laboratory. The CAST experiment at CERN gives the strongest constraint on the coupling. With an eye to the next generation axion helioscope, IAXO, CAST has improved detectors, and this improvement gives more stringent constraint: $g_{a\gamma\gamma} < 6.6 \times 10^{-11}\text{GeV}^{-1}$ for $m_a < 0.02\text{eV}$ [78].

1.7 Halo Axion

The most famous strategy is the one which exploits microwave cavities [79-81]. A strong magnetic field is applied in the cavity, and axions are detected as microwaves. This cavity experiment is suitable for a mass range $10^{-7}\text{eV} \lesssim m_a \lesssim 10^{-5}\text{eV}$. The leading group of this technique is ADMX. Although the mass range is very limited, $(2.81 - 3.31)\mu\text{eV}$, this experiment gives the strongest constraint on the coupling: $g_{a\gamma\gamma} \lesssim 3 \times 10^{-16}\text{GeV}^{-1}$ [82]. A method has been proposed to target heavier axions than ADMX by using dielectric plates, the so-called MADMAX experiment [83, 84]. By contrast, in the case of lighter mass range, $10^{-9}\text{eV} \lesssim m_a \lesssim 10^{-7}\text{eV}$, a method which uses LC circuits instead of the cavity has been devised, e.g., the ABRACADABRA experiment [85].

A detailed understanding of the halo is very important for dark matter detection, but it is also true that there are still many uncertainties. The ADMX's constraints [82] assume both

the isothermal model, which is called the standard halo model and the N-body model, which incorporate baryonic physics by using N-body simulation.

1.8 Axion Produced in the Laboratory

There is also an attempt to detect axions which are produced from the laser by applying a magnetic field in the laboratory. Axions, unlike photons, can pass through a wall. Thus, if a magnetic field is applied again over the other side of the wall, and if photons are detected, then it means detection of axions. This observation is a pure experiment which is complete in the laboratory, so it is independent of the physical condition in the Sun. However, these experiments give less constraint for parameter space than the observation of the solar axion. This is due largely to less intensity of the axion flux. This experiment is known as a collective name, shining light through walls, e.g., ALPS experiment at DESY: $g_{a\gamma\gamma} < 7 \times 10^{-8} \text{GeV}^{-1}$ for $m_a < 5 \times 10^{-4} \text{eV}$ [86], and the OSQAR experiment at CERN: $g_{a\gamma\gamma} < 3.5 \times 10^{-8} \text{GeV}^{-1}$ for $m_a < 3 \times 10^{-4} \text{eV}$ [87]. ALPS is now in preparation for the upgrade experiment [88], and it will go down $g_{a\gamma\gamma} < 2 \times 10^{-11} \text{GeV}^{-1}$ for $m_a < 10^{-4} \text{eV}$.

1.9 Other Approaches

There are many more ongoing or future experiments which I could not mention. All of the above are experiments using the interaction between axions and photons, but there are also experiments which use the interaction with fermions. The details of other experiments can be found, e.g., in Refs. [44,46].

Chapter 2

Introduction of the Theta Term in the Lagrangian

$$\begin{aligned} F^{\mu\nu} &\equiv \partial^\mu A^\nu - \partial^\nu A^\mu + ig[A^{\mu\nu}, A^\nu] \\ &= D^\mu A^\nu - D^\nu A^\mu, \end{aligned} \tag{2.1}$$

and $\nabla_\mu K^\mu = *d*K$

$$\begin{aligned} *K &= \text{tr}[A \wedge F - \frac{1}{3}A \wedge A \wedge A] \\ (*F)_{\mu\nu} F^{\mu\nu} &= *\text{tr}[F \wedge F], \quad F = dA + A \wedge A, \quad dF + A \wedge F - F \wedge A = 0. \end{aligned}$$

$$\begin{aligned} d*K &= \text{tr}[dA \wedge F - A \wedge dF - dA \wedge A \wedge A] \\ &= \text{tr}[(F - A \wedge A) \wedge F - A \wedge (-A \wedge F + F \wedge A) \\ &\quad - (F - A \wedge A) \wedge A \wedge A] \\ &= \text{tr}[F \wedge F - A \wedge A \wedge F + A \wedge A \wedge F \\ &\quad - A \wedge F \wedge A - F \wedge A \wedge A + A \wedge A \wedge A \wedge A] \\ &= \text{tr}[F \wedge F + A \wedge A \wedge A \wedge A]. \end{aligned}$$

$$\text{tr}[A \wedge F \wedge A + F \wedge A \wedge A] = 0$$

$$\begin{aligned} \text{tr}[A \wedge A \wedge A \wedge A] &= \text{tr}[T_a T_b T_c T_d] A^a \wedge A^b \wedge A^c \wedge A^d \\ &= \text{tr}[[T_a, T_b][T_c, T_d]] A^a \wedge A^b \wedge A^c \wedge A^d \\ &= f_{abe} f_{ecd} A^a \wedge A^b \wedge A^c \wedge A^d \\ &= -(f_{cae} f_{ebd} + f_{bce} f_{ead}) A^a \wedge A^b \wedge A^c \wedge A^d \\ &= -2f_{abe} f_{ecd} A^a \wedge A^b \wedge A^c \wedge A^d. \end{aligned} \tag{2.2}$$

or

$$\begin{aligned} \text{tr}A^4 &= \text{tr}[T_a T_b T_c T_d] A^a A^b A^c A^d \\ &= -\text{tr}[T_a T_b T_c T_d] A^b A^c A^d A^a \\ &= -\text{tr}[T_b T_c T_d T_a] A^b A^c A^d A^a \\ &= -\text{tr}A^4. \end{aligned} \tag{2.3}$$

proved: $\text{tr}[F \wedge F] = d*K$, or $\text{Tr}\{\star F_{\mu\nu} F^{\mu\nu}\} = \partial^\mu K_\mu$.

where: $K_\mu = \epsilon_{\mu\nu\rho\sigma} \text{Tr}\{A^\nu F^{\rho\sigma} - \frac{2}{3}igA^\nu A^\rho A^\sigma\}$.

reading :Instantons Vacuum

Now Let us consider a vacuum-to-vacuum transition given by the functional integral

$$\langle 0 | \exp(-iHt) | 0 \rangle \sim \int \mathcal{D}A_\mu \exp \left[i \int d^4x (\mathcal{L}_{\text{YM}} + \mathcal{L}_G + \mathcal{L}_{\text{FP}}) \right].$$

It remains to be understood what one means by 'vacuum' for the transition given by above. Firstly, we must pay some attention to the possible boundary conditions on the fields in the integral. We are interested in transitions from one classical vacuum to another so we require that the initial $t \rightarrow -\infty$ and final $t \rightarrow +\infty$ field configurations are of the pure gauge form. Actually, we shall require that gauge potentials tend to a pure gauge at large distances in all four directions

$$A_\mu \xrightarrow{|x_\mu| \rightarrow \infty} U^{-1} \partial_\mu U.$$

For the time being, we leave the gauge unspecified. Let us now consider the integral

$$Q = - (1/16\pi^2) \int d^4x \text{Tr} \left[\tilde{G}^{\mu\nu} G_{\mu\nu} \right],$$

where

$$\tilde{G}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} G_{\rho\sigma}.$$

Q is gauge-invariant and is called the topological charge or the Pontryagin index. It is stationary with respect to any local variation δA_μ whether or not the equation of motion is satisfied:

$$\delta Q \sim \frac{1}{2} \int d^4x \text{Tr} \left[\tilde{G}_{\mu\nu} (D^\mu \delta A^\nu - D^\nu \delta A^\mu) \right] = \int d^4x \text{Tr} \left[\tilde{G}_{\mu\nu} D^\mu \delta A^\nu \right].$$

The first term vanishes from the Bianchi identities and no surface terms arise from the second term; Q is a topological invariant. Since

$$- (1/16\pi^2) \text{Tr} \left[G_{\mu\nu} \tilde{G}^{\mu\nu} \right] = \partial_\mu K^\mu,$$

$$K_\mu = - (1/16\pi^2) \varepsilon^{\mu\alpha\beta\gamma} \text{Tr} \left[G_{\alpha\beta} A_\gamma + \frac{2}{3} A_\alpha A_\beta A_\gamma \right],$$

we can write Q as an integral over the surface S at infinity:

$$Q = \int dS_\mu K^\mu.$$

For potentials A_μ , Q may be written in terms of U by inserting the asymptotic form of K_μ [See Winding Number !]:

$$Q = - (1/24\pi^2) \int dS_\mu \varepsilon^{\mu\alpha\beta\gamma} \text{Tr} \left[U^{-1} \partial_\alpha U U^{-1} \partial_\beta U U^{-1} \partial_\gamma U \right].$$

To make contact with the discussion earlier in this section, let us take the gauge $A_0 = 0$. Then from above equation:

$$Q = \int d^3x K^0 \Big|_{-\infty}^{\infty} = N(+\infty) - N(-\infty).$$

Using the remaining gauge freedom we can set $N(-\infty) = 0$. Therefore, in this gauge, the topological charge Q can be interpreted as the winding number of the pure gauge configuration to which A_i tends (we always also assume, which is necessary for the integral to be finite), and the functional integral, with the integral restricted to the Q sector, as the transition amplitude between topological vacua $|N=0\rangle$ and $|N=Q\rangle$, or more generally between $|N\rangle$ and $|N+Q\rangle$.

In the gauge $A_0 = 0$, we are now ready to construct the functional integral corresponding to the $|\Theta\rangle \rightarrow |\Theta\rangle$ transition. The result turns out to have the gauge-invariant form for which we are looking. We have

$$\begin{aligned}
\langle \Theta | \exp(-iHt) | \Theta' \rangle &= \sum_{m,n} \exp[i(n-m)\Theta] \exp[in(\Theta' - \Theta)] \langle m | \exp(-iHt) | n \rangle \\
&= 2\pi\delta(\Theta - \Theta') \sum_Q \exp(-iQ\Theta) \langle N+Q | \exp(-iHt) | N \rangle \\
&\sim \sum_Q \exp(-iQ\Theta) \\
&\quad \times \int \mathcal{D}A_Q \exp \left\{ i \int d^4x \left[\mathcal{L}_{\text{YM}} + \mathcal{L}_G + \mathcal{L}_{\text{FP}} + (1/16\pi^2) \Theta \text{Tr} \left[G_{\mu\nu} \tilde{G}^{\mu\nu} \right] \right] \right\}. \\
\mathcal{L} &= \bar{\Psi}_L i D \psi_L + \psi_R i D \Psi_R - m \bar{\psi}_L \psi_R + h.c. \\
U(1)_L : \quad \Psi_L &\rightarrow e^{i\alpha_L} \Psi_L \quad U(1)_R : \Psi_R \rightarrow e^{i\alpha_R} \Psi_R \\
U(1)_V : \alpha_L &= \alpha_R = \alpha_V \quad \Psi \rightarrow e^{i\alpha_V} \Psi \quad \Psi = \left(\begin{pmatrix} \Psi_L \\ \Psi_R \end{pmatrix} \right) \\
U_1(A) : \alpha_R &= -\alpha_L = \alpha_A; \psi \rightarrow e^{i\gamma^5 \alpha_A} \psi \\
\mathcal{L}' &= \bar{\psi}_L i \not{D} \psi_L + \bar{\psi}_R i \not{D} \psi_R - m e^{-i\gamma^5 \alpha_L} e^{-i\gamma^5 \alpha_R} \bar{\Psi}_L \psi_R \\
\delta \mathcal{L} &= \mathcal{L} - \mathcal{L}' = - \left(1 - (-2i\gamma^5 \alpha_1) \right) m \bar{\psi}_L \psi_R \\
\delta \mathcal{L} &= -2i\alpha_A m \bar{\psi} \gamma^5 \psi \\
y_i &= a_{i1}x_1 + \cdots a_{ip} \\
\frac{\partial y_i}{\partial x_j} &= a_{ij} \Rightarrow \\
J &= |A| \\
dy &= |A| dx \\
j^{A\mu} &= \frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \delta \psi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \bar{\psi})} = i \bar{\psi} \gamma^\mu (i\alpha_A \gamma^5 \psi) \\
\delta \psi &= i\alpha_A \gamma^5 \psi; \delta \bar{\psi} = -i\alpha_A \gamma^5 \bar{\psi} \\
\therefore j^{A\mu} &= -\alpha_A \bar{\psi} \gamma^\mu \gamma^5 \psi \\
\partial_\mu j^{A\mu} &= \delta \mathcal{L} = 2im \bar{\psi} \gamma^\mu \psi.
\end{aligned}$$

$$G_n(x_1, \dots, x_n) = \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle.$$

Here $T\{\dots\}$ is the time-ordering operator for which orders the field operators so that earlier time field operators appear to the right of later time field operators. By transforming the fields and states into the interaction picture, this is rewritten as

$$G_n(x_1, \dots, x_n) = \frac{\langle 0 | T \{ \phi(x_1) \dots \phi(x_n) e^{iS[\phi]} \} | 0 \rangle}{\langle 0 | e^{iS[\phi]} | 0 \rangle}.$$

Let's make the change of integration variables

$$\Psi(x) \rightarrow e^{-i\alpha(x)} \psi(x) \quad \bar{\psi}(x) \rightarrow e^{i\alpha(x)} \bar{\Psi}(x).$$

$$\langle \psi(x_1) \bar{\psi}(x_2) \rangle = \int D\psi D\bar{\psi} e^{i \int d^4x \mathcal{L}} \psi(x_1) \bar{\psi}(x_2)$$

$$\Psi(x) = \Delta(x) \Psi(x); \quad \bar{\psi}(x) = \Delta_c(x) \bar{\psi}(x)$$

$$J_c = \det(\Delta_c) = e^{\text{tr}(\ln \Delta_c)} = e^{\int d^4x \langle x | \text{tr}(\ln \Delta_c) | x \rangle}$$

$$\text{for } U_1(A) : \alpha_R = -\alpha_L = \alpha_A; \psi \rightarrow e^{i\gamma^5 \alpha_A} \psi.$$

$$\bar{\psi} \rightarrow \overline{\Delta \psi} = \psi^\dagger \Delta^\dagger \gamma^0 = \psi^\dagger e^{-i\beta \gamma^5} \gamma^0 = \bar{\psi} e^{i\beta \gamma^5}$$

$$\begin{aligned}
U(1)_A &= \Delta = \Delta_c = e^{i\beta(x)\gamma^5}, \\
J &= J_c = e^{i \int d^4x \text{Tr}(\langle x | \beta(\hat{x}) | x \rangle)}, \\
\text{Tr}(\langle x | \beta(x) | x \rangle) &= \lim_{\Lambda \rightarrow \infty} \text{Tr} \left[\left\langle x \left| \beta(x) \gamma^5 e^{\frac{\not{p}^2}{\Lambda^2}} \right| x \right\rangle \right], \\
\Pi^\mu &= p^\mu - eA^\mu(\hat{x}) \quad \mathbb{A}\Pi = \Pi^\mu \gamma_\mu, \\
D^2 &= D_\mu^2 + \frac{e}{2} F_{\mu\nu} \sigma^{\mu\nu} \Rightarrow \mathbb{A}\Pi^2 = \Pi^2 - \frac{e}{2} \sigma_{\mu\nu} F^{\mu\nu}(\hat{x})
\end{aligned}$$

Because we can prove that :

$$\begin{aligned}
&\text{we know } \sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu], \\
F^{\mu\nu} &= A^{\nu,\mu} - A^{\mu,\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -H^3 & H^2 \\ E^2 & H^3 & 0 & -H^1 \\ E^3 & -H^2 & H^1 & 0 \end{pmatrix}, \\
\psi &= (i\gamma^\nu \partial_\nu - A_\nu \gamma^\nu) (i\gamma^\mu \partial_\mu - A_\mu \gamma^\mu) \psi = \\
&(-\gamma^\nu \gamma^\mu \partial_\nu \partial_\mu - iA_\nu \gamma^\nu \gamma^\mu \partial_\mu - i\gamma^\nu A_\mu \gamma^\mu \partial_\nu - i\gamma^\nu \gamma^\mu A_{\mu,\nu} + A_\nu A_\mu \gamma^\nu \gamma^\mu) \psi = \\
&\left(-\frac{1}{2} (\gamma^\nu \gamma^\mu + \gamma^\mu \gamma^\nu) \partial_\nu \partial_\mu - iA_\nu \gamma^\nu \gamma^\mu \partial_\mu - i\gamma^\mu A_\nu \gamma^\nu \partial_\mu - i\gamma^\nu \gamma^\mu A_{\mu,\nu} + \right. \\
&\left. \frac{1}{2} (A_\nu A_\mu \gamma^\nu \gamma^\mu + A_\mu A_\nu \gamma^\mu \gamma^\nu) \right) \psi = \\
&\left(-g^{\mu\nu} \partial_\nu \partial_\mu - 2iA_\nu g^{\mu\nu} \partial_\mu - \frac{i}{2} (\gamma^\nu \gamma^\mu A_{\mu,\nu} + \gamma^\mu \gamma^\nu A_{\nu,\mu}) + A_\nu A_\mu g^{\mu\nu} \right) \psi = \\
&\left(-\partial^\mu \partial_\mu - 2iA^\mu \partial_\mu - \frac{i}{2} (\gamma^\nu \gamma^\mu A_{\mu,\nu} + (2g^{\mu\nu} - \gamma^\nu \gamma^\mu) A_{\nu,\mu}) + A^\mu A_\mu \right) \psi = \\
&\left(-\partial^\mu \partial_\mu - 2iA^\mu \partial_\mu - \frac{i}{2} (\gamma^\nu \gamma^\mu (A_{\mu,\nu} - A_{\nu,\mu}) + 2A_{\mu\mu}^\mu) + A^\mu A_\mu \right) \psi = \\
&\left(-\partial^\mu \partial_\mu - 2iA^\mu \partial_\mu - \frac{i}{4} (\gamma^\nu \gamma^\mu - \gamma^\mu \gamma^\nu) F_{\nu\mu} - iA_{\mu,\mu}^\mu + A^\mu A_\mu \right) \psi = \\
&\left(-\partial^\mu \partial_\mu - 2iA^\mu \partial_\mu - iA_{\mu,\mu}^\mu + A^\mu A_\mu - \frac{1}{2} F_{\nu\mu} \sigma^{\nu\mu} \right) \psi.
\end{aligned}$$

Lattice QCD perspective:

$$\begin{aligned}
\text{Tr}(\langle x | \beta(x) | x \rangle) &= \lim_{\Lambda \rightarrow \infty} \beta(x) \left\langle x \left| \text{Tr} \left[\gamma^5 \exp \frac{(\hat{p} - eA(x))^2 - \frac{e}{2} \sigma_{\mu\nu} F^{\mu\nu}}{\Lambda^2} \right] \right| x \right\rangle, \\
\text{Tr}(\gamma^\mu \gamma^\nu \gamma^\alpha \gamma^\beta \gamma^5) &= -4i\epsilon^{\mu\nu\alpha\beta}, \\
&= -\frac{e^2}{2} \epsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \lim_{\Lambda \rightarrow \infty} \left[\frac{1}{\Lambda^4} \left\langle x \left| e^{\frac{(\hat{p} - eA)^2}{\Lambda^2}} \right| x \right\rangle + O\left(\frac{1}{\Lambda^5}\right) \right], \\
\hat{p}|k\rangle &= k|k\rangle, \langle k | x \rangle \propto e^{-ikx}, \\
\frac{1}{\Lambda^4} \left\langle x \left| e^{\hat{p}^2/\Lambda^2} \right| x \right\rangle \sum_k |k\rangle \langle k| &= \frac{1}{\Lambda^4} \left\langle x \left| e^{\frac{(\hat{p} - (A)^2)}{\pi^2}} \right| x \right\rangle \sum_k |k\rangle \langle k|, \\
t = \frac{k}{\Lambda} &= \frac{i}{\Lambda^4} \int \frac{d^4k}{(2\pi)^4} e^{-\frac{k^2}{\Lambda^2}} = \frac{i}{16\pi^2} \simeq O(\Lambda^0).
\end{aligned}$$

background, to be integrated later. We have

$$Z(A) \equiv \int \mathcal{D}\Psi \mathcal{D}\bar{\Psi} e^{iS(A)},$$

where

$$S(A) \equiv \int d^4x \bar{\Psi} i \not{D} \Psi$$

is the Dirac action, $i \not{D} = i\gamma^\mu D_\mu$ is the Dirac wave operator, and

$$D_\mu = \partial_\mu - igA_\mu$$

is the covariant derivative. Here A_μ is either the U(1) gauge field, or the matrix-valued nonabelian gauge field), depending on the theory under consideration. Our notation allows us to treat both cases simultaneously. We can formally evaluate above eq as a functional determinant,

$$Z(A) = \det(i \not{D}).$$

However, this expression is not useful without some form of regularization. We will take up this issue shortly.

Now consider an axial U(1) transformation of the Dirac field, but with a spacetime dependent parameter $\alpha(x)$:

$$\Psi(x) \rightarrow e^{-i\alpha(x)\gamma_5} \Psi(x),$$

$$\bar{\Psi}(x) \rightarrow \bar{\Psi}(x) e^{-i\alpha(x)\gamma_5}.$$

We can think of above eqs. as a change of integration variable ; then $Z(A)$ should be independent of $\alpha(x)$. The corresponding change in the action is

$$S(A) \rightarrow S(A) + \int d^4x j_A^\mu(x) \partial_\mu \alpha(x).$$

We can integrate by parts to write this as

$$S(A) \rightarrow S(A) - \int d^4x \alpha(x) \partial_\mu j_A^\mu(x).$$

If we assume that the measure $\mathcal{D}\Psi \mathcal{D}\bar{\Psi}$ is invariant under the axial U(1) transformation, then we have

$$\begin{aligned} Z(A) &\rightarrow \int \mathcal{D}\Psi \mathcal{D}\bar{\Psi} e^{iS(A)} e^{-i \int d^4x \alpha(x) \partial_\mu j_A^\mu(x)} \\ J_c &= \exp \left[-i \int d^4x \beta(x) \frac{e^2}{32\pi^2} \epsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \right] \\ &\int D\bar{\psi} D\psi DA \exp \left[i \int d^4x L_{QED} \right] \psi \bar{\psi} \\ &\psi \rightarrow e^{i\beta(x)\gamma_5} \\ &\int D\bar{\psi} D\psi DA \exp \left[i \int d^4x L_{QED} + \beta \frac{e^2}{16\pi^2} \epsilon^{\mu\nu\alpha\beta} F_{\mu\nu} F_{\alpha\beta} \right] \\ (\det J)^{-2} &= \exp \left[-\frac{ig^2}{16\pi^2} \int d^4x \alpha(x) \epsilon^{\mu\nu\rho\sigma} \text{Tr} F_{\mu\nu}(x) F_{\rho\sigma}(x) \right]. \end{aligned}$$

Including the transformation of the measure, in the transformation of the path integral, above then yields

$$Z(A) \rightarrow \int \mathcal{D}\Psi \mathcal{D}\bar{\Psi} e^{iS(A)} e^{-i \int d^4x \alpha(x) [(g^2/16\pi^2) \epsilon^{\mu\nu\rho\sigma} \text{Tr} F_{\mu\nu}(x) F_{\rho\sigma}(x) + \partial_\mu j_A^\mu(x)]}$$

$$\mathcal{L} > -\theta_{QCD} \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \cdot \tilde{G}^{a\mu\nu}$$

Now in the massless term (i.e: if we assume massless quark) of the lagrangian then there is an anomalous U(1)A under which

$$\mathcal{L} \rightarrow \mathcal{L} - \alpha_A \frac{g_s^2}{16\pi^2} G_{\mu\nu}^a \tilde{G}^{a\mu\nu}$$

$$\mathcal{L} = \bar{\Psi}_L i \not{D} \psi_L + \psi_R i \not{D} \Psi_R - m \bar{\psi}_L \psi_R + hc.$$

which is similar with the expression of the theta term and when quarks are massive that is

it has non zero mass term then $U(1)_A$ ($\alpha_R = -\alpha_L = \alpha_A$; $\psi \rightarrow e^{i\gamma^5 \alpha_A} \psi$) already explicitly broken by the quark mass term. There are two things happening while we're doing $U(1)_A$ axial transformation.

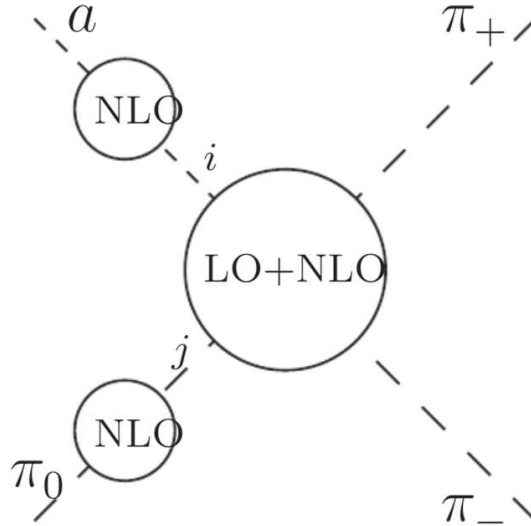
$m\bar{\psi}_L\psi_R \rightarrow m(e^{-i\gamma^5 \alpha_L} \bar{\psi}_L)(e^{i\gamma^5 \alpha_R} \psi_R) \rightarrow m(e^{i\gamma^5 \alpha_A} \bar{\psi}_L)(e^{i\gamma^5 \alpha_A} \psi_R) \rightarrow m(e^{2i\gamma^5 \alpha_A} \bar{\psi}_L)\psi_R$ There are two things happening here :

Considering mass as a complex parameter with argument $\theta_f = \arg(m)$ changes to $\rightarrow \theta_f + 2\alpha_A$ and the massless term gaining phase due to anomaly $\theta_{QCD} \rightarrow \theta_{QCD} + 2\alpha_A$ but the phase say $\bar{\theta} = \theta_{QCD} + \theta_f$ remains invariant under axial transformation independent of basis (therefore physical) $\bar{\theta}$ is the physical CP violating phase. But for massless quark only $\theta_{QCD} \rightarrow \theta_{QCD} + 2\alpha_A$ change is unphysical because there is a basis where $\theta_{QCD} = 0$. In the literature there are a lot of debate one can perhaps set the up quark mass to be 0 and that will immediately solve the strong CP problem but lattice QCD shows that up quark mass is not zero with a great significance therefore ruling out the possibility. For many flavours the combination $\bar{\Theta} = \Theta_{QCD} - \arg(\det(M))$ is invariant under $U(1)_A$ and basis independent.

Back to

$$\mathcal{L} > -\theta_{QCD} \frac{g_s^2}{32\pi^2} G_{\mu\nu}^a \cdot \tilde{G}^{a\mu\nu} \propto \partial_\mu K^\mu$$

In fact the surface term or total derivative in the lagrangian does not contribute to the Feynman diagram or the perturbation theory of any order because when we derive vertex feynman rule for local operator is proportional to the total momentum, and momentum is conserved at the vertex.



2.1 QCD-Axion Lagrangian and Axion-pion Lagrangian at leading order(LO):

The 2-flavour axion effective Lagrangian in terms of quarks and gluons reads

$$\begin{aligned} \mathcal{L}_a^{\text{QCD}} = & \frac{1}{2} (\partial_\mu a)^2 - \frac{1}{2} m_{a,0}^2 a^2 + \frac{\alpha_s}{8\pi} \frac{a}{f_a} G\tilde{G} - \bar{q}_L M_q q_R + \text{h.c.} \\ & + \frac{\partial_\mu a}{2f_a} \bar{q} c_q^0 \gamma^\mu \gamma^5 q + \frac{1}{4} g_{a\gamma}^0 a F\tilde{F} \end{aligned}$$

where $q = (u, d)^T$, $M_q = \text{diag}(m_u, m_d)$, $G\tilde{G} \equiv \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} G_{\mu\nu}^A G_{\rho\sigma}^A$ and $F\tilde{F} \equiv \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$, with $\epsilon^{0123} = -1$. For the QCD axion $m_{a,0}^2 = 0$, while $m_{a,0}^2 \neq 0$ for the ALP case. The couplings $c_q^0 = \text{diag}(c_u^0, c_d^0)$ and $g_{a\gamma}^0$ are model-dependent. For instance, $c_{u,d}^0 = 0$ and $g_{a\gamma}^0 = 0$ in the KSVZ model, while $c_u^0 = \frac{1}{3} \cos^2 \beta$, $c_d^0 = \frac{1}{3} \sin^2 \beta$ and $g_{a\gamma}^0 = \alpha / (2\pi f_a) 8/3$ in the DFSZ model (with $\tan \beta = v_u/v_d$ the ratio between the vacuum expectation values of the two Higgs doublets present in the DFSZ model).

Upon an anomalous axial rotation of the quark doublet

$$q \rightarrow e^{i\gamma_5 \frac{a}{2f_a} Q_a} q$$

with $\text{Tr } Q_a = 1$, the $aG\tilde{G}$ term in E is shifted away, and the Lagrangian becomes

$$\mathcal{L}_a^{\text{QCD}} = \frac{1}{2} (\partial_\mu a)^2 + \frac{1}{2} m_{a,0}^2 a^2 - (\bar{q}_L M_a q_R + \text{h.c.}) + \frac{\partial_\mu a}{2f_a} \bar{q} c_q \gamma^\mu \gamma_5 q + \frac{1}{4} g_{a\gamma} a F \tilde{F},$$

where we have redefined the parameters as

$$M_a = e^{i\frac{a}{2f_a} Q_a} M_q e^{i\frac{a}{2f_a} Q_a}, \quad c_q = c_q^0 - Q_a, \quad g_{a\gamma} = g_{a\gamma}^0 - \frac{3\alpha}{2\pi f_a} \text{Tr} (Q_a Q_{\text{EM}}^2)$$

with $Q_{\text{EM}} = \text{diag}(2/3, -1/3)$.

2.2 Axion-pion Lagrangian at LO

At energies $\lesssim 1\text{GeV}$, the axion-QCD effective Lagrangian is replaced by the axion chiral Lagrangian, which at the LO reads

$$\mathcal{L}_a^{\chi(\text{LO})} = \frac{1}{2} (\partial_\mu a)^2 - \frac{1}{2} m_{a,0}^2 a^2 + \frac{f_\pi^2}{4} \text{Tr} \left[(D^\mu U)^\dagger D_\mu U + U \chi_a^\dagger + \chi_a U^\dagger \right] + \frac{\partial^\mu a}{2f_a} \text{Tr} [c_q \sigma^a] J_{A,\mu}^a \Big|_{\text{LO}},$$

where $f_\pi = 92.21\text{MeV}$, $\chi_a = 2B_0 M_a$ (with B_0 denoting the quark condensate) and σ^a ($a = 1, 2, 3$) the Pauli matrices. $U = e^{i\pi^a \sigma^a / f_\pi}$ is the pion Goldstone matrix, with

$$\pi^a \sigma^a = \begin{pmatrix} \pi_0 & \sqrt{2}\pi_+ \\ \sqrt{2}\pi_- & -\pi_0 \end{pmatrix}$$

The pion axial current, $J_{A,\mu}^a$, reads at the LO

$$J_{A,\mu}^a \Big|_{\text{LO}} = \frac{i}{4} f_\pi^2 \text{Tr} \left[\sigma^a \left\{ U, (D_\mu U)^\dagger \right\} \right],$$

defined in terms of the covariant derivative $D_\mu U = \partial_\mu U - i r_\mu U + i U l_\mu$, with $r_\mu = r_\mu^a \sigma^a / 2$ and $l_\mu = l_\mu^a \sigma^a / 2$ external fields which can be used to include electromagnetic or weak effects. The matching of the derivative axion term with the corresponding one has been obtained by rewriting

$$\bar{q}_i [c_q]_{ij} \gamma^\mu \gamma_5 q_j = \frac{1}{2} (\text{Tr} [c_q] \underbrace{\bar{q} \gamma^\mu \gamma_5 q}_{\text{iso-singlet}} + \text{Tr} [c_q \sigma^a] \underbrace{\bar{q} \gamma^\mu \gamma_5 \frac{\sigma^a}{2} q}_{\text{iso-triplet}}),$$

where the Fierz identity $\sigma_{ij}^a \sigma_{kl}^a = 2(\delta_{il} \delta_{jk} - \frac{1}{2} \delta_{ij} \delta_{kl})$. The iso-singlet current is associated to the heavy η' and it can be neglected for our purposes, while the iso-triplet quark axial current is replaced with the pion axial current.

In the following, set $Q_a = M_q^{-1} / \text{Tr } M_q^{-1}$, so that terms linear in a (including a π^0 mass mixing) drop out and the only linear axion term arise from the derivative interaction with the pion axial current. Explicitly, the derivative axion coupling reads

$$\text{Tr} [c_q \sigma^a] = \left(\frac{m_u - m_d}{m_u + m_d} + c_u^0 - c_d^0 \right) \delta^{a3}.$$

Expanding the pion axial current $J_{A,\mu}^a \Big|_{\text{LO}} = f_\pi \partial_\mu \pi^a - \frac{1}{f_\pi} \pi^2 \partial_\mu \pi^a - \frac{3}{2f_\pi} \pi^a \partial_\mu \pi^2 + \dots$, with $\pi = \sqrt{\pi_0 \pi_0 + 2\pi_+ \pi_-}$, the axion-pion derivative terms are given by

$$\begin{aligned} \frac{\partial^\mu a}{2f_a} \text{Tr} [c_q \sigma^a] J_{A,\mu}^a \Big|_{\text{LO}} &\simeq -\frac{1}{2} \left(\frac{m_d - m_u}{m_u + m_d} + c_d^0 - c_u^0 \right) \frac{f_\pi}{f_a} \partial_\mu a \partial^\mu \pi_0 \\ &+ \frac{1}{3} \left(\frac{m_d - m_u}{m_u + m_d} + c_d^0 - c_u^0 \right) \frac{1}{f_a f_\pi} \partial_\mu a (2\partial^\mu \pi_0 \pi_+ \pi_- - \pi_0 \partial^\mu \pi_+ \pi_- - \pi_0 \pi_+ \partial^\mu \pi_-) \end{aligned}$$

The first operator introduces a kinetic mixing between the axion and the neutral pion, parametrized

by the coefficient

$$\epsilon \equiv -\frac{1}{2} \frac{f_\pi}{f_a} \left(\frac{m_d - m_u}{m_d + m_u} + c_d^0 - c_u^0 \right).$$

At the quadratic level the $a - \pi^0$ Lagrangian reads

$$\mathcal{L}_{a-\pi^0}^{\text{quad}} = \frac{1}{2} \begin{pmatrix} \partial_\mu a & \partial_\mu \pi_0 \end{pmatrix} \mathcal{K}_{\text{LO}} \begin{pmatrix} \partial^\mu a \\ \partial^\mu \pi_0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} a & \pi_0 \end{pmatrix} \mathcal{M}_{\text{LO}}^2 \begin{pmatrix} a \\ \pi_0 \end{pmatrix},$$

with

$$\mathcal{K}_{\text{LO}} = \begin{pmatrix} 1 & \epsilon \\ \epsilon & 1 \end{pmatrix}, \quad \mathcal{M}_{\text{LO}}^2 = \begin{pmatrix} m_a^2 & 0 \\ 0 & m_\pi^2 \end{pmatrix}$$

and $m_a^2 = m_{a,0}^2 + m_{a,\text{QCD}}^2$, where

$$m_{a,\text{QCD}}^2 = \frac{m_u m_d}{(m_u + m_d)^2} \frac{m_\pi^2 f_\pi^2}{f_a^2} \simeq 5.7 \left(\frac{10^{12} \text{GeV}}{f_a} \right) \mu\text{eV}$$

is the QCD axion mass squared at the LO. The procedure in order to diagonalize the quadratic Lagrangian consists of three steps: *i*) diagonalization of the kinetic term by an orthogonal transformation, *ii*) re-scaling of the fields to have a canonical kinetic term and *iii*) diagonalization of the mass matrix (rotated and re-scaled after steps *i*) and *ii*). The first orthogonal rotation

$$R_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

gives

$$R_1^T \mathcal{K}_{\text{LO}} R_1 = \begin{pmatrix} 1 - \epsilon & 0 \\ 0 & 1 + \epsilon \end{pmatrix}$$

Therefore the re-scaling is given by (fields need to be multiplied by the inverse of W)

$$W = \begin{pmatrix} \frac{1}{\sqrt{1-\epsilon}} & 0 \\ 0 & \frac{1}{\sqrt{1+\epsilon}} \end{pmatrix}$$

2.3 Axion Electrodynamics

consider the following system:

$$S = \int d^4x \left[-\frac{1}{2} (\partial_\mu a \partial^\mu a + m_a^2 a^2) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} g_{a\gamma\gamma} a F_{\mu\nu} \tilde{F}^{\mu\nu} \right]$$

The field strength $F_{\mu\nu}$ of the electromagnetic field A_μ and its dual $\tilde{F}_{\mu\nu}$ are given by

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \tilde{F}_{\mu\nu} \equiv \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$$

From the action, we can derive the equations of motion for the axion

$$(\square - m_a^2) a = \frac{1}{4} g_{a\gamma\gamma} F_{\mu\nu} \tilde{F}^{\mu\nu}$$

and for the photon

$$\partial_\mu F^{\mu\nu} = -g_{a\gamma\gamma} \tilde{F}^{\rho\nu} \partial_\rho a$$

These equations can be rewritten by using the potential $A_\mu = [-\phi, \vec{A}]$ and the electric and magnetic field,

$$\vec{E} = -\partial_t \vec{A} - \nabla \phi, \quad \vec{B} = \nabla \times \vec{A}$$

We can get the equations for the axion

$$(\square - m_a^2) a = g_{a\gamma\gamma} \vec{E} \cdot \vec{B}$$

and for the electromagnetic fields

$$\begin{cases} \square\phi = -g_{a\gamma\gamma}\vec{B} \cdot (\nabla a) \\ \square\vec{A} = g_{a\gamma\gamma}\left[(\partial_t a)\vec{B} + (\nabla a) \times \vec{E}\right] \end{cases} \quad (3.1.1)$$

Here, we have chosen the Lorenz gauge:

$$\nabla \cdot \vec{A} + \partial_t \phi = 0$$

Equation (3.1.1) are basic equations of the axion electrodynamics. For the full analysis, we need to resort to numerical calculations. Here, we use the perturbative analysis.

2.4 Perturbative Equations

Let us consider the situation that there is a static and uniform magnetic field in the background. We divide the fields into background and perturbation as follows:

$$\begin{cases} a(z, t) = 0 + \delta a(z, t) \\ \vec{B}(z, t) = \vec{B}_0 \text{ (const.)} + \delta \vec{B}(z, t) \\ \vec{E}(z, t) = 0 + \delta \vec{E}(z, t) \end{cases}$$

A coordinate basis is introduced so that the propagation is in the direction $\vec{e}_z = [0, 0, 1]$, one of the rests is parallel to the magnetic field $\vec{e}_\parallel = [1, 0, 0]$, and the other is $\vec{e}_\perp = [0, 1, 0]$. That is to say, the background magnetic field is represented as follows:

$$\vec{B}_0 = [B_0, 0, 0]$$

Since background equations trivially hold, we focus simply on perturbative equations in what follows. Linearizing Eq. (3.1.1), we obtain

$$\begin{cases} (\square - m_a^2) \delta a = -g_{a\gamma\gamma} B_0 (\partial_t \delta A_\parallel) \\ \square \delta A_\parallel = g_{a\gamma\gamma} B_0 (\partial_t \delta a) \\ \square \delta A_\perp = 0 \end{cases} \quad (3.2.1)$$

Here, we choose the radiation gauge:

$$\delta\phi = 0 \quad \text{and} \quad \nabla \cdot \delta\vec{A} = 0$$

since the source term of the equation for the scalar potential vanishes.

we can see that axions and photons have each other's components in the source term. Going one step further, what is called "axion photon conversion" usually denotes the equation, which is reduced to the firstorder differential equation [9].

2.5 Axion Photon Conversion

If the mass of the axion m_a is sufficiently smaller than the energy of the photons, the energy of the system can be considered to be determined by $\omega \simeq k$. In this case, we can assume axion and photon as follows:

$$\delta a(z, t) \equiv e^{i\omega(z-t)} \delta a(z) \delta A_\parallel(z, t) \equiv i e^{i\omega(z-t)} \delta A_\parallel(z)$$

Here, we have shifted the phase only for the photon in order to make a later equation easier to see. The left hand side of the equation for axion can be rewritten as follows:

$$\left(\frac{\partial}{\partial z}\right)^2 [e^{i\omega(z-t)} \delta a(z)] = e^{i\omega(z-t)} \left[\frac{\partial}{\partial z} + i\omega\right]^2 \delta a(z) \quad (2.4)$$

$$\simeq e^{i\omega(z-t)} \left[i2\omega \frac{d}{dz} - \omega^2\right] \delta a(z) \quad (3.3.1)$$

we use following formula:

$$\left(\frac{\partial}{\partial z}\right)^n [e^{i\omega z} f(z)] = e^{i\omega z} \left[\frac{\partial}{\partial z} + i\omega\right]^n f(z) \quad (3.3.2)$$

where $f(z)$ is an arbitrary function. Considering that $|i2\omega\delta a| \gg |d\delta a/dz|$ holds, the formula on the second line is obtained. By applying the same deformation to Eq. (3.2.3), the following equations can be obtained:

$$\begin{cases} i\frac{d\delta a}{dz} = \left(\frac{m_a^2}{2\omega}\right)\delta a - \frac{1}{2}g_{a\gamma\gamma}B_0\delta A_{\parallel} \\ i\frac{d\delta A_{\parallel}}{dz} = -\frac{1}{2}g_{a\gamma\gamma}B_0\delta a \end{cases} \quad (3.3.3)$$

As a result, we obtain the following Schrödinger-like equation,

$$i\frac{d}{dz}\vec{\psi} = \begin{bmatrix} \frac{m_a^2}{2\omega} & -\frac{1}{2}g_{a\gamma\gamma}B_0 \\ -\frac{1}{2}g_{a\gamma\gamma}B_0 & 0 \end{bmatrix} \vec{\psi}, \quad \vec{\psi} \equiv \begin{bmatrix} \delta a(z) \\ \delta A_{\parallel}(z) \end{bmatrix} \quad (3.3.4)$$

and this equation is the fundamental equation of axion photon conversion. The matrix in the right hand side is called the mixing matrix. The mass term of axion is denoted by

$$\Delta_a \equiv \frac{m_a^2}{2\omega} \quad (3.3.3)$$

and the mixing term is denoted by

$$\Delta_{a\gamma} \equiv -\frac{1}{2}g_{a\gamma\gamma}B_0 \quad (3.3.4)$$

As a side note, if we assumed

$$\delta A_{\parallel}(z, t) = e^{i\omega(z-t)}\delta A_{\parallel}(z)$$

then the mixing matrix would be a Hermitian matrix(analogous to Hamiltonian!).

2.6 Single Domain Helical Model:

We consider the situation where the direction of the magnetic field changes within a single domain in terms of the fixed $x - y$ coordinate system:

$$\vec{B}_{\text{helical}} \equiv B_0 \cos(\alpha z)\vec{e}_x + B_0 \sin(\alpha z)\vec{e}_y \quad (3.7.25)$$

In this case, the linearized equations of motion is as follows:

$$\begin{cases} (\square - m_a^2)\delta a = -g_{a\gamma\gamma}B_0[(\partial_t\delta A_x)\cos(\alpha z) + (\partial_t\delta A_y)\sin(\alpha z)] \\ \square\delta A_x = g_{a\gamma\gamma}B_0(\partial_t\delta a)\cos(\alpha z), \\ \square\delta A_y = g_{a\gamma\gamma}B_0(\partial_t\delta a)\sin(\alpha z) \end{cases} \quad (3.7.26)$$

we can reduce the first-order differential equations:

$$i\frac{d}{dz}\begin{bmatrix} \delta a(z) \\ \delta A_x(z) \\ \delta A_y(z) \end{bmatrix} = \begin{bmatrix} \Delta_a & \Delta_{a\gamma}\cos(\alpha z) & \Delta_{a\gamma}\sin(\alpha z) \\ \Delta_{a\gamma}\cos(\alpha z) & \Delta_{xx} & \Delta_{xy} \\ \Delta_{a\gamma}\sin(\alpha z) & \Delta_{yx} & \Delta_{yy} \end{bmatrix} \begin{bmatrix} \delta a(z) \\ \delta A_x(z) \\ \delta A_y(z) \end{bmatrix} \quad (3.7.27)$$

with

$$\begin{aligned} \Delta_{xx} &= \Delta_{\parallel}\cos^2(\alpha z) + \Delta_{\perp}\sin^2(\alpha z) \\ \Delta_{xy} &= \Delta_{yx} = (\Delta_{\parallel} - \Delta_{\perp})\cos(\alpha z)\sin(\alpha z) \\ \Delta_{yy} &= \Delta_{\parallel}\sin^2(\alpha z) + \Delta_{\perp}\cos^2(\alpha z) \end{aligned}$$

Here, $\Delta_{\parallel}$ and $\Delta_{\perp}$ represent effective photon mass. The correction to Δ_{QED} due to the magnetic field becoming z dependent can be ignored.

We perform a unitary transformation $\mathbf{U}$,

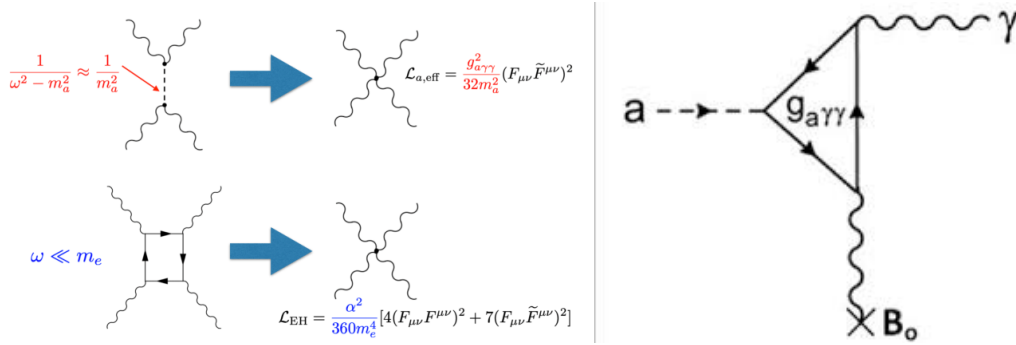
$$\begin{bmatrix} \delta a(z) \\ \delta A_{\parallel}(z) \\ \delta A_{\perp}(z) \end{bmatrix} = \mathbf{U} \begin{bmatrix} \delta a(z) \\ \delta A_x(z) \\ \delta A_y(z) \end{bmatrix}, \quad \mathbf{U} \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha z) & \sin(\alpha z) \\ 0 & -\sin(\alpha z) & \cos(\alpha z) \end{bmatrix}$$

and change the coordinate basis from the $x - y$ coordinate to the $\parallel - \perp$ coordinate.

$$i \frac{d}{dz} \vec{\Psi} = \mathbf{M}_{\text{helical}} \vec{\Psi}, \quad \mathbf{M}_{\text{helical}} \equiv \begin{bmatrix} \Delta_a & \Delta_{a\gamma} & 0 \\ \Delta_{a\gamma} & \Delta_{\parallel} & i\alpha \\ 0 & -i\alpha & \Delta_{\perp} \end{bmatrix}, \quad \vec{\Psi} \equiv \begin{bmatrix} \delta a(z) \\ \delta A_{\parallel}(z) \\ \delta A_{\perp}(z) \end{bmatrix} \quad (3.7.28)$$

2.7 Calculation without any constraints:

We repeat the basic equations below. The equation of motion for the axion is as follows:



$$(\square - m_a^2) a = g_{a\gamma\gamma} \vec{E} \cdot \vec{B} \quad (3.1.7)$$

and for electromagnetic fields,

$$\begin{cases} \square \phi = -g_{a\gamma\gamma} \vec{B} \cdot (\nabla a) \\ \square \vec{A} = g_{a\gamma\gamma} [(\partial_t a) \vec{B} + (\nabla a) \times \vec{E}] \end{cases} \quad (3.1.8)$$

with the Lorenz gauge:

$$\nabla \cdot \vec{A} + \partial_t \phi = 0 \quad (3.1.1.9)$$

We divide the fields into the background and the perturbation as follows:

$$\begin{cases} a(z, t) = a_0(t) & + \delta a(z, t) \\ \vec{B}(z, t) = \vec{B}_0 \text{ (const.)} & + \delta \vec{B}(z, t) \\ \vec{E}(z, t) = \vec{E}_0(t) & + \delta \vec{E}(z, t) \end{cases} \quad (4.0.1)$$

Background Equations

We assume that both a static magnetic field,

$$\vec{B}_0 = [B_0, 0, 0] \quad (4.1.1)$$

and a coherent oscillating axion $a_0(t)$ exist in the background. The coordinate bases are the same as for Sec. 3.2: the propagation direction $\vec{e}_z = [0, 0, 1]$, the direction parallel to the magnetic field $\vec{e}_{\parallel} = [1, 0, 0]$ and the direction normal to the magnetic field $\vec{e}_{\perp} = [0, 1, 0]$. The background equation for the axion is

$$\partial_t^2 a_0(t) + m_a^2 a_0(t) = -g_{a\gamma\gamma} B_0 E_{0\parallel}(t) \quad (4.1.2)$$

and for the photon

$$\partial_t E_{0\parallel}(t) = g_{a\gamma\gamma} B_0 [\partial_t a_0(t)] \quad (4.1.3)$$

Since the scalar potential $\phi(t)$ equation has no source term

$$\square\phi(t) = 0 \quad (4.1.4)$$

we choose the radiation gauge

$$\phi(t) = 0, \quad \nabla \cdot \vec{A}(\vec{x}) = 0 \quad (4.1.5)$$

We can see that the axion oscillation induces the electric field, which is parallel to the direction of the background magnetic field.

$$E_{0\parallel}(t) = g_{a\gamma\gamma} B_0 a_0(t) \quad (4.1.6)$$

Substituting we can get

$$\ddot{a}_0(\tau) + \Omega_\beta^2 a_0(\tau) = 0 \quad (4.1.7)$$

where we replace the variable t with $\tau \equiv m_a t$, and express a derivative with respect to τ by a dot. Here, we also introduced new dimensionless parameters β and Ω_β as follows:

$$\beta \equiv \frac{g_{a\gamma\gamma} B_0}{m_a} = 1.95 \times 10^{-9} \left(\frac{10^{-22} \text{eV}}{m_a} \right) \left(\frac{g_{a\gamma\gamma}}{10^{-11} \text{GeV}^{-1}} \right) \left(\frac{B_0}{10^{-9} \text{G}} \right) \quad (4.1.8)$$

$$\Omega_\beta \equiv \sqrt{1 + \beta^2} \quad (4.1.9)$$

As a result, there are uniform static magnetic field, oscillating axion field and oscillating electric field in the background:

$$a_0(\tau) = \bar{a} \cos(\Omega_\beta \tau) \quad (4.1.10)$$

$$E_{0\parallel}(\tau) = m_a \beta \bar{a} \cos(\Omega_\beta \tau) \quad (4.1.11)$$

We introduce energy density ρ which is the sum of axion and induced electric field:

$$\rho \equiv \frac{1}{2} (\partial_t a_0)^2 + \frac{1}{2} m_a^2 a_0^2 \Big|_{\text{present}} + \frac{1}{2} E_{0\parallel}^2 = \frac{1}{2} \bar{a}^2 m_a^2 \Omega_\beta^2 \quad (4.1.12)$$

then background energy density ρ_{BG} is given by

$$\rho_{\text{BG}} \equiv \rho + \frac{1}{2} B_0^2 \quad (4.1.13)$$

We determine the axion amplitude $\bar{a}$ by the energy density ρ ,

$$\bar{a} = \frac{\sqrt{2\rho}}{m_a \Omega_\beta} \quad (4.1.14)$$

Thus, we found the following expressions

$$a_0(\tau) = \frac{\sqrt{2\rho}}{m_a \Omega_\beta} \cos(\Omega_\beta \tau) \quad (4.1.15)$$

$$E_{0\parallel}(\tau) = \frac{\beta \sqrt{2\rho}}{\Omega_\beta} \cos(\Omega_\beta \tau) \quad (4.1.16)$$

4.2 Perturbative Equations

Now we deal with perturbative equations as follows: for the axion,

$$(\square - m_a^2) \delta a = g_{a\gamma\gamma} [\delta \vec{E} \cdot \vec{B}_0 + \vec{E}_0 \cdot \delta \vec{B}] \quad (4.2.1)$$

and for the photon,

$$\begin{cases} \square \delta \phi = -g_{a\gamma\gamma} \vec{B}_0 \cdot (\nabla \delta a) \\ \square \delta \vec{A} = g_{a\gamma\gamma} \left[(\partial_t \delta a) \vec{B}_0 + (\partial_t a_0) \delta \vec{B} + (\nabla \delta a) \times \vec{E}_0 \right] \end{cases} \quad (4.2.2)$$

We consider the situation where the magnetic field has the only x component, and the propagating direction of the axion and the photon is the z -axis. In this case, the scalar potential equation has no source term

$$\delta \phi(z, t) = 0 \quad (4.2.3)$$

Thus, we can choose the radiation gauge

$$\delta \phi(z, t) = 0, \quad \nabla \delta \vec{A}(z, t) = 0 \quad (4.2.4)$$

Here is the equation written with components,

$$(\square - m_a^2) \delta a(z, t) = g_{a\gamma\gamma} \left[\delta E_{\parallel}(z, t) B_0 - E_{0\parallel}(t) \frac{\partial \delta A_{\perp}(z, t)}{\partial z} \right] \quad (4.2.5)$$

$$\square \delta A_{\parallel}(z, t) = g_{a\gamma\gamma} \left[[\partial_t \delta a(z, t)] B_0 - [\partial_t a_0(t)] \frac{\partial \delta A_{\perp}(z, t)}{\partial z} \right] \quad (4.2.6)$$

$$\square \delta A_{\perp}(z, t) = g_{a\gamma\gamma} \left[[\partial_t a_0(t)] \frac{\partial \delta A_{\parallel}(z, t)}{\partial z} + \frac{\partial \delta a(z, t)}{\partial z} E_{0\parallel}(t) \right]. \quad (4.2.7)$$

Due to the time dependent oscillating axion field, the time translational symmetry is broken, but the system still has the spatial translation invariance. Hence, it is useful to perform the Fourier transformation,

$$\delta a(z, t) = \int \frac{dk}{2\pi} \delta a(k, t) e^{ikz} \quad (4.2.8)$$

$$\delta A_i(z, t) = \int \frac{dk}{2\pi} \delta A_i(k, t) e^{ikz} \quad (4.2.9)$$

$$\delta E_i(z, t) = - \int \frac{dk}{2\pi} [\partial_t \delta A_i(k, t)] e^{ikz} \quad (4.2.10)$$

where the subscript i denotes $\parallel$ or $\perp$. The equations are rewritten as follows:

$$\partial_t^2 \delta a(k, t) + (k^2 + m_a^2) \delta a(k, t) = g_{a\gamma\gamma} B_0 [\partial_t \delta A_{\parallel}(k, t)] + i g_{a\gamma\gamma} k E_{0\parallel}(t) \delta A_{\perp}(k, t) \quad (4.2.11)$$

$$\partial_t^2 \delta A_{\parallel}(k, t) + k^2 \delta A_{\parallel}(k, t) = -g_{a\gamma\gamma} B_0 [\partial_t \delta a(k, t)] + i g_{a\gamma\gamma} k [\partial_t a_0(t)] \delta A_{\perp}(k, t) \quad (4.2.12)$$

$$\partial_t^2 \delta A_{\perp}(k, t) + k^2 \delta A_{\perp}(k, t) = -i g_{a\gamma\gamma} k [\partial_t a_0(t)] \delta A_{\parallel}(k, t) - i g_{a\gamma\gamma} k E_{0\parallel}(t) \delta a(k, t). \quad (4.2.13)$$

In addition, substituting the background solutions, we get the following equations:

$$\delta \ddot{a}(k, \tau) + [1 + \kappa^2] \delta a(k, \tau) = \beta \delta \dot{A}_{\parallel}(k, \tau) + i \frac{\beta \epsilon}{\Omega_{\beta}} \cos(\Omega_{\beta} \tau) \delta A_{\perp}(k, t) \quad (4.2.14)$$

$$\delta \ddot{A}_{\parallel}(k, \tau) + \kappa^2 \delta A_{\parallel}(k, \tau) = -\beta \delta \dot{a}(k, \tau) - i \epsilon \sin(\Omega_{\beta} \tau) \delta A_{\perp}(k, t) \quad (4.2.15)$$

$$\delta \ddot{A}_{\perp}(k, \tau) + \kappa^2 \delta A_{\perp}(k, \tau) = i \epsilon \sin(\Omega_{\beta} \tau) \delta A_{\parallel}(k, \tau) - i \frac{\beta \epsilon}{\Omega_{\beta}} \cos(\Omega_{\beta} \tau) \delta a(k, \tau) \quad (4.2.16)$$

where we introduced dimensionless parameters κ and ϵ as follows:

$$\kappa \equiv \frac{k}{m_a}, \quad \epsilon \equiv \frac{g_{a\gamma\gamma} \sqrt{2\rho} k}{m_a^2} = \frac{g_{a\gamma\gamma} \sqrt{2\rho}}{m_a} \kappa. \quad (4.2.17)$$

From now on, for simplicity, we neglect higher order terms in β and ϵ . Taking into account only the first order in β and ϵ , we obtain

$$\delta\ddot{a}(k, \tau) + [1 + \kappa^2] \delta a(k, \tau) = \beta \delta \dot{A}_{\parallel}(k, \tau) \quad (4.2.18)$$

$$\delta\ddot{A}_{\parallel}(k, \tau) + \kappa^2 \delta A_{\parallel}(k, \tau) = -\beta \delta \dot{a}(k, \tau) - i\epsilon \sin(\tau) \delta A_{\perp}(k, \tau) \quad (4.2.19)$$

$$\delta\ddot{A}_{\perp}(k, \tau) + \kappa^2 \delta A_{\perp}(k, \tau) = i\epsilon \sin(\tau) \delta A_{\parallel}(k, \tau) \quad (4.2.20)$$

In the case of $\epsilon = 0$, the conventional axion photon conversion refetched.

Taking the circular polarization basis

$$\vec{e}_{L/R} = \frac{1}{\sqrt{2}} [\vec{e}_{\parallel} \mp i\vec{e}_{\perp}] \quad (4.2.21)$$

$$\delta\ddot{a}(k, \tau) + [1 + \kappa^2] \delta a(k, \tau) = \frac{\beta}{\sqrt{2}} [\delta \dot{A}_L(k, \tau) + \delta \dot{A}_R(k, \tau)] \quad (4.2.22)$$

$$\delta\ddot{A}_{L/R}(k, \tau) + [\kappa^2 \pm \epsilon \sin(\tau)] \delta A_{L/R}(k, \tau) = -\frac{\beta}{\sqrt{2}} \delta \dot{a}(k, \tau) \quad (4.2.23)$$

In the case of $\beta = 0$, Eqs. (4.2.23) end up the Mathieu equation and describe photon propagation in the presence of only axion dark matter [96]. Here, we should mention the previous work [97]. They investigated a similar system, but they neglected the parametric resonance. In our study, we consider the Mathieu type terms and deal with their resonance instability.

2.8 Correct form of the axion field

$$a(\mathbf{x}, t) = \text{Re} \left(\frac{\sqrt{\rho_{\text{DM}}(x)}}{m_a} \sum_i \alpha_i \sqrt{f(\mathbf{v}_i, x)} (\Delta v)^3 e^{i(\mathbf{k}_{a,i} \cdot \mathbf{x} - \omega_i t + \phi_i)} \right) \quad (1)$$

where f is the velocity distribution and the sum is over all possible velocities in the distribution.

The volume element $(\Delta v)^3$ is chosen small enough such that the f within the volume is almost constant.

For each velocity $\mathbf{v}_i$, α_i is a random number chosen from the Rayleigh distribution $P[\alpha] = \alpha e^{-\alpha^2/2}$ and ϕ_i is a random phase chosen uniformly from $[0, 2\pi]$.

As usual $\mathbf{k}_{a,i} = \frac{m\mathbf{v}_i}{h}$ (since the axions are all non-relativistic) and

$$w_i = \frac{\sqrt{m_a^2 c^4 + \hbar^2 k_{a,i}^2 c^2}}{h} = \frac{m_a c^2}{h} \sqrt{1 + (v_i/c)^2}.$$

Now since we adopt a velocity distribution with a cut off, the maximum speed that any DM particle will have in the frame of the earth is of the order of a few hundred kilometres. So the maximum amount that w_i varies from $\frac{mc^2}{h}$, call it Δw , will be such that

$$\frac{\Delta w}{w} = \frac{1}{2} \left(\frac{v_{\text{max}}}{c} \right)^2 \sim 10^{-6}.$$

If we assume that the resolution of our detector is not smaller than this (as we do from now on), then all of the $w_i \approx \frac{mc^2}{h} = w$ and the axion field is given by

$$a(x, t) = A(x) e^{-i\omega t} \quad (2)$$

$$\text{where } A(x) = \frac{\sqrt{\rho_{\text{DM}}(x)}}{m_a} \sum_i \alpha_i \sqrt{f(\mathbf{v}_i, x)} (\Delta v)^3 e^{i(\mathbf{k}_{a,i} \cdot \mathbf{x} + \phi_i)}$$

Also according to [1], the coherence length and time of the field are given by

$$\lambda_c \sim \frac{h}{m_{\text{DM}} v_0}, \quad \tau \sim \frac{h}{m_{\text{DM}} \bar{v} v_0} \quad (3)$$

where $\bar{v} \sim v_0 \sim 10^5$ m/s

So for $m_a = 10^{-10} \text{ eV}$ (we will come to why we are focusing on such a mass range later)

coherence length $\lambda_c \approx 10^4$ km

and coherence time $\tau \approx 100$ s.

This means that as long as we are looking at a region of space with length scale much smaller than around 10^4 km, equation (1) is a valid form of the field within it.

I interpreted the coherence time to mean that the random numbers involved in the field need

to be ‘rechosen’ from their respective distributions after every time τ , to accurately represent the field. So if we are doing our detection over a time much longer than τ and let’s say some function $g(\alpha_i, \phi_i)$ factors out of the calculated flux, then it can be replaced by the expectation value.

2.9 Power calculation

We carry out a calculation using the axion electrodynamics equations onwards.

If we have a volume V with a background magnetic field $\vec{B}_0(x)$ (and 0 electric field) and a bunch of dark matter represented by an axion field a , it turns out that the volume acts as a source for electromagnetic fields with a current density given by

$$\vec{j}_a(x, t) = -g\vec{B}_0(x)\frac{\partial a}{\partial t} \quad (4)$$

where g is the coupling constant in the lagrangian

So if we go to the Lorentz gauge $\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$

our equation for the vector potential becomes

$$\left(-\nabla^2 + \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) \vec{A}(x, t) = \vec{j}_a(x, t)$$

We assume that $\epsilon = \mu = 1$ all throughout space and mention later how the result will be modified if that is not the case.

Now we know that the solution is given by the retarded green’s function

$$A(x, t) = \frac{\mu_0}{4\pi} \int_V \frac{j_a(x', t - |x - x'|/c)}{|x - x'|} d^3x'$$

If we use the form in equation 2

$$j_a(x, t) = iwg\vec{B}_0(x)A(x)e^{-iwt}$$

$$A(x, t) = \frac{i\mu_0wg}{4\pi} \int \frac{A(x')\vec{B}_0(x')e^{-iwt}e^{i\omega|x-x'|/c}}{|x-x'|} d^3x'$$

If we make the approximation that we are in the radiation zone with $|x| \gg$

$|x'|$ for all x' in the volume V then we have

$$|x - x'| \approx |x| - \hat{x} \cdot x'$$

$$\text{So } A(x, t) = \frac{i\mu_0wge^{-iwt}e^{i(w/c)|x|}}{4\pi|x|} \int A(x') \vec{B}_0(x') e^{-i(w/c)\hat{x} \cdot x'} d^3x'$$

Now if we call $(w/c)\hat{x} = \vec{k}$ and $|x| = r$ then the above can be written as

$$A(x, t) = \frac{i\mu_0wge^{-iwt}e^{ikr}}{4\pi r} \int A(x') \vec{B}_0(x') e^{-i\vec{k} \cdot x'} d^3x'$$

Substituting $A(x')$ from equation 2 we get

$$A(x, t) = \frac{i\mu_0wge^{-iwt}e^{ikr}}{4\pi r} \int \frac{\sqrt{\rho_{DM}(x')}}{m_a} \vec{B}_0(x') \sum_i \alpha_i \sqrt{f(\mathbf{v}_i, x')} (\Delta v)^3 e^{i(\mathbf{k}_a, i \cdot \mathbf{x}' + \phi_i)} e^{-i\mathbf{k} \cdot \mathbf{x}'} d^3x'$$

$$A(x, t) = \frac{i\mu_0wge^{-iwt}e^{ikr}}{4\pi m_a r} \sum_i \alpha_i e^{i\phi_i} \int_V \vec{B}_0(x') \sqrt{\rho_{DM}(x')} \sqrt{f(\mathbf{v}_i, x')} (\Delta v)^3 e^{i(\mathbf{k}_a, i - \mathbf{k}) \cdot \mathbf{x}'} d^3x' \quad (5)$$

Looking at the above expression, $|k_{a,i}| = m|v_i|/\hbar$ and $|k| = mc/\hbar$ So

$(k_{a,i} - k) \cdot x' \approx -k \cdot x'$ almost everywhere in the volume as long as it is not some sort of thin disk whose plane is perpendicular to the direction of observation. Hence

$$A(x, t) = \frac{i\mu_0wge^{-iwt}e^{ikr}}{4\pi m_a r} \sum_i \alpha_i e^{i\phi_i} \int_V \vec{B}_0(x') \sqrt{\rho_{DM}(x')} \sqrt{f(\mathbf{v}_i, x')} (\Delta v)^3 e^{-i\mathbf{k} \cdot \mathbf{x}'} d^3x'$$

We should keep in mind that if the permittivity $\epsilon \neq 1$ then $k \rightarrow \sqrt{\epsilon}k$ which makes it smaller and comparable to $k_{a,i}$ which can make the above approximation break down (especially for smaller masses). Anyhow we accept equation 6 for now.

The behaviour of the above integral is clearly determined by the length scale $1/|k| = \hbar/mc \approx 10$ km for $m_a = 10^{-10}$ eV. In the allowed range of axion masses of 10^{-6} to 10^{-12} eV this is always smaller than about 1000 km. This means that we can already rule out axion-photon conversions in galactic magnetic fields because those, along with all the other quantities involved in the above integral vary on much larger kPc scales.

A volume of interest for us for example, would be one where B_0 is especially large for a distance $\ll 1/|k|$ and then rapidly falls off, slowly varying outside. The whole integral can then be approximated as just that over the volume of length scale $\ll 1/|k|$ where B_0 is high (as the

contribution from everything else would basically be zero). We call this volume V' . Now since this scale is much smaller than 1000 km in any case, the DM density and velocity distribution both do not vary over it and can be taken outside the integral.

Hence

$$A(x, t) = \frac{i\mu_0\omega g e^{-i\omega t} e^{ikr}}{4\pi m_a r} \left(\sum_i \alpha_i e^{i\phi_i} \sqrt{f(\mathbf{v}_i) (\Delta v)^3} \right) \sqrt{\rho_{DM}} \int_{V'} \vec{B}_0(x') d^3x' \quad (7)$$

For, say $m_a = 10^{-12} \text{eV}$ $1/k$ is around a 1000 km so if we can find a region of space of 10–100 km with a large magnetic field (like a pulsar magnetosphere, but that probably has other issues like the curved-space time and the plasma which make our equations invalid) the above form of the vector potential will be valid at the Earth (assuming the region is far enough so that we are in the radiation zone). Luckily we do not have to worry about the coherence length of the field within that volume because its always going to be about a factor of 10^3 larger than $1/k$, so the form of the axion field we have used is valid.

For large distances

$$B = \nabla \times A(x, t) = \frac{-\mu_0 k \omega g e^{-i\omega t} e^{ikr}}{4\pi m_a r} \left(\sum_i \alpha_i e^{i\phi_i} \sqrt{f(\mathbf{v}_i) (\Delta v)^3} \right) \sqrt{\rho_{DM}} \hat{r} \times \int_{V'} \vec{B}_0(x') d^3x' \quad (8)$$

Now we can calculate the time averaged power per unit solid angle in the standard way
 $\frac{dP}{d\Omega} = \frac{1}{2} \text{Re}(E^* \times B) \cdot r^2 \hat{r} = (c/2) |B|^2 r^2$
Hence

$$\frac{dP}{d\Omega} = \frac{c}{2} \left(\frac{\mu_0 k \omega g}{4\pi m_a} \right)^2 \rho_{DM} \left(\sum_i \alpha_i e^{i\phi_i} \sqrt{f(\mathbf{v}_i) (\Delta v)^3} \right)^2 \left| \hat{r} \times \int_{V'} \vec{B}_0(x') d^3x' \right|^2 \quad (9)$$

If our volume of strong magnetic field V' is at a distance R_0 from earth then we can show that the number flux we expect due to DM-photon conversions is

$F = \frac{c}{2\hbar\omega} \left(\frac{\mu_0 k \omega g}{4\pi m_a R_0} \right)^2 \rho_{DM} < \left(\sum_i \alpha_i e^{i\phi_i} \sqrt{f(\mathbf{v}_i) (\Delta v)^3} \right)^2 > \left| \hat{r} \times \int_{V'} \vec{B}_0(x') d^3x' \right|^2$
where $\hat{r}$ denotes the unit vector towards the volume from Earth and the triangle brackets denote the expectation value of the random variable expression (given that the flux is computed from a total count spanning many coherence times). As usual ρ and f are the density and velocity distribution at V' respectively.

For $\epsilon \neq 1$ the only change would be $c \rightarrow c/\sqrt{\epsilon}$ and $k \rightarrow \sqrt{\epsilon}k$ and the integral would be significant even if the magnetic field is inhomogenous at a scale $< 1/\sqrt{\epsilon}k$, which gives us even more possible regions of space to look at.

Take away

To summarise, here is the main takeaway: In order for us to see a flux of photons originating from DM along any line of sight, there must be some region of space along that line of sight where the magnetic field is inhomogenous at a scale much smaller than $1/k$ (say extremely high for some distance $\ll 1/k$ and then rapidly falling off outside or vice versa). If this is the case, then the flux will be given by equation 10 where V' is the region where this happens.

This result is more significant for the smaller masses in the allowed range $10^{-10} - 10^{-12} \text{eV}$ which have $1/k$ in the range of 10–1000 km because we can probably find such fields as described above (pulsars or whatever) especially with the $1/\sqrt{\epsilon}$ correction that allows even bigger regions of inhomogeneity.

Next reference Hook's paper <https://arxiv.org/pdf/1804.03145>

2.10 Effective Mass of Photon

So far, we assumed that only axions have a very small mass, but photons can also have an effective mass by some effects. In this section, we consider them.

The first effect is the plasma effect. The dispersion relation of the electromagnetic waves propagating in the plasma is corrected as follows:

$$\omega^2 = k^2 + \omega_p^2, \quad \omega_p^2 \equiv 4\pi\alpha_{\text{EM}} \frac{n_e}{m_e} \quad (3.7.1)$$

where α_{EM} is the fine structure constant, and n_e is the electron density, and m_e is the electron mass. We choose the natural Lorentz-Heaviside units. If one assumes that all the hydrogen atoms existing in the Universe are ionized by reionization and if one chooses electron density in outer space, $n_e \simeq 10^{-7} \text{ cm}^{-3}$, then the effective mass is approximately

$$\omega_p = 1.17 \times 10^{-14} \text{ eV} \left(\frac{n_e}{10^{-7} \text{ cm}^{-3}} \right) \quad (3.7.2)$$

The second effect is the QED effect. The corrections in the limit of $\omega \sim k \ll m_e$ is given by Euler-Heisenberg lagrangian:

$$\mathcal{L}_{\text{EH}} = \frac{\alpha_{\text{EM}}^2}{90m_e^4} \left[(F_{\mu\nu}F^{\mu\nu})^2 + \frac{7}{4} (\tilde{F}_{\mu\nu}F^{\mu\nu})^2 \right] \quad (3.7.3)$$

Incorporating the above two effects, the left hand side formula is corrected as follows:

$$(\square + 7\varrho_{\text{QED}}\omega^2 B_0^2 - \omega_p^2) \delta A_{\parallel} = g_{a\gamma\gamma} B_0 (\partial_t \delta a) \quad (3.7.4)$$

$$(\square + 4\varrho_{\text{QED}}\omega^2 B_0^2 - \omega_p^2) \delta A_{\perp} = 0 \quad (3.7.5)$$

Thus, we can define following quantities:

$$\Delta_{\text{plasma}} \equiv \frac{\omega_p^2}{2\omega} \quad (3.7.6)$$

$$\Delta_{\text{QED}}^{\parallel} \equiv -\frac{7}{2}\omega\varrho_{\text{QED}}B_0^2 \quad (3.7.7)$$

$$\Delta_{\text{QED}}^{\perp} \equiv -2\omega\varrho_{\text{QED}}B_0^2 \quad (3.8.8)$$

as components of the mixing matrix which represents the effective mass of photons. Here, we define

$$\varrho_{\text{QED}}B_0^2 \equiv \frac{\alpha_{\text{EM}}}{45\pi} \left(\frac{e}{m_e^2} \right)^2 B_0^2 = \frac{\alpha_{\text{EM}}}{45\pi} \left(\frac{B_0}{B_{\text{cr}}} \right)^2 \quad (3.8.9)$$

with

$$B_{\text{cr}} \equiv \frac{m_e^2}{e} = 4.42 \times 10^{13} \text{ G} \quad (3.8.10)$$

2.11 Conversion Probability

Let us evaluate a conversion probability by solving the equation,

$$i \frac{d}{dz} \vec{\psi} = \mathbf{M}_{\parallel} \vec{\psi}, \quad \mathbf{M}_{\parallel} \equiv \begin{bmatrix} \Delta_a & \Delta_{a\gamma} \\ \Delta_{a\gamma} & \Delta_{\parallel} \end{bmatrix} \quad (3.5.1)$$

where $\Delta_{\parallel}$ is the mass term of photon, i.e., Δ_{plasma} and/or $\Delta_{\text{QED}}^{\parallel}$. Since the mixing matrix $\mathbf{M}_{\parallel}$ is a symmetric matrix, it can be diagonalized by using an orthogonal matrix $\mathbf{O}$,

$$\mathbf{O}^{-1}\mathbf{M}_{\parallel}\mathbf{O} = \begin{bmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{bmatrix}, \quad \mathbf{O} \equiv \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad (3.7.2)$$

where θ is called the mixing angle, and $\lambda_{\pm}$ are eigenvalues,

$$\lambda_{\pm} \equiv \frac{1}{2} \left[(\Delta_a + \Delta_{\parallel}) \pm \sqrt{(\Delta_a - \Delta_{\parallel})^2 + (2\Delta_{a\gamma})^2} \right] \quad (3.7.3)$$

Therefore, we can solve Eq. (3.7.1) as follows:

$$\psi_i(z) = \sum_{j=1}^2 \mathbf{O}_{ij} \left[\mathbf{O}^{-1} \vec{\psi}(0) \right]_j e^{-\lambda_j z} \quad (3.7.4)$$

where $\lambda_1 \equiv \lambda_+$ and $\lambda_2 \equiv \lambda_-$; that is,

$$\delta a(z) = [\cos^2 \theta e^{-i\lambda_+ z} + \sin^2 \theta e^{-i\lambda_- z}] \delta a(0) + \cos \theta \sin \theta [e^{-i\lambda_+ z} - e^{-i\lambda_- z}] \delta A_{\parallel}(0) \quad (3.7.5)$$

$$\delta A_{\parallel}(z) = \cos \theta \sin \theta [e^{-i\lambda_+ z} - e^{-i\lambda_- z}] \delta a(0) + [\sin^2 \theta e^{-i\lambda_+ z} + \cos^2 \theta e^{-i\lambda_- z}] \delta A_{\parallel}(0) \quad (3.7.6)$$

For example, when we choose $\delta a(0) = 0$ and $\delta A_{\parallel}(0) = 1$, the conversion probability of the photon into the axion after traveling a distance z is given as follows:

$$P_0(z) \equiv (\sin 2\theta)^2 \sin^2 \left[\frac{\sqrt{(\Delta_a - \Delta_{\parallel})^2 + (2\Delta_{a\gamma})^2}}{2} z \right] \quad (3.7.7)$$

We refer to

$$\Delta_{\text{osc}} \equiv \lambda_+ - \lambda_- = \sqrt{(\Delta_a - \Delta_{\parallel})^2 + (2\Delta_{a\gamma})^2} \quad (3.7.8)$$

as the oscillation length. If one realizes the following relation,

$$\sin 2\theta = \frac{2\Delta_{a\gamma}}{\Delta_{\text{osc}}} \quad (3.7.9)$$

then we can rewrite Eq. (3.7.7) as follows:

$$P_0(z) = (\Delta_{a\gamma} z)^2 \frac{\sin^2 \left(\frac{\Delta_{\text{osc}}}{2} z \right)}{\left(\frac{\Delta_{\text{osc}}}{2} z \right)^2} \quad (3.7.10)$$

The axion photon conversion is the most effective when the following condition is satisfied:

$$\Delta_{a\gamma} \gg |\Delta_a - \Delta_{\parallel}|, \tan 2\theta_s = \frac{2\Delta_{a\gamma}}{\Delta_a - \Delta_{\parallel}} \gg 1 \quad (3.7.11)$$

In this case, $\theta_s \sim \pi/4$ and $\Delta_{\text{osc},s} \simeq 2\Delta_{a\gamma}$. Here, the subscript "s" is added to clearly indicate that this is the case for strong coupling. When $\Delta_{\text{osc}} \ll 1$ holds, Eq. (3.7.10) can be further simplified:

$$P_0(z) \simeq (\Delta_{a\gamma} z)^2 \quad (3.7.12)$$

2.12 Effects on the Polarization of Photons

Since only one of the photon's two components can mix with the axion, the axion photon conversion can affect the polarization of photons. The photon polarization is described in the context of Stokes parameters (I, Q, U, V),

$$\begin{cases} I(z) & \equiv \delta A_{\parallel} \delta A_{\parallel}^* + \delta A_{\perp} \delta A_{\perp}^* \\ Q(z) & \equiv \delta A_{\parallel} \delta A_{\parallel}^* - \delta A_{\perp} \delta A_{\perp}^* \\ U(z) & \equiv \delta A_{\parallel} \delta A_{\perp}^* + \delta A_{\perp} \delta A_{\parallel}^* \\ V(z) & \equiv i \left[\delta A_{\parallel} \delta A_{\perp}^* - \delta A_{\perp} \delta A_{\parallel}^* \right] \end{cases} \quad (3.8.1)$$

They satisfy a following equation.

$$I^2(z) = Q^2(z) + U^2(z) + V^2(z) \quad (3.8.2)$$

Note that $I(z)$ and $V(z)$ are invariant under the coordinate transformation, but $Q(z)$ and $U(z)$ depend on the coordinate system. Using the intensity of the photon $I(z)$, we can define the degree of circular polarization

$$\Pi_C = \frac{|V(z)|}{I(z)} \quad (3.8.3)$$

and the degree of linear polarization,

$$\Pi_L = \frac{\sqrt{Q^2(z) + U^2(z)}}{I(z)} \quad (3.8.4)$$

Since $\delta A_{\perp}$ which do not participate in the conversion must be included in the calculation of Stokes parameters, we redefine the mixing matrix $\mathbf{M}$ as follows:

$$i \frac{d}{dz} \vec{\Psi} = \mathbf{M} \vec{\Psi}, \quad \mathbf{M} \equiv \begin{bmatrix} \Delta_a & \Delta_{a\gamma} & 0 \\ \Delta_{a\gamma} & \Delta_{\parallel} & 0 \\ 0 & 0 & \Delta_{\perp} \end{bmatrix}, \quad \vec{\Psi} \equiv \begin{bmatrix} \delta a(z) \\ \delta A_{\parallel}(z) \\ \delta A_{\perp}(z) \end{bmatrix} \quad (3.8.5)$$

Stokes parameters can be calculated by using Eq. (3.7.5) and Eq. (3.7.6), but they have a simpler form in the case of $\Delta_{\parallel} = \Delta_{\perp} = \Delta_{\text{plasma}}$, because the polarization depends only on $(\Delta_a - \Delta_{\text{plasma}})$. The Stokes parameters do not change when we redefine the field as $\Psi_i e^{-i\Delta_{\text{plasma}} z}$. While the mixing matrix is transformed as

$$\mathbf{M} = \begin{bmatrix} \Delta_a - \Delta_{\text{plasma}} & \Delta_{a\gamma} & 0 \\ \Delta_{a\gamma} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (3.8.6)$$

Thus, we can consider only the case $\Delta_{\text{plasma}} = 0$ without loss of generality. The result in the case $\Delta_{\text{plasma}} \neq 0$ can be obtained by replacing Δ_a with $(\Delta_a - \Delta_{\text{plasma}})$. Under the assumption $\Delta_{\parallel} = \Delta_{\perp} = \Delta_{\text{plasma}}$ and an initial condition: $\delta a(0) = 0$ and $V_0 = 0$, we can evaluate polarizations as follows:

$$I(z) = I_0 - (I_0 + Q_0) \left(\frac{\Delta_M}{\Delta_{osc}} \right)^2 [1 - \cos(\Delta_{osc} z)] \quad (3.8.7)$$

$$Q(z) = Q_0 - (I_0 + Q_0) \left(\frac{\Delta_M}{\Delta_{osc}} \right)^2 [1 - \cos(\Delta_{osc} z)] \quad (3.8.8)$$

$$U(z) = \frac{U_0 \Delta_M^2}{\Delta_{osc}} \left[\frac{\cos(\lambda_+ z)}{\lambda_+} - \frac{\cos(\lambda_- z)}{\lambda_-} \right] \quad (3.8.9)$$

$$\Pi_C(z) = \left| \frac{U_0 (\Delta_M^2 / \Delta_{osc})}{I_0 - (I_0 + Q_0) (\Delta_M / \Delta_{osc})^2 [1 - \cos(\Delta_{osc} z)]} \left[\frac{\sin(\lambda_+ z)}{\lambda_+} - \frac{\sin(\lambda_- z)}{\lambda_-} \right] \right| \quad (3.8.10)$$

where I_0, Q_0, U_0 represent the initial Stokes parameters.

Chapter 3

Axion–Photon Coupling in Our Galaxy

3.1 Solution in Uniform Magnetic Field:

- The ALP Langrangian has a term of the form $a \frac{\mathbf{E} \cdot \mathbf{B}}{M}$. If we now have a monochromatic beam of frequency w propagating along the z-axis in the presence of some arbitrary field $\mathbf{B}$, let us accept that this leads to an equation of motion that looks like:

$$(wI + \begin{pmatrix} \Delta_\gamma & \Delta_F & \Delta_{\gamma ax} \\ \Delta_F & \Delta_\gamma & \Delta_{\gamma ay} \\ \Delta_{\gamma ax} & \Delta_{\gamma ay} & \Delta_a \end{pmatrix}) \begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix} = i\partial_z \begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix}$$

where $\begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix}$ is the regular 3- component wave function of the particle ie. the probability of finding it in an interval dz around z as a photon polarised in the x direction is $|\gamma_x(z)|^2 dz$, as a photon polarised in the y-direction is $|\gamma_y(z)|^2 dz$, and as an axion is $|a(z)|^2 dz$

- Let us say we are in a region where the magnetic field is uniform for some length along the z-axis, say L . We may now choose the y-axis along the projection of $\mathbf{B}$ perpendicular to the z-axis .
- Defining the terms from the first slide, $\Delta_\gamma = -w_{pl}^2/2w$ where w_{pl} is the plasma frequency and $w_{pl}^2 = 4\pi n_e/m_e$ where n_e is the free electron density. $\Delta_{\gamma ax} = B_x/2M$, $\Delta_{\gamma ay} = B_y/2M$, and $\Delta_a = -m_a^2/w$ and Δ_F has to do with mixing of the two photon polarisation states which we do not care about and hence will be setting to 0. Also by the choice of our axes, $B_x = 0$. Therefore $\Delta_F = \Delta_{\gamma ax} = 0$ and the simplified equations are

$$(wI + \begin{pmatrix} \Delta_\gamma & 0 & 0 \\ 0 & \Delta_\gamma & \Delta_{\gamma ay} \\ 0 & \Delta_{\gamma ay} & \Delta_a \end{pmatrix}) \begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix} = i\partial_z \begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix}$$

- This leads to $\dot{\gamma}_x = -i(w + \Delta_\gamma)\gamma_x$ and $\begin{bmatrix} \dot{\gamma}_y(z) \\ \dot{a}(z) \end{bmatrix} = -i \begin{pmatrix} w + \Delta_\gamma & \Delta_{\gamma ay} \\ \Delta_{\gamma ay} & w + \Delta_a \end{pmatrix} \begin{bmatrix} \gamma_y(z) \\ a(z) \end{bmatrix}$
- If we now further assume that the free electron density n_e is constant in this region of length L then these equations can be solved explicitly given the initial conditions. So let us say that we have an axion produced at $z=0$ ie. we know for sure that if we look at $z=0$ our particle is going to be an axion, not a photon. $a(0) = 1, \gamma_x(0) = 0, \gamma_y(0) = 0$. The first equation clearly implies that $\gamma_x(z) = 0$. To solve the second we invoke some general results on diagonalising 2*2 symmetric matrices:

If we have a matrix $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ then we know it gets diagonalised ie. $A = PDP^{-1}$ by

$P = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$ where $\tan(2\theta) = 2c/(a-b)$ and $D = \text{diag.}(\lambda_1 = (a+b+\Delta)/2, \lambda_2 =$

$(a + b - \Delta)/2$ where $\Delta = ((b - a)^2 + 4c^2)^{1/2}$

So the most general solution is $c_1 \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} e^{\lambda_1 z} + c_2 \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix} e^{\lambda_2 z}$

- In our case $\tan(2\theta) = \frac{2\Delta_{\gamma ay}}{\Delta_\gamma - \Delta_a}$, $\lambda_1 = -i(2w + \Delta_\gamma + \Delta_a + \Delta)/2$, $\lambda_2 = -i(2w + \Delta_\gamma + \Delta_a - \Delta)/2$ where $\Delta = ((\Delta_\gamma - \Delta_a)^2 + 4\Delta_{\gamma ay}^2)^{1/2}$
- So our solution must be $\begin{bmatrix} \gamma_y(z) \\ a(z) \end{bmatrix} = \sin(\theta) \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} e^{\lambda_1 z} + \cos(\theta) \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix} e^{\lambda_2 z}$ as that is the only choice of constants which will satisfy the initial conditions .
- Now the probability that the axion produced at $z=0$ is found to be a photon (either x or y polarised) by the time it reaches $z=L$ is $|\gamma_y(L)|^2 + |\gamma_x(L)|^2 = |\gamma_y(L)|^2 = \sin^2(\theta)\cos^2(\theta)(2 - 2\cos(\Delta L)) = 2\sin^2(\theta)\cos^2(\theta)2\sin^2(\frac{\Delta L}{2}) = \sin^2(2\theta)\sin^2(\frac{\Delta L}{2})$
- If we further assume that the ALP is massless, $\Delta_a = 0$ so $\tan(2\theta) = \frac{2\Delta_{\gamma ay}}{\Delta_\gamma}$ and $\Delta = ((\Delta_\gamma)^2 + 4\Delta_{\gamma ay}^2)^{1/2}$ which means that $\cos(2\theta) = \frac{\Delta_\gamma}{\Delta} = \frac{w_{pl}^2}{2w\Delta}$. Hence $\Delta = \frac{w_{pl}^2}{2w\cos(2\theta)}$

3.2 Estimating the average probability of ALP Conversion from a galaxy :

- From the previous slide it follows that $P_{a \rightarrow \gamma}(z = L) = \sin^2(2\theta)\sin^2(\frac{k}{\cos(2\theta)})$ where $k = \frac{w_{pl}^2 L}{4w}$ and $\tan(2\theta) = \frac{2\Delta_{\gamma ay}}{\Delta_\gamma} = \frac{2B_\perp w}{Mw_{pl}^2}$ where $B_\perp$ is the constant magnetic field in the direction perpendicular to the propagation of the ALP in a coherence region of length L .
- When θ and k are both small as is the case for most of the Milky Way for w corresponding to a 3.55 keV beam (the one we have detected and are interested in) , the probability reduces to $4\theta^2 k^2 = 4B_\perp^2 w^2 / (M^2 w_{pl}^4) * w_{pl}^4 L^2 / 16w^2 = \frac{B_\perp^2 L^2}{4M^2}$

Now if an axion is produced from dark matter decaying in some region of a galaxy, the probability that it turns to a photon by the time of reaching Earth technically depends on the path it takes (what the magnetic field, plasma frequency and as we will see later hydrogen density look like on its way). The path it takes to reach the detector is determined by where the axion was formed, so we cannot say that every axion produced in a galaxy or galaxy cluster will convert to a photon with the same probability (it is a function of position). However as a rough estimate of the average, we may use the probability of a 'typical' axion produced somewhere in the central region of the galaxy because the axions produced 'further' away from it will travel through a larger region of magnetic field and have a greater probability of conversion while the ones produced closer will have a smaller probability. Now for this typical axion say R away from us, if we divide its path to us into (R/L) regions of length L such that the magnetic field perpendicular to the path remains fairly uniform in each region of L (called the coherence length) and has an average value $\bar{B}_\perp$ then the probability it becomes a photon before reaching is the sum of the probabilities that it becomes a photon in each of the domains $= (R/L) * \frac{\bar{B}_\perp^2 L^2}{4M^2} = \frac{\bar{B}_\perp^2 LR}{4M^2}$, which is the key relation we wanted to motivate.

3.3 Photon Fluxes and estimating the decay time :

Defined as the number of counts per unit solid angle per unit area per unit time . It is obviously a function of the direction (θ, ϕ) in which the detector is pointed. So for a given direction $1/\tau$ axions are produced by a DM particle per second on average so $\frac{1}{\tau} \left(\frac{P_{DM} \Omega r^2 dr}{m_{DM}} \right)$ axions formed in

the volume shown if A is area of detector then the total flux in the detector along the line of sight is given as:

$$\int_0^\infty \frac{1}{\tau} \left(\frac{P_{DM} \Omega r^2 dr}{m_{DM}} \right) \left(\frac{A}{4\pi r^2} \right) \rho(r, \theta, \varphi) dr$$

- Now it was found from the observations of the flux of the 3.55 keV line from a 'sample of stacked galaxy clusters' (clarify what this means) that the flux is such that to produce the same counts, dark matter would have to decay directly to photons with a rate of $5 * 10^{27} s$
- If we assume that a good estimate of the average probability of an axion produced anywhere in this collection of galaxy clusters converting to a photon is the probability of a photon produced in the central $1Mpc^3$ of Coma (technically an average again, but the central $1Mpc^3$ of something $100Mpc$ away looks almost like a point) $\frac{\bar{B}_\perp^2 LR}{4M^2}$ as derived above this turns out to be $10^{-3} \left(\frac{10^{13}}{M_{GeV}} \right)^2$. Will have to understand the stacking process better to see why this assumption is necessary. Roughly, there is a weighted 'modified column density' (some sort of average of $\int P \rho dl$ over the line of sight for each galaxy cluster) so assuming the same average P for each cluster is necessary to get it out in common.
- Using this average probability in the integral for the flux, $\frac{1}{\tau_{DM \rightarrow a}} 10^{-3} \left(\frac{10^{13}}{M_{GeV}} \right)^2 = \frac{1}{\tau_{DM \rightarrow \gamma}} = \frac{1}{5 \times 10^{27}}$ which means $\tau_{DM \rightarrow a} = 5 \times 10^{24} \left(\frac{10^{13}}{M_{GeV}} \right)^2$. This is the estimate we desired and finding the product $\tau_{DM \rightarrow \gamma} M^2$ is the best we can do from the data. We need to know this decay time (at least in terms of the coupling constant) to predict the flux from different directions in the Milky Way (that is the final goal).

3.4 Improvement of finding $P(r, \theta, \phi)$

- It turns out that the earlier propagation equation only applies to a pure electron plasma but taking into account the photoelectric absorption leads to the new solution, in a region of constant magnetic field, electron density, and hydrogen density (say $0 < z < L$) becoming

$$\rho(z) = e^{-iHz} \rho(0) e^{iH^\dagger z} \text{ where } \rho(z) \text{ is the density matrix } \begin{bmatrix} \gamma_x(z) \\ \gamma_y(z) \\ a(z) \end{bmatrix} [\gamma_x(z) \quad \gamma_y(z) \quad a(z)]^*$$

and

$$H = \begin{pmatrix} \Delta_\gamma & \Delta_F & \Delta_{\gamma ax} \\ \Delta_F & \Delta_\gamma & \Delta_{\gamma ay} \\ \Delta_{\gamma ax} & \Delta_{\gamma ay} & \Delta_a \end{pmatrix} - i \begin{pmatrix} \Gamma/2 & 0 & 0 \\ 0 & \Gamma/2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where $\Gamma = \sigma_{eff}(n_H + 2n_{H2})$. $n_H + 2n_{H2}$ is the total density of hydrogen atoms and σ_{eff} is some constant.

- Practically we would break up our path into regions of length δz where the matrix H is almost constant and iteratively compute the density matrix in the center of each domain. $\rho_k = e^{-iH_k \delta z} \rho_{k-1} e^{iH_k^\dagger \delta z}$ where H_k is the H matrix from above in the center of the k th domain.

- More precisely, we would first set $\rho(r) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and then break up the straight line path

from (r, θ, ϕ) to $(0, 0, 0)$ into domains of length δz , compute H_k in the center of each domain using models of the magnetic field, free electron density, and hydrogen density. Using these we would arrive at the density matrix $\rho(0)$ and the sum of the first two diagonal entries of this, $|\gamma_x(0)|^2 + |\gamma_y(0)|^2$, will give us the probability of an axion produced at (r, θ, ϕ) converting to a photon by the time it reaches Earth, ie. $P(r, \theta, \phi) = \rho(0)_{11} + \rho(0)_{22}$.

- Another simplification we will make is assuming that most axions originating from further than 30kpc away do not convert before coming within 30kpc since the magnetic fields are weak. So $P(r > 30kpc, \theta, \phi) = P(30kpc, \theta, \phi)$
- Once we have $P((r, \theta, \phi))$ the expected photon flux in a particular direction due to $DM \rightarrow a \rightarrow \gamma$ decays can be easily found using the equation derived above, $F(\theta, \phi) = \frac{1}{4\pi\tau_{DM \rightarrow a}} \int_0^\infty \frac{\rho_{DM}}{m_{DM}} P(r, \theta, \phi) dr$, after we subscribe to a model of dark matter density like the NFW profile. Also note that this will be in units of M^2 if we employ the estimate of the dark matter to axion decay time which we found earlier. Normalising by M^2 will result in a plot of expected flux as

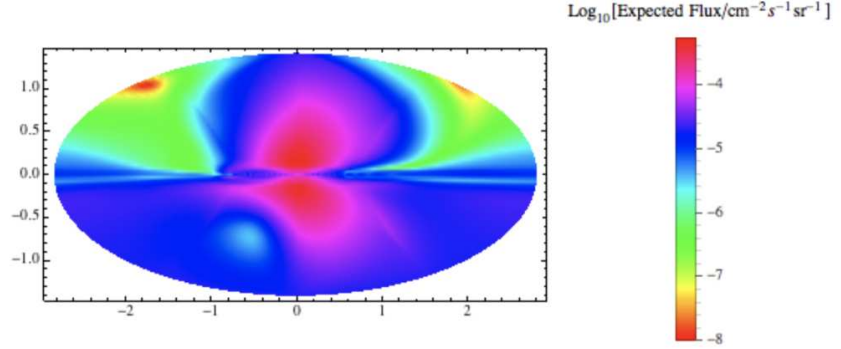


Figure 3. The expected 3.55 keV line flux across the sky in the ALP scenario, using the NFW dark matter density and an electron density scale height $h = 1.1$ kpc.

shown in the paper:

3.5 Some Comments:

- θ is on the y-axis and ϕ on the x-axis.
- We have some freedom in choosing the model for the electron density (in the sense that we do not know which one is more accurate) so calculations are repeated with both possible models
- Same goes with the dark matter profile. Calculations are repeated with the NFW and Einasto profiles.
- The highest expected flux of the 3.55 keV line in any direction is found to be in the range of $10^{-4} cm^{-2} s^{-1} sr^{-1}$ (with M set to $10^{13} GeV$ probably) from the Milky Way. From this if we deduce the count rate for a typical instrument (using its area and field of view) it turns out to be around $4 * 10^{-8}$ per second which cannot be detected. So a failure to detect this line in the Milky Way in fact supports our hypothesis that its origin is $DM \rightarrow a \rightarrow \gamma$ conversions rather than just $DM \rightarrow \gamma$ because in the latter case our expected flux would be higher by a factor of 1000 and the lines would be visible (using the acceptable range of DM to photon decay times I guess)
- Another reason our model is consistent is because we are able to observe the 3.55 keV line from a cluster like Coma where we had predicted an average conversion probability of 10^{-3} for $M = 10^{13} GeV$ whereas we are not able to see the line from the Milky Way where even for axions originating from 30 kpc (its likely to be maximum at this distance) away the maximum probability of conversion is still only $\approx 10^{-6}$ as shown in this plot:

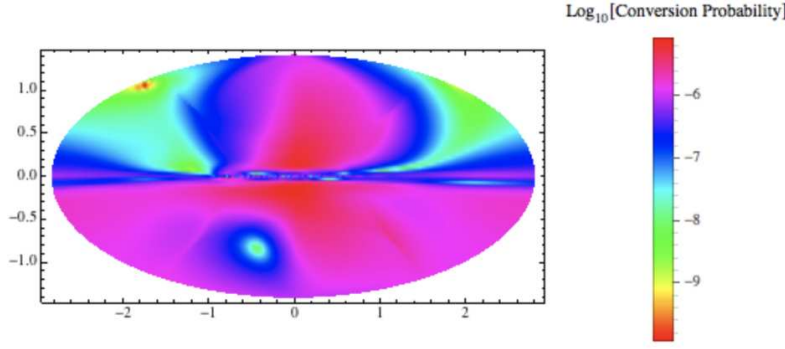


Figure 2. The ALP to photon conversion probability for an ALP with $M = 10^{13}$ GeV starting at 30 kpc from the Earth and propagating to Earth using a vertical scale height $h = 1.1$ kpc for the electron density. Galactic longitude increases to the right and galactic latitude increases vertically. The centre of the plot corresponds to the direction of the galactic centre.

3.6 Evidence from the Andromeda Galaxy and Perseus cluster for the sterile neutrino model

The results of [https://arxiv.org/pdf/1402.4119.pdf] shown:

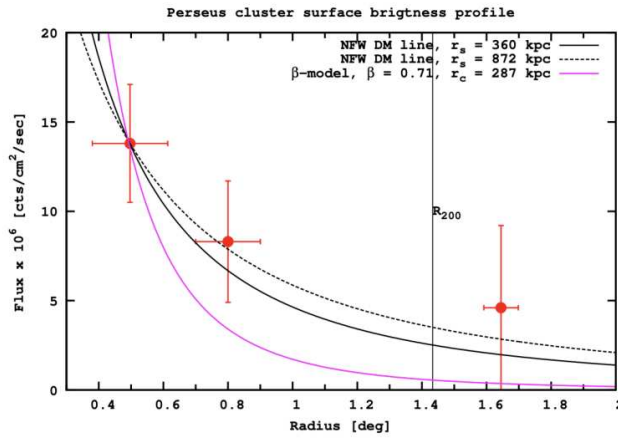
Dataset	Exposure [ksec]	$\chi^2/\text{d.o.f.}$	Line position [keV]	Flux [10^{-6} cts/sec/cm 2]	$\Delta\chi^2$	Significance
M31 ON-CENTER	978.9	97.8/74	3.53 ± 0.03	$4.9^{+1.6}_{-1.3}$	13.0	3.2σ
M31 OFF-CENTER	1472.8	107.8/75	$3.50 - 3.56$	< 1.8 (2σ)	...	
PERSEUS CLUSTER (MOS)	628.5	72.7/68	3.50 ± 0.04	$7.0^{+2.6}_{-2.6}$	9.1	2.6σ
PERSEUS CLUSTER (PN)	215.5	62.6/62	3.46 ± 0.04	$9.2^{+3.1}_{-3.1}$	8.0	2.4σ
PERSEUS (MOS)	1507.4	191.5/142	3.52 ± 0.02	$8.6^{+2.3}_{-2.3}$ (Perseus)	25.9	4.4σ
+ M31 ON-CENTER				$4.6^{+1.4}_{-1.4}$ (M31)	(3 dof)	
BLANK-SKY	15700.2	33.1/33	$3.45 - 3.58$	< 0.7 (2σ)	...	

- Firstly we note that M31 is roughly an on-edge galaxy ie. Earth is in the plane of its disk (actually there is a ≈ 12 degree tilt but we neglect that in further analysis). The On-Center flux from M31 is some sort of average of the flux obtained from 29 lines of sight within $1.5'$ of the center and the off-center flux is the average of 20 lines of sight between $23'$ and $56'$ of the center. The data for Perseus is an average of 16 off-center observations.
- The flux in these data is not normalised to per unit solid angle, so $F = \frac{1}{4\pi\tau_{DM \rightarrow \gamma} m_{DM}} \Omega_{fov} \int \rho_{DM} dl$ if we propose that the origin of the 3.55 keV line is the direct decay of dark matter to photons. Ω_{fov} is the solid angle subtended by the field of view. The integral is technically along the entire line of sight but if we are confident of the dark matter being mainly localised only within the galaxy/cluster then we might as well take it only along the interior. Let us call the integral along any particular line of sight the column density, $S_{DM} = \int \rho_{DM} dl$. Now if we go by the model where DM is entirely composed of sterile neutrinos of mass m which decay into photons of mass $\frac{m}{2}$ then if the line was originating from this decay, m_{DM} would have to be 7.1 keV. Using the NFW model of M31 we also know that the average column density over a few lines going through the on-centre region of M31 is $600 \frac{M_{\odot}}{\text{pc}^2}$. Also the field of view of XMM Newton is a $30'$ cone, which subtends a solid angle of $\pi(15')^2 = 225\pi \text{ arcmin}^2$ (although the data shows a $\text{FOV} \approx 500 \text{ arcmin}^2$?) With the appropriate conversions, the flux is given by:

$$F_{DM} \approx 2.0 \times 10^{-6} \frac{\text{cts}}{\text{cm}^2 \cdot \text{sec}} \left(\frac{\Omega_{fov}}{500 \text{ arcmin}^2} \right) \times \left(\frac{S_{DM}}{500 M_{\odot}/\text{pc}^2} \right) \frac{10^{29} \text{ s}}{\tau_{DM}} \left(\frac{\text{keV}}{m_{DM}} \right).$$

Using the average on-centre flux from M31 of $4.9 \times 10^{-6} \text{ cts/sec/cm}^2$ this gives us $\tau_{DM \rightarrow \gamma} \approx 9.75 \times 10^{27} \text{ s}$

- This is now consistent for 2 reasons
 - The expected average column density of a line of sight through Perseus is also in the range of 200-600 $\frac{M_\odot}{pc^2}$ so this value of the decay constant leads to a roughly correct prediction of the flux (although slightly lower than what we observe). For it to be completely consistent, column density through Perseus should have been higher as the observed flux from Perseus is larger.
 - As discussed earlier, <https://arxiv.org/pdf/1402.2301.pdf> used the data from a stacked sample of galaxy clusters to conclude that if the origin of the 3.55 keV line is sterile neutrino decays then their lifetime must be $5 \times 10^{27}s$. This is quite close considering all the possible errors and came from a completely independent analysis. Our estimate came from data from Andromeda whereas they used data from 72 galaxy clusters (≈ 100 times further away)
-
- Further evidence comes from the following plot (drawn using the above estimate of decay time):

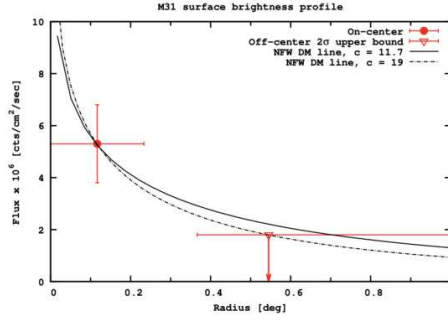


- * Setting $r_s = 872kpc$ and choosing the other parameter in the NFW profile ρ_0 such that the flux passes through the first data point makes the correct predictions for further off-centre observations such as from around 50' off-centre.

This can be verified from the following table which has divided the 16 observations from Perseus into 3 radial bins

Range of offsets	Exposure [ksec]	Flux [cts/sec/cm ²]
23 – 37'	400	13.8 ± 3.3
42' – 54'	230	8.3 ± 3.4
96' – 102'	56	4.6 ± 4.6

- The centre of the first bin is $30' = 0.5^\circ$ where the flux is ≈ 13.8 and that's the data point we have fitted our profile to. But the encouraging sign is that at the centre of the second bin, $48' = 0.8^\circ$ the predicted flux matches up almost exactly with the observed 8.3. There is also an explanation for the large mismatch of the last data point at a 100' offset. It is that the line of sight through that goes through a region further away than R_{200} (radius at which the average density inside is 200 times the mean density of the universe) from the centre and the NFW profile is not really valid there (in this case it makes an underestimate of the column density).

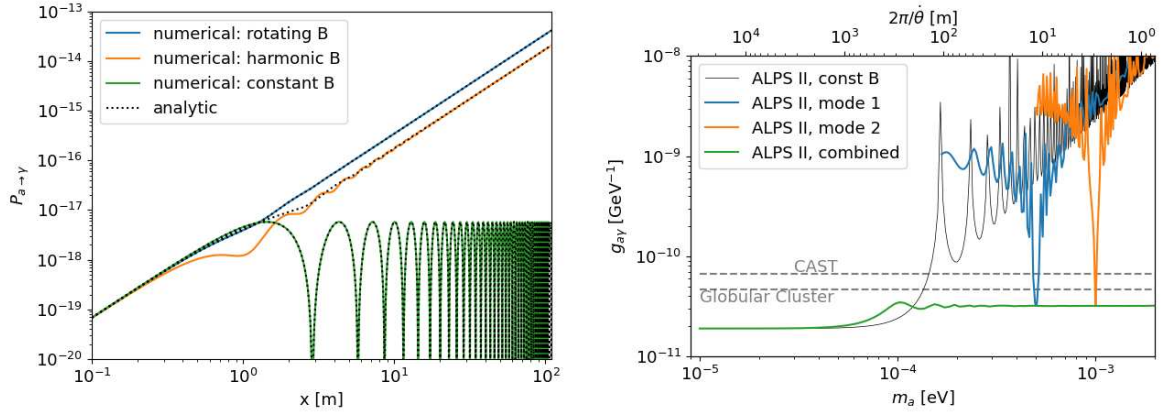


3.7 Evidence for the axion decay model

- Two major points in favour of this are
- The similarity of the on-centre flux from M31 ($\approx 4.9 \times 10^{-6} \text{ cts/cm}^2/\text{sec}$) to that from Perseus ($7 \times 10^{-6} \text{ cts/cm}^2/\text{sec}$) which is atypical of galaxies. This supports the axion model because if we look at the regular magnetic field in M31 it is known to be a spiral with a large magnitude ($\sim 50 \mu\text{G}$) in the central 1 kPc , $5 \mu\text{G}$ between $6 - 14 \text{ kPc}$ and we guess that it has a similar structure all the way up to 25 kPc after which it falls off until 35 kPc (its radius). Clearly the magnitude and coherence are both higher in M31 than in the Milky Way, for example. Also because M31 is edge-on, this spiral magnetic field is always perpendicular to the path of axions originating near its centre. As an estimate if we neglect the large central magnetic field and model the magnetic field of M31 as a constant $5 \mu\text{G}$ spiral until 20 kPc from its centre then the probability that an axion originating at its centre converts to a photon by the time it leaves M31 (roughly the probability that it converts before reaching Earth) is $\frac{B_{\perp}^2 L^2}{4M^2} =$ (we also invoke the small angle approximations)

$$= P \approx 2.3 \times 10^{-8} \left(\frac{L}{1 \text{ kpc}} \frac{R}{1 \text{ kpc}} \right) \left(\frac{B_{\perp}}{1 \mu\text{G}} \frac{10^{13} \text{ GeV}}{M} \right)^2, = 2.3 \times 10^{-4} \times \left(\frac{10^{13} \text{ GeV}}{M} \right)^2.$$
 Now if we stick by the assumption we had made before (that the typical probability of conversion of axions originating in galaxy clusters is the same as that from the central region of Coma and equal to $10^{-3} \times \left(\frac{10^{13} \text{ GeV}}{M} \right)^2$) then the results are perfectly consistent because the column densities of on-centre lines of sight of M31 and Perseus are similar. So for the fluxes to be similar (as they are) the average probability of conversion of axions along the line of sight should also be similar. In fact the factor of 4 smaller probability from M31 coupled with higher column density in M31 perfectly accounts for the factor of ~ 2 greater flux from Perseus.
- Also the stark fall off in the flux when looking off-centre of M31 compared to on-centre can be explained by the highly regular spiral magnetic field. For most of the path of axions which originate somewhere near the edge of the galaxy, the spiral field will be parallel and hence make no contribution to the average $\bar{B}_{\perp}$. The random field in the plane, although larger in magnitude, will effectively become smaller when we average its component perpendicular to the path and also have a much smaller coherence length. So the probability of these axions converting to photons will be smaller. Note that we have neglected out of plane components of the magnetic field, if any.
- This might be reading too much into the data but this plot

shows that assuming the sterile neutrino model leads to an overestimate of the flux at large offset angles (at least at $c=11.7$). This might be because in addition to the fall in the column density as we move off-centre there is also a fall in probability of conversion which it does not take into account (the modified SD decreases more rapidly with offset as compared to the regular SD). A similar plot which assumes the axion decay model can be made and compared with data for Andromeda. We would have to calculate the 'modified column densities' $\int P(l, \theta) \rho(l, \theta) dl$ for each angle of offset θ .



3.8 Magnetic field model of the Milky Way

following <https://arxiv.org/pdf/1204.3662.pdf>

A Galactic magnetic field and electron density model A model of the coherent component of the Galactic magnetic field in the disk and halo of Milky way was developed by Jansson et al. [78]. Magnetic field in the Galactic halo at a radius r and height l can be written into toroidal (B^{tor}) and poloidal (B^{pol}) component in terms of step function $L(l, h, w) = (1 + e^{-2(|l|-h)/w})^{-1}$. The toroidal component is separated into the north (B_n) and south (B_s) component as [78]

$$B^{\text{tor}}(r, l) = e^{-|l|/l_0} L(l, h_{\text{disk}}, w_{\text{disk}}) \times \begin{cases} B_n (1 - L(r, r_n, w_h)) & l > 0, \\ B_s (1 - L(r, r_s, w_h)) & l < 0, \end{cases}$$

The step function $L(l, h, w)$ goes to zero for $l \rightarrow 0, h \gg w$ and unity at $l \gg h$. The first factor of L means that we restrict the toroidal component to outside the disk of height h_{disk} and the second factor of L makes the field diminish outside a radius of r_n, r_s for the northern and southern regions of the Galaxy respectively.

$$B^{\text{pol}}(r, l) = B_X e^{-r_p/r_X} \times \begin{cases} \left(\frac{r_p}{r}\right), & \text{with } r_p = r - |l|/\tan(\Theta_X^0) \text{ } r > r_X \\ \left(\frac{r_p}{r}\right)^2, & \text{with } r_p = \frac{rr_X^c}{r_X^c + |l|/\tan(\Theta_X^0)} \text{ } r < r_X \& \\ \Theta_X(r, l) = \tan^{-1}\left(\frac{|l|}{r - r_p}\right) \end{cases}$$

The following best fit parameters for the toroidal and poloidal components of the magnetic field are fitted by [78]: $l_0 = 5.3 \pm 1.6 \text{ kpc}$, $r_n = 9.22 \pm 0.08 \text{ kpc}$, $r_s > 16.7 \text{ kpc}$, $w_h = 0.2 \pm 0.02 \text{ kpc}$, $h_{\text{disk}} = 0.4 \pm 0.03 \text{ kpc}$, $w_{\text{disk}} = 0.27 \pm 0.08 \text{ kpc}$, $B_n = 1.4 \pm 0.1 \mu\text{G}$, $B_s = -1.1 \pm 0.1 \mu\text{G}$, $B_X = 4.6 \mu\text{G}$, $\Theta_X^0 = 49 \pm 1^\circ$, $r_X^c = 4.8 \pm 0.2 \text{ kpc}$, $r_X = 2.9 \pm 0.1 \text{ kpc}$.

The electron density decreases exponentially with increasing distance from the Galactic plane [121, 122, 128]. The electron density in the Galactic halo can be modeled as $\text{sech}^2(|l|/H)$ [122].

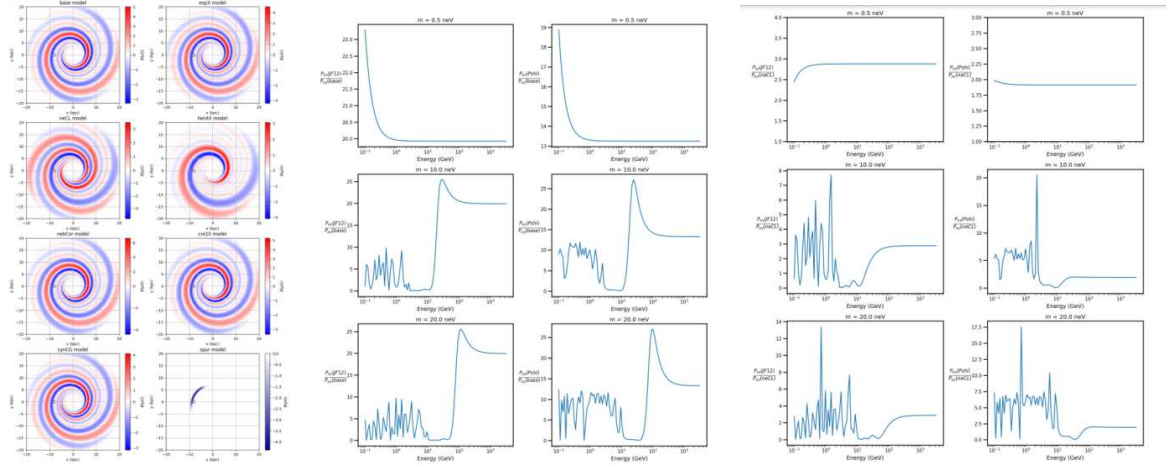
$$n_e(r, l) = n_1 \left[\frac{\cos(\pi r/2A_1)}{\cos(\pi R_\odot/2A_1)} \right] \text{sech}^2(|l|/H) U(r - A_1)$$

where, $U(x)$ is a step function. The value of vertical scale height of $H = 0.95 \text{ kpc}$ was used by Cordes et al. [122], which was later modified to $H = 1.8 \text{ kpc}$ by Gaensler et al. [128] and $A_1 = 17 \text{ kpc}$, $n_1 = 0.035 \text{ cm}^{-3}$. This indicates a much higher electron density at high latitudes than the previous analysis [122]. Similar to the paper by Jansson et al. [78] (which provides the model for the magnetic field), we use the model of electron density given by Cordes et al. [122] with the improved model parameters from Gaensler et al. [128].

The estimation of photon-ALP conversion depends on the model of the magnetic field and electron density in the high latitudes. In particular, for the toroidal component, the magnetic field strength drops exponentially with a scale height of $l = 5.3 \text{ kpc}$. The electron density also decreases exponentially with increasing distance from the Galactic plane [121, 122, 128]. Current observations do not provide a good probe for the electron density in the Galactic halo. However, from the upcoming mission SKA [132] an accurate observation of electron density can improve the constraints on the value of n_e and B . In this paper, we will assume the electron density

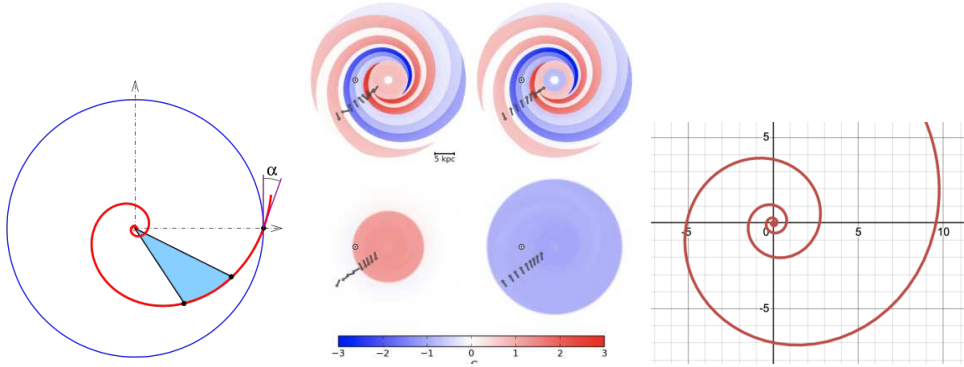
model of Cordes et al. [122] with the parameter values according to Gaensler et al. [128].

- We break it up into a large-scale regular field and a random field. The random field will have a small coherence length and almost no contribution to our probabilities of conversion. Hence we neglect it.
- The regular field has 3 components
 - The disc field
 - Toroidal halo field
 - Out of plane 'X' field
- We set up our coordinates with the origin at the galactic centre and the sun at (-8.5 kpc,0,0).
- The disc field
 - The first part of this is a purely azimuthal magnetic field between 3kpc and 5kpc with a field strength of b_{ring} . This is independent of the z-coordinate (for now) and hence symmetric about the mid-plane.
 - In the region between 5 kpc and 20 kpc, we draw 8 similar logarithmic spirals. These have equations in polar coordinates $r = r_{-x}e^{\phi \tan \alpha}$ where α is the constant angle made by the tangent to the spiral with the tangent to the circle centred at the origin intersecting the spiral at that point. different model of magnetic field of our galaxy shown here :



- r_{-x} are the distances at which the spirals intersect the negative x-axis and are 5.1, 6.3, 7.1, 8.3, 9.8, 11.4, 12.7, 15.5 kpc respectively. Note that $\phi = 0$ on the negative x-axis so the conversion to Cartesian coordinates will require a negative sign.
- Now clearly these spirals will divide the plane into 8 regions (between spiral 1-2, 2-3, ..., 7-8 of the same 'turn' and 8th spiral of one turn-1st spiral of next turn) and the algorithm to find which region a point (r, ϕ) lies in will be as follows. (Actually it will depend on the numerical values of the scaling factors and α if we can divide into regions consistently but we will show later that for the values proposed by this model it is consistent) . If $5 < r < 20 \text{ kpc}$ then find the largest n such that $5.1e^{\tan \alpha (\phi + 2n\pi)} \leq r$. Call $\psi = \phi + 2n\pi$.

- If $6.3e^{\psi \tan \alpha} > r$ then declare that point (r, ϕ) is in region 1. Else if $7.1e^{\psi \tan \alpha} > r$ declare that point is in region 2. And so on until we have to check the condition $15.5e^{\psi \tan \alpha} > r$. If this is true then the point is in region 7 and if not then it is in region 8 (between the 8th spiral and the 1st spiral of the next turn). Note how we used the fact that for any ϕ the radius of the 8th spiral of a given turn is smaller than that of the 1st spiral of the next turn for this algorithm to work. Mathematically this translates to $e^{2\pi \tan \alpha} > \frac{r_{\text{largest}}}{r_{\text{smallest}}}$. This model proposes $\alpha = 11.5^\circ$ and $\frac{r_{\text{largest}}}{r_{\text{smallest}}} = 15.5/5.1 \approx 3$ for which this is valid.
- Finally this model says that for any point between 5 and 20 kpc from the galactic centre, the magnetic field magnitude will depend only on r and the region it lies in and it will be parallel to the spirals in the sense that its direction will be the same as that of the tangent to the similar spirals at that ϕ . Since we know that the tangent to any of the spirals at ϕ makes an angle α with the tangent to the circle centred at the origin at that ϕ , the direction of the disc magnetic field is always $\cos \alpha \hat{\phi} + \sin \alpha \hat{r}$. As for the magnitude, the magnitude of the field at 5kpc in each of the regions b_{1-8} are taken as parameters in the model and the field is known to fall off as $1/r$. b_{1-7} are free parameters and b_8 is inferred from them based on conservation of flux. So for example if I wanted the field at a point (r, ϕ) in region 5 that would be $b_5 \frac{5}{r} (\cos \alpha \hat{\phi} + \sin \alpha \hat{r})$



(Why should there be a point in every region at $r=5\text{kpc}$ (at any radius in fact) ? Because the circle $r=5$ can be easily established to intersect each spiral once and all regions can hence be attained on the circle. For example region 1 and 8 are on either side of the point of intersection of the circle with spiral 1, region 2 is on one side of its intersection with spiral 2, and so on till spiral 7.)

- Now obviously this field should be valid only inside the disc so we must involve some sort of dependence on z so that it falls off outside. This is done by the means of a transition function $L(z, h, w) = (1 + e^{-2\frac{|z|-h}{w}})^{-1}$. For small w this is clearly a step function which goes from 0 at $|z| < h$ to 1 at $|z| > h$. So we multiply our disc field by $(1 - L(z, h_{\text{disc}}, w_{\text{disc}}))$ where h_{disc} and w_{disc} are free parameters in the model denoting the height of the disc and the width of the transition region (technically the height of the disc would be $2h_{\text{disc}}$ as the transition happens at $|z| = h$). Even though now there is a dependence on z our disc field is still symmetric about the mid plane because L involves only $|z|$.
- The halo field
 - We know this to be a purely azimuthal field which the disc field transitions into. It also falls off with increasing distance from the mid-plane on either side. However it is not symmetric in z . These are all encapsulated by its functional form

$$B_\phi^{\text{halo}}(r, z) = L(z, h_{\text{disc}}, w_{\text{disc}}) e^{-\frac{|z|}{z_0}} B_n (1 - L(r, r_n, w_h)) \quad \text{for } z > 0,$$

$$B_\phi^{\text{halo}}(r, z) = L(z, h_{\text{disc}}, w_{\text{disc}}) e^{-\frac{|z|}{z_0}} B_s (1 - L(r, r_s, w_h)) \quad \text{for } z < 0.$$

- The northern and southern radial extent is controlled by $r_{n/s}$ and the transition functions are appended to ensure the fields fall off with the end of the halo just like before. w_h again controls the width of the region in which the halo field cuts off.
- The model proposes that this has no azimuthal component (or this is absorbed in the other components) so at any point specifying the magnitude of the field along with its inclination with the mid-plane is sufficient to describe the field. At any height z the field starts perpendicular to the plane at $r=0$ and its inclination is supposed to decrease as $\tan\theta = \frac{\tan(\theta_0)}{r}(r_C^X + \frac{|z|}{\tan(\theta_0)})$ where r_C^X and θ_0 are parameters. If we define r_p to be the radius at which the field line at any point intersects the mid-plane then $\tan(\theta) = |z|/(r - r_p) \Rightarrow r_p = r - \frac{|z|}{\tan(\theta)} = \frac{rr_C^X}{r_C^X + \frac{|z|}{\tan(\theta_0)}}$. So we can see that at any height, at the same radius that r_p linearly increases to r_C^X the angle made by the field line at that point with the mid-plane becomes θ_0 .
- Hence this model says that within this region ($r < r_C^X + \frac{|z|}{\tan(\theta_0)}$) $\tan\theta = \frac{\tan(\theta_0)}{r}(r_C^X + \frac{|z|}{\tan(\theta_0)})$ and outside this region θ becomes constant at θ_0 . As for the magnitudes, we define a function $b_X(r) = B_X e^{-\frac{r}{r_X}}$ where B_X and r_X are parameters. It turns out that at a point (r, z) within the region: $B = b_X(r_p)(\frac{r_p}{r})^2$ where $r_p = \frac{rr_C^X}{r_C^X + \frac{|z|}{\tan(\theta_0)}}$ and it is inclined at $\tan(\theta) = \tan(\theta_0)(\frac{r_C^X}{r_p})$. Whereas for a point outside the region $B = b_X(r_p)(\frac{r_p}{r})$ where $r_p = r - \frac{|z|}{\tan(\theta_0)}$ and the field is inclined at $\tan(\theta) = \tan(\theta_0)$

3.9 Questions

- Have I understood the out of plane field correctly? Page 6 in the paper says that the elevation angle θ_X varies linearly with radius in the inner region but that is not consistent with the expression for r_p given in equation 9. What is going on here? I have gone by the equations. Also the magnitude of the field does not seem continuous on the boundary of the two regions.
- In the disc field, the equation of spirals given in the paper is $r = r_{-x} e^{\phi \tan(90-\alpha)}$ where $\alpha = 11.5^\circ$. If we want the field to be parallel to the spirals then this is not consistent with the direction of magnetic field which they have specified at any point to be $(\cos\alpha\hat{\phi} + \sin\alpha\hat{r})$. Also the spirals described by $r = r_{-x} e^{\phi \tan(90-\alpha)}$ look nothing like the plot given in the paper:
- Whereas if we take the equations to be $r = r_{-x} e^{\phi \tan(\alpha)}$ and look at the first spiral with $r_{-x} = 5.1$:
then the first time it crosses the +ve x-axis after crossing the -ve x axis at -5.1 is at around 9.5 kpc. Looking at the figure above, the inner boundary of the first red region (the $r_{-x} = 5.1$ spiral) also seems to cross the +ve x axis at around twice the radius at which it crossed the -ve x axis. So I have gone ahead and assumed that this was a typo and the equations are actually $r = r_{-x} e^{\phi \tan(\alpha)}$ with $\alpha = 11.5^\circ$. Is this correct?
- Given the equations, have I got the algorithm for identifying which spiral region a point is in correct?

3.10 Changes :

1. The function $P(r, \theta, \phi)$ (ie. the probability that a particle that is an axion at (r, θ, ϕ) converts to a photon when it reaches Earth) that we computed earlier is no longer useful because instead

of knowing all the sources of axions we now know the exact number density of axions at any point along the line of sight. So using this function, say some sort of multiplying the axion flux*area*P and summing all over the line of sight would clearly be overcounting, because at some point A on the line of sight flux*area*P (at A) by definition gives us the photons we expect per unit time due to axions that crossed A (at any time). But some of these same axions will also cross a closer point B on the line of sight. So flux*area*P (at A)+flux*area*P (at B) will count the photons produced due to conversions from that set of axions twice. It all adds up because letting the decay time $\tau \rightarrow 0$ (viewing DM as axions) in the previous prescription would be equivalent to doing this sum, and it gives a divergent answer. [The average along the l.o.s I had proposed earlier was ad-hoc (so there's no contradiction really), and it's actually all wrong. Please ignore it from here on.]

2. What we need is a computation that given the axion flux and magnetic field in a certain volume, gives us the total number of photons produced in that volume per unit time moving in a particular direction. Then we can just perform this computation for the volume of sky covered by our detector (determined by the solid angle) and we will have our answer. Exactly this has already been done in the basic case of a constant flux (in space) of axions that are all of the same momenta $\vec{k}_a$. I extend this calculation in 3 ways:

- i. Deal with the non-constant flux of axions of a particular momentum (over the cone in the sky that our detector is looking at)
- ii. Deal with all the possible momenta the axions can have and their distribution, given the velocity distribution of DM in the galactic centre frame
- iii. Compute the final answer for our particular set up with one planar detector facing in some direction, as compared to the conventional set up of a cavity detecting all the photons produced inside of it.

Also note that the typical plasma frequency in the ISM of the milky way is $\approx 10^{-12}eV$ and so for axion masses even as small as $10^{-11}eV$ (far from our expected $10^{-5}eV$) we can show that the electric permittivity is almost constant in space at 1. So we do not have to worry about it. We set $\epsilon = \mu = 1$ from here on.

Also, we don't use the standard assumption of the magnetic field varying on a much larger scale compared to axion and photon wavelengths.

3.10.1 Elementary case :

The axion field is taken to be $a(x, t) = Ae^{i(\vec{k}_a \cdot \vec{x} - wt)}$ to model the constant number flux $F = \frac{1}{2\hbar^2 c} |A|^2 \vec{k}_a$ of $\vec{k}_a$ momentum axions throughout the volume.

(where $w = c\sqrt{k_a \cdot k_a + m_a^2 c^2 / \hbar^2}$)

It then turns out that the number of photons produced 'in' or due to the volume V, headed towards direction $\hat{n}$ per unit solid angle is

$$\frac{dN}{d\Omega} = \frac{kw^2}{32\pi^2 \hbar c^2 M^2} |\hat{n} \times \int_V Ae^{i(\vec{k}_a - \vec{k}) \cdot x} \vec{B}_0(x) d^3x|^2$$

where $\vec{k} = \frac{w}{c} \hat{n}$

written more simply in terms of the flux F

$$\frac{dN}{d\Omega} = \frac{\hbar}{16\pi^2 c^2} \frac{w^3}{k_a M^2} \left| \int_V \sqrt{F} e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3x \right|^2$$

3.11 Generalization to Non-Constant Flux:

We could try breaking up our volume into smaller parts and using the constant flux result on each part and then adding up to get the total photons produced but it turns out that it is not additive (why?) and the result for a small volume ΔV goes as ΔV^2 so such a result would depend on how we discretise our domain (a common approximate answer is usually reached by doing this according to the coherence length of the magnetic field). Instead we modify our axion field for the case where the flux varies in space, by letting the amplitude A and hence flux F be a function of position.

The above result then generalises to

$$\frac{dN}{d\Omega} = \frac{\hbar}{16\pi^2 c^2} \frac{w^3}{k_a M^2} \left| \int_V \sqrt{F_{k_a}(x)} e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3x \right|^2$$

where $F_{k_a}(x)$ is the magnitude of the number flux of axions of momentum $\vec{k}_a$.

3.11.1 Generalising to various momentum axions

For us this is given by $F_{k_a}(x) = \frac{\rho(x)}{m_a} n(\vec{k}_a, x) d^3 k_a \frac{\hbar}{m_a} |k_a|$ (number density*velocity) where $n(\vec{k}_a, x) d^3 k_a$ is the fraction of DM with momentum vector in the neighbourhood of k_a at position x .

Now $n(\vec{k}_a, x) = f(\frac{\hbar \vec{k}_a}{m_a}, x) (\frac{\hbar}{m_a})^3$ where $f(v, x)$ is the velocity distribution function in the frame of the earth at position x .

$$\begin{aligned} \implies F_{k_a}(x) &= \frac{\rho(x)}{m_a} (\frac{\hbar}{m_a})^4 f(\frac{\hbar \vec{k}_a}{m_a}, x) |\vec{k}_a| d^3 k_a \\ \implies \frac{dN}{d\Omega} &= \int_{k_a} \frac{\hbar}{16\pi^2 c^2} \frac{w^3}{k_a M^2} \frac{1}{m_a} (\frac{\hbar}{m_a})^4 k_a |\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2} d^3 k_a \\ &= \frac{\hbar^5}{16\pi^2 c^2} \frac{1}{M^2 m_a^5} \int_{k_a} w^3 |\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2} d^3 k_a \end{aligned}$$

Since we want the full details of the spectrum, the number of photons corresponding to energies between $|k_a|$ and $|k_a| + d|k_a|$ detected per unit solid angle in direction $\hat{n}$ will be

$$= \frac{\hbar^5}{16\pi^2 c^2} \frac{1}{M^2 m_a^5} \int (d\Omega_{k_a} w^3 |\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2} |k_a|^2 d|k_a|$$

Therefore, number of photons between energy $E = \hbar c \sqrt{|k_a|^2 + \frac{m_a^2 c^2}{\hbar^2}}$ and $E + dE$, using $dE = (\hbar c)^2 \frac{|k_a| d|k_a|}{E}$ is

$$\begin{aligned} &= \frac{\hbar^3}{16\pi^2 c^4} \frac{1}{M^2 m_a^5} \int (d\Omega_{k_a} w^3 |\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2} |k_a| E dE \\ \implies \frac{1}{d\Omega_k} \frac{dN}{dE} &= \frac{1}{16\pi^2 c^4} \frac{E^4 |k_a|}{M^2 m_a^5} \int (|\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2) d\Omega_{k_a} \\ &\text{evaluated at } |k_a| = \sqrt{(\frac{E}{\hbar c})^2 - (\frac{m_a c}{\hbar})^2} \end{aligned}$$

3.11.2 Calculating photons received on detector:

Following from above, photons of energy between E and $E+dE$ received on the detector per unit time,

$$\frac{dN}{dE}_{detector} = \frac{1}{16\pi^2 c^4} \frac{E^4 |k_a|}{M^2 m_a^5} \int' d\Omega_k (\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2) d\Omega_{k_a})$$

evaluated at

$$|k_a| = \sqrt{(\frac{E}{\hbar c})^2 - (\frac{m_a c}{\hbar})^2}$$

$$\text{and } |k| = \frac{E}{\hbar c}$$

where V is the volume of the sky that our detector looks at, the primed integral over $d\Omega_k$ is over the average solid angle subtended by the detector in this volume, and the unprimed integral over $d\Omega_{k_a}$ is over all solid angles. So if we only consider conversions in the Milky Way and a detector that looks at a solid angle Ω along a particular line of sight, then our volume V would be the cone with inside solid angle Ω extending out to about 30kpc and our average solid angle to integrate over would be approximately $\frac{A}{15kpc^2}$ where A is the area of the detector.

3.11.3 Calculating for our specific problem

Now we accept the expression

$$\frac{dN}{dE} = \frac{1}{16\pi^2 c^4} \frac{E^4 |k_a|}{M^2 m_a^5} \int' d\Omega_k (\int_V \sqrt{\rho(x) f(\frac{\hbar \vec{k}_a}{m_a}, x) e^{i(\vec{k}_a - \vec{k}) \cdot x} \hat{n} \times \vec{B}_0(x) d^3 x|^2) d\Omega_{k_a})$$

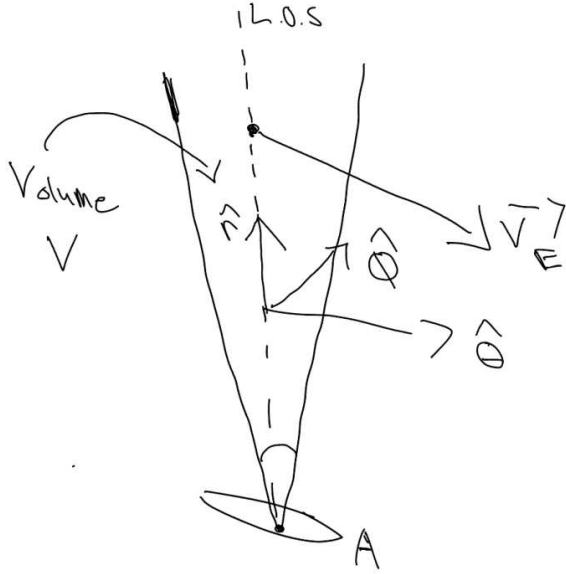
and try to make simplifications for our specific problem. We also add on the specific form of the velocity distribution function.

First off, we know that $f_{earth}(\vec{v}) = f_{g.c.}(\vec{v} + \vec{v}_e)$ where $\vec{v}_e$ is the velocity of the earth in the frame of the galactic centre

Hence $f_{earth}(\vec{v}) = f_{g.c.}(\sqrt{|\vec{v}|^2 + |\vec{v}_e|^2 + 2|\vec{v}||\vec{v}_e|\cos\theta})$ where θ is the angle between $\vec{v}$ and $\vec{v}_{earth}$. We of course assume that the velocity distribution function in the galactic centre frame depends only upon the magnitude of the velocity (we add that it is Maxwell-Boltzmann later).

For now we neglect the motion of the earth around the sun and assume that $\vec{v}_e$ is the constant velocity of the sun in the GC frame.

For a particular line of sight (say along θ_0, ϕ_0), the volume we are interested in looks like the following:



where the cone is of small solid angle Ω and extends out to say some distance R .

First thing we do is replace the integral over $d\Omega_k$, since it is over a really small angle A/R_0^2 (where R_0 is the sort of intermediate distance from the detector to the volume that we can set later).

$d\Omega_k \rightarrow A/R_0^2$, $\vec{k} = -\frac{E}{\hbar c} \hat{r}$, and $\hat{n}$ (which is the direction of $\vec{k}$) = $-\hat{r}$

$\hat{n} \times B_0 = -\hat{r} \times B_0 = B_\phi \hat{\theta} - B_\theta \hat{\phi}$

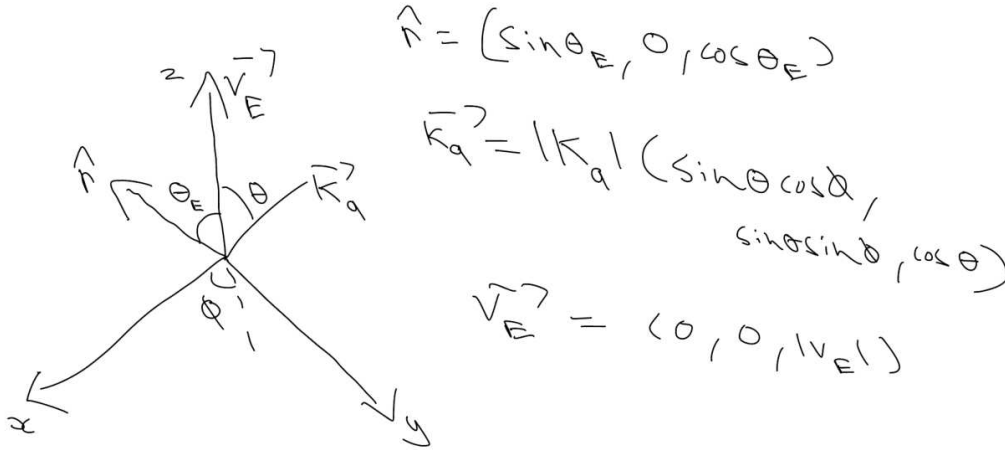
and $d^3x \rightarrow \Omega r^2 dr$

$$\Rightarrow \frac{dN}{dE} = \frac{1}{16\pi^2 c^4} \frac{E^4 |k_a| A}{M^2 m_a^5 R_0^2} \left(\int_0^R \sqrt{\rho(r) f\left(\frac{\hbar \vec{k}_a}{m_a}, r\right)} e^{i \vec{k}_a \cdot \hat{r} r} e^{\frac{i E r}{\hbar c}} B_\phi(r) \Omega r^2 dr \right)^2 + \left(\int_0^R \sqrt{\rho(r) f\left(\frac{\hbar \vec{k}_a}{m_a}, r\right)} e^{i \vec{k}_a \cdot \hat{r} r} e^{\frac{i E r}{\hbar c}} B_\theta(r) \Omega r^2 dr \right)^2$$

The line of sight θ_0, ϕ_0 dependence of all functions is left implicit.

To evaluate the $d\Omega_{k_a}$ integral we set the z-axis along $\vec{v}_e$ and choose the x-axis in the plane of $\hat{r}$ and $\vec{v}$ as shown below.

Note that the angle θ_E is also dependent on the line of sight direction ie. $\theta_E(\theta_0, \phi_0)$



Hence $\vec{k}_a \cdot \vec{v}_e = |k_a| |v_e| \cos(\theta)$

and $\vec{k}_a \cdot \hat{r} = |k_a| (\sin \theta \cos \phi \sin \theta_E + \cos \theta \cos \theta_E)$

So $f\left(\frac{\hbar \vec{k}_a}{m_a}, r\right) = f_{g.c.} \left(\sqrt{\left(\frac{\hbar |k_a|}{m}\right)^2 + |v_e|^2} + 2 \left|\frac{\hbar |k_a|}{m}\right| |v_e| \cos \theta, r \right)$

Also since our DM is non-relativistic, $|k_a| \ll \frac{E}{\hbar c}$

Hence $e^{i \vec{k}_a \cdot \hat{r} r} e^{\frac{i E r}{\hbar c}} = e^{i r (|k_a| (\sin \theta \cos \phi \sin \theta_E + \cos \theta \cos \theta_E) + \frac{E}{\hbar c})} \approx e^{\frac{i r E}{\hbar c}}$

Now substituting the form of the velocity distribution in and using this approximation:

$$\Rightarrow \frac{dN}{dE} = \frac{1}{16\pi^2 c^4} \frac{E^4 |k_a| A \Omega^2}{M^2 m_a^5 R_0^2} \int \sin \theta d\theta d\phi$$

$$\begin{aligned}
& (|\int_0^R \sqrt{\rho(r)f_{g.c.}(\sqrt{(\frac{\hbar|k_a|}{m})^2 + |v_e|^2 + 2\frac{\hbar|k_a|}{m}|v_e|\cos\theta, r)}e^{\frac{iEr}{\hbar c}} B_\phi(r)r^2 dr|^2 \\
& + |\int_0^R \sqrt{\rho(r)f_{g.c.}(\sqrt{(\frac{\hbar|k_a|}{m})^2 + |v_e|^2 + 2\frac{\hbar|k_a|}{m}|v_e|\cos\theta, r)}e^{\frac{iEr}{\hbar c}} B_\theta(r)r^2 dr|^2) \\
& = \frac{1}{8\pi c^4} \frac{E^4|k_a|A\Omega^2}{M^2m_a^5R_0^2} \int_{-1}^1 dt \\
& (|\int_0^R \sqrt{\rho(r)f_{g.c.}(\sqrt{(\frac{\hbar|k_a|}{m})^2 + |v_e|^2 + 2\frac{\hbar|k_a|}{m}|v_e|t, r)}e^{\frac{iEr}{\hbar c}} B_\phi(r)r^2 dr|^2 \\
& + |\int_0^R \sqrt{\rho(r)f_{g.c.}(\sqrt{(\frac{\hbar|k_a|}{m})^2 + |v_e|^2 + 2\frac{\hbar|k_a|}{m}|v_e|t, r)}e^{\frac{iEr}{\hbar c}} B_\theta(r)r^2 dr|^2) \\
& \text{again, to be evaluated at } |k_a| = \sqrt{(\frac{E}{\hbar c})^2 - (\frac{m_a c}{\hbar})^2}
\end{aligned}$$

3.12 Using the MB Distribution:

We know that in any direction θ_0, ϕ_0

$$f_{g.c.}(v, r) = b(r, \theta_0, \phi_0) e^{-a(r, \theta_0, \phi_0)v^2}$$

if $v < c(r, \theta_0, \phi_0)$

Else it is 0.

$a(r, \theta, \phi)$ is given by $\frac{1}{\sigma(R)^2}$ where R is the distance of point (r, θ, ϕ) from the galactic centre.

and $\sigma(R) = (C\rho(R)R^{1.875})^{\frac{1}{3}}$ where C is a constant

$c(r, \theta, \phi) = \sqrt{2|\Phi(R)|}$ where

$$\Phi(R) = \frac{-4\pi G\rho_0 R_s^3}{R} \ln(1 + \frac{R}{R_s})$$

$$\rho_0 = \rho_\odot \frac{R_\odot}{R_s} (1 + \frac{R_\odot}{R_s})^2$$

$$\text{And } b(r, \theta, \phi) = \frac{1}{4\pi \int_0^{c(r, \theta, \phi)} v^2 e^{-a(r, \theta, \phi)v^2} dv}$$

So we first compute the functions a,b,c and then define $f(v, r, \theta, \phi)$ accordingly with an if statement.

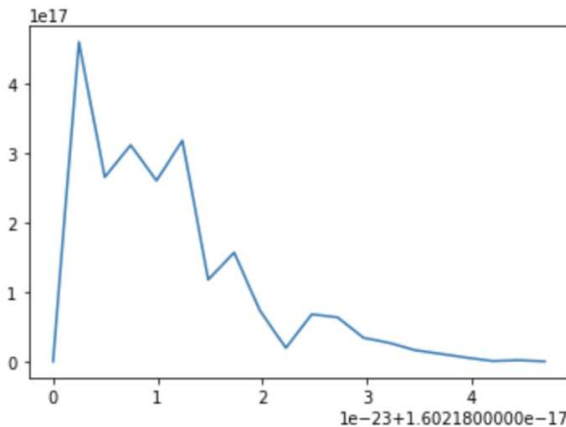
3.13 Some results

For any axion mass m and any given line of sight we have to look at the spectrum between $E_0 = mc^2$ and $E_1 = mc^2 + \frac{1}{2}mv_{max}^2$ where v_{max} is the maximum earth frame speed of any particle along that line of sight. We know $v_{max} = v_{earth} + v_{esc, max}$ where $v_{esc, max}$ is the maximum escape velocity in the GC frame along that line of sight. Instead we just use the maximum escape velocity throughout the Milky way which turns out $52514061.34 \text{ cm/s} \approx 5.25 \times 10^7 \text{ cm/s}$ and $v_e = 220 \text{ km/s} = 2.2 \times 10^7 \text{ cm/s}$ to get $E_1 = m(c^2 + \frac{1}{2}5.5 \times 10^{15})$

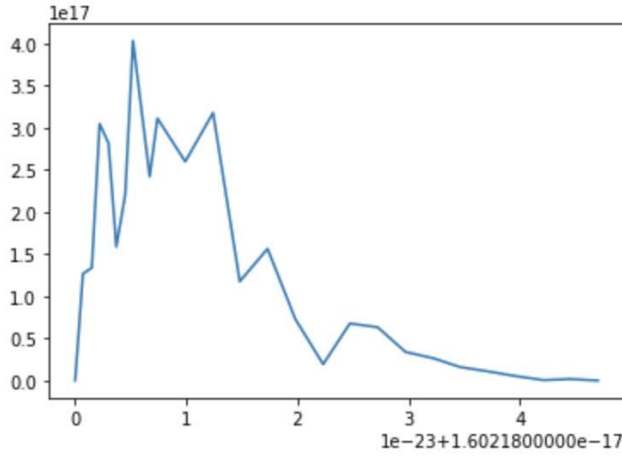
For $m = 10^{-5} \text{ eV} = 1.77 \times 10^{-38} \text{ g}$

$E_0 = 1.60218 \times 10^{-17}$ and $E_1 = 1.602184948987703 \times 10^{-17} \text{ esu}$

Here is what the spectrum looks like roughly for the direction $(\pi/2, 0)$ directly towards the galactic centre and the above mass:



Including some closer spaced points in the start we get



If I can find out the actual energy resolution of the telescopes we have at this $\approx 2.4GHz$ energy then I can make the correct plot to compare it with.

If it is more than $E_1 - E_0 \approx 7.5kHz$ (it probably is but I don't know) then the details of the spectrum do not even matter. The detector will just count the total photons of any energy between E_0 and E_1 . So for a given direction we have to compute

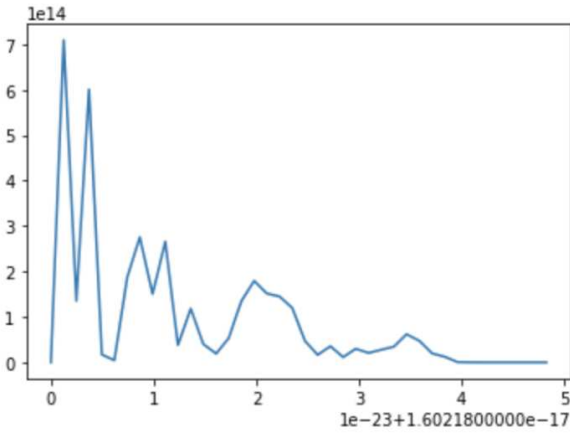
$$\int_{E_0}^{E_1} \frac{dN}{dE} dE \text{ and compare that result.}$$

We try to do this for the direction towards the galactic centre numerically and this value (up to multiplying constants) approximately comes out to be:

$$5.067132022591682 \times 10^{-6}$$

As a sanity check we also do this now for the direction away from the galactic centre, expecting a smaller value because of the overall lower density and magnetic field and it turns out $4.588331598864404 \times 10^{-9}$

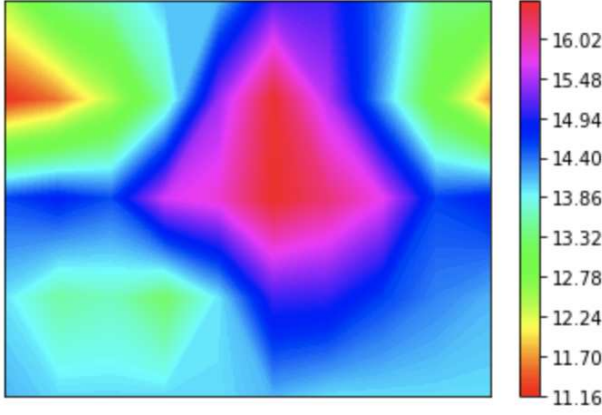
And the spectrum looks like the following



Probably needs a few more points in the start but anyway,

We should plot the integral of $\frac{dN}{dE}$ from E_0 to E_1 now over all directions, but doing the integral for each direction takes about an hour. So for roughly 1000 directions that is way too long. How can I make it faster?

An estimate we can make for now is that the integral is probably proportional to the peak value of $\frac{dN}{dE}$ and looking at the two directions we have the peak seems to come between E_0 and $E_0 + (E_1 - E_0)/10$. So we look at 5 values of $\frac{dN}{dE}$ between these 2, and attribute the maximum of them to that direction. On doing so with 50 directions in the sky, we get the following:



(on a log scale)

3.14 CMB Photon -axion conversion in our Galaxy :

$$\cos(2\theta) = \frac{\Delta_a - \Delta_e}{\Delta_{\text{osc}}}$$

$$\Delta_{\text{osc}}^2 = (\Delta_a - \Delta_e)^2 + 4\Delta_{\gamma a}^2$$

$$\Delta_e \equiv (n - 1)\omega$$

and n is the refractive index for photon propagation through matter. As we see below, for high electron densities photon has real effective mass, $\Delta_e \propto -m_\gamma^2 < 0$ and $\Delta_a \propto -m_a^2 < 0$ and in the limit $|\Delta_e| \gg |\Delta_a|, |\Delta_{\gamma a}|$ mixing is suppressed ($\sin 2\theta \rightarrow 0$) and we get $\cos(2\theta) \rightarrow 1$. Similarly in vacuum, when $\Delta_e \rightarrow 0, |\Delta_a| \gg |\Delta_{\gamma a}|$, again the mixing is suppressed ($\sin 2\theta \rightarrow 0$) but we have $\cos 2\theta \rightarrow -1$. Photon axion mixing is maximum at resonance, $\Delta_e = \Delta_a$ giving $2\theta = \pi/2, \sin 2\theta = 1$.

For astrophysical matter densities, $|n - 1| \ll 1$ and can be approximated (away from resonance frequencies ω_i of atoms) by

$$n - 1 \approx \frac{-m_\gamma^2}{2\omega^2} \approx \frac{2\pi\alpha}{m_e\omega^2} \left(-n_e + n_H \sum_i \frac{f_i^H \omega^2}{H\omega_i^2 - \omega^2} + n_{\text{He}} \sum_i \frac{f_i^{\text{He}} \omega^2}{\text{He}\omega_i^2 - \omega^2} \right)$$

where m_γ is the effective mass of the photon, α is the fine structure constant, ω is the angular frequency, n_e, n_H and n_{He} are the number the density of free electrons, neutral hydrogen, and neutral helium respectively, f_i^α is the oscillator strength of element $\alpha \in \{\text{H}, \text{He}\}$ for transitions with energy ω_i from the ground state. For free electrons, there is no resonant frequency and the effective mass squared is positive. For neutral atoms, such as hydrogen, the effective mass squared is negative below the resonant frequency. Neutral atoms exist only after recombination and in ground state with the minimum resonant frequency of 10.2 eV corresponding to Ly- α transition of hydrogen. For CMB we will therefore always have $\omega \ll \omega_i$ and we can approximate Δ_e as (ignoring helium and heavier elements).

$$\Delta_e \approx \frac{\omega_p^2}{2\omega} \left[-1 + 7.3 \times 10^{-3} \frac{n_H}{n_e} \left(\frac{\omega}{\text{eV}} \right)^2 \right]$$

$$= -2.6 \times 10^6 \left(\frac{n_e}{10^{-5} \text{ cm}^{-3}} \right) \left(\frac{100 \text{ GHz}}{\nu} \right) \left[1 - 7.3 \times 10^{-3} \frac{n_H}{n_e} \left(\frac{\omega}{\text{eV}} \right)^2 \right] \text{ Mpc}^{-1}$$

where $\omega_p^2 = 4\pi\alpha n_e / (m_e)$ is the plasma frequency. For the range of parameters of interest we also have

$$\Delta_{\gamma a} \equiv \frac{g_{\gamma a} |B_T|}{2} = 15.2 \left(\frac{g_{\gamma a}}{10^{-11} \text{ GeV}^{-1}} \right) \left(\frac{B_T}{\mu\text{G}} \right) \text{ Mpc}^{-1}$$

$$\Delta_a \equiv -\frac{m_a^2}{2\omega} = -1.9 \times 10^4 \left(\frac{m_a}{10^{-14} \text{ eV}} \right) \left(\frac{100 \text{ GHz}}{\nu} \right) \text{ Mpc}^{-1}$$

$g_{\gamma a}$ is the photon-ALP coupling and $\omega = 2\pi\nu$. The photon polarization state orthogonal to

B_T is unaffected. Thus initially unpolarized light propagating through a magnetic field will become polarized as intensity in one of the linear polarizations is decreased due to photonALP oscillation. These results apply only if both the electron density and the magnetic field are homogeneous. The departure from BlackBody (BB) spectrum for the affected polarization can be quantified by

$$\mathcal{I}^{\gamma a} = \frac{\Delta I_{\nu}^{\gamma a}}{I_{\nu}} \equiv \frac{I_{\nu}^{\text{obs}} - I_{\nu}}{I_{\nu}} = -\bar{P}(\gamma \rightarrow a)$$

where, $I_{\nu} = (h\nu^3/c^2) / (e^{h\nu/k_B T_{\text{CMB}}} - 1)$ is the Planck spectral form for single polarization in physical units, h is the Planck's constant, c is the speed of light and k_B is Boltzmann constant. We see that it is a fast oscillating function of frequency (as well as distance s). In any experiment with reasonable frequency and angular resolution we will only detect the result of average over many oscillations. We can therefore replace the oscillating factor with its average over an oscillation giving

$$P(\gamma \rightarrow a) = \frac{2\Delta_{\gamma a}^2}{\Delta_e^2}$$

if the magnetic field is nearly coherent over the scales of size $s \gg \ell_{\text{osc}} \equiv 2\pi/\Delta_{\text{osc}}$.

In reality both the electron density and magnetic fields are inhomogeneous. In particular the primordial magnetic fields (because of stochastic initial conditions) as well as small scale Galactic magnetic fields (because of turbulence) are expected to be stochastic. In this case, – 5 as a toy model, we can approximate the magnetic fields as composed of independent domains of size d_0 such that the magnetic field and electron density are homogeneous inside each domain but in different domains the magnetic field has different random orientations but same strength for simplicity. In the limit of large number of domains we can obtain an analytical solution for the conversion probability of the total unpolarized intensity given b.

$$\bar{P}(\gamma \rightarrow a)(r) = \frac{1}{3} (1 - e^{(-3P(\gamma \rightarrow a)r/2d_0)}) \quad r \gg d_0$$

where r is the size of the turbulent region, $P(\gamma \rightarrow a)$ is the probability of conversion inside each domain and $\bar{P}$ is the average conversion probability over the whole turbulent region. In this case, we have a spectral distortion but no polarization. In the limit $r \rightarrow \infty$ we saturate the probability with $1/3^{\text{rd}}$ of the photons getting converted into ALPs.

The true situation will be in between the above two extreme limits and both the magnetic field and electron density would vary along the photon geodesic. In particular, when photons and ALPs propagate through the inhomogeneous medium, there is the possibility of resonance when the effective mass of photon becomes equal to the mass of the ALP,

$$\Delta_e = \Delta_a$$

For inhomogeneous matter distribution and magnetic fields, relevant for considering the CMB - ALP conversions, the conversion probability is sensitive to how fast the matter/electron density and the magnetic fields, and therefore the mixing angle 2θ , change compared to the oscillation length, $\ell_{\text{osc}} = 2\pi/\Delta_{\text{osc}}$. We therefore define an adiabaticity parameter (with ∇ denoting the spatial derivative with respect to the physical distance along the line of sight or proper time),

$$\begin{aligned} \gamma_{\text{ad}} &= \left| \frac{\pi}{\ell_{\text{osc}} \nabla \theta} \right| \\ &= \left| \frac{\Delta_{\text{osc}}}{\sin(2\theta) \cos(2\theta) \nabla (\ln \Delta_{\gamma a}) + \sin(2\theta) \Delta_e / \Delta_{\text{osc}} \nabla (\ln \Delta_e)} \right| \end{aligned}$$

with the adiabatic limit defined as $\gamma_{\text{ad}} \gg 1$. The propagation is adiabatic when the length scale over which the mixing angle changes, $1/\nabla \theta$ is much larger than the oscillation length ℓ_{osc} . In the adiabatic limit, the final conversion probability depends only on the initial mixing angle when the photon is emitted, θ_0 , and the final mixing angle at the detector θ irrespective of whether there is a resonance.

$$P(\gamma \rightarrow a) = \frac{1}{2} (1 - \cos 2\theta_0 \cos 2\theta)$$

The first term in the denominator arises due to the inhomogeneous magnetic fields and the second term due to inhomogeneity in the matter distribution and ionization fraction. At resonance $\sin(2\theta) = 1; \cos(2\theta) = 0$ and the first term in the denominator due to the inhomogeneous magnetic field vanishes. Therefore, for resonant conversion only the inhomogeneity in the matter is relevant and the adiabaticity parameter becomes

$$\gamma_{\text{ad}}(\text{resonance}) = \frac{4\Delta_{\gamma a}^2}{|\nabla\Delta_e|}$$

For the non-trivial solutions to exist, the determinant of the operator on the left hand side must vanish. This gives us two dispersion relations defining the two eigenstates ($m_{\text{eff}}^2 = \omega^2 - k^2 \approx 2\omega(\omega - k)$) of the system,

$$\begin{aligned} 2\omega(\omega - k) &= -\omega(\Delta_e + \Delta_a) \pm \omega\Delta_{\text{osc}} \\ &= \frac{m_a^2 + m_\gamma^2}{2} \pm \left[\left(\frac{m_a^2 - m_\gamma^2}{2} \right)^2 + \omega^2 g_{\gamma a}^2 B_T^2 \right]^{1/2} \end{aligned}$$

The resonance happens when $m_a = m_\gamma$. If the resonance is adiabatic, $\gamma_{\text{ad}} \gg 1$, the system stays in the instantaneous eigenstate on the same branch of the dispersion relation. In particular in this case a photon produced at high density away from the resonance would follow the upper branch as the density of the medium decreases and we will have a full conversion to ALPs at sufficiently low densities¹. This can also be seen with $\cos 2\theta_0 \approx 1, \cos 2\theta \approx -1$ giving $P(\gamma \rightarrow a) \approx 1$. If the density of the medium changes rapidly compared to the oscillation.

length, ℓ_{osc} , there is a non-zero probability that the quantum system would make a transition between the two eigenstates or branches of the dispersion relation. In the limit that the change in density near the resonance is linear, the transition probability is given by the Landau-Zener formula .

$$p = e^{-\pi\gamma_{\text{ad}}/2}$$

and the conversion probability is

$$P(\gamma \rightarrow a) = \frac{1}{2} (1 - (1 - 2p) \cos 2\theta_0 \cos 2\theta)$$

For detection at earth, we should take into account the fact that solar system is inside a local hot bubble with electron density $\sim 5 \times 10^{-3} \text{ cm}^{-3}$ with radius $\sim 100 \text{ pc}$. The ionization fraction is high enough that we can ignore the neutral hydrogen contribution to the photon mass, yielding $m_\gamma^{\text{local}} \approx 2.6 \times 10^{-12} \text{ eV}$. For ALP masses far from resonance, we have the oscillation length $\ell_{\text{osc}} \lesssim 7 \times 10^{-3} (\nu/150 \text{ GHz}) \text{ pc}$. Therefore, for $m_a \gg m_\gamma^{\text{local}}$ we are detecting the photons in vacuum while for $m_a \ll m_\gamma^{\text{local}}$ we are detecting the photons at high electron density. Previous analysis of cosmological CMB constraints has assumed the cosmic average electron density of $n_e \approx 2 \times 10^{-7} \text{ cm}^{-3}$ for calculating m_γ at the detection point .

3.14.1 Resonant Case:

of going from initial state $|\psi_i(0)\rangle$ to final state $|\psi_j(1)\rangle$ after crossing one resonance,

$$|\langle\psi_i(1) | \psi_j(0)\rangle|^2 = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$$

In case of N resonances we get

$$\begin{aligned} |\langle\psi_i(N) | \psi_j(0)\rangle|^2 &= |\langle\psi_i(N) | \cdots | \psi_k(a)\rangle \langle\psi_k(a) | \psi_l(a-1)\rangle \langle\psi_l(a-1) | \cdots | \psi_j(0)\rangle|^2 \\ &= \prod_{a=1}^N \begin{pmatrix} 1-p_a & p_a \\ p_a & 1-p_a \end{pmatrix} \equiv \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix} \end{aligned}$$

where p is the final level crossing probability after N resonances which can be written as

$$p = \frac{1}{2} \left(1 - \prod_{a=1}^N (1 - 2p_a) \right)$$

In the above calculation we have ignored the interference between different resonances and treated the level crossing probabilities as classical probabilities. This is justified if there is decoherence in the wavefunction evolution, for example, due to propagation through the stochastic primordial magnetic fields in-between the resonances. The photon to axion probability is equal to the probability that starting with an approximately pure photon case on the upper eigenstate at high electron densities we end up on the axion line far from the resonance. It is therefore given by

$$P(\gamma \rightarrow a) = \begin{cases} p : N \text{ even} \\ 1 - p : N \text{ odd} \end{cases}$$

We note that we do not have a choice to independently choose N even or odd and at the same time require final detection at high density or in vacuum. Specifying one condition automatically fixes the other.

For the frequency range $\nu \gtrsim 150\text{GHz}$, we will always have two resonances during the dark ages for small axion masses on account of the effective mass of photon getting dominated by neutral gas and becoming imaginary and then real again once reionization starts. In this case of two resonances, the conversion to axion would occur if at one of the resonances we cross level (with probability p_i) but fail to do so at the other resonance (with probability $(1 - p_j)$), where p_i is the level crossing probability for i th resonance. We therefore have the total conversion probability for two resonances

$$P(\gamma \rightarrow a) = p = p_1(1 - p_2) + p_2(1 - p_1)$$

This result is true for all axion masses, including $10^{-14} \leq m_a \leq 5 \times 10^{-13}$. It is also clear that if one of resonances is more adiabatic than the other (larger $(1 - p_i)$), then that resonance will dominate the overall probability. Using $P(\gamma \rightarrow a) = p$ (instead of $P(\gamma \rightarrow a) = 1 - p$ used by [58]) for this mass range yields the correct constraint which is similar to the mass range just outside this interval and much weaker.

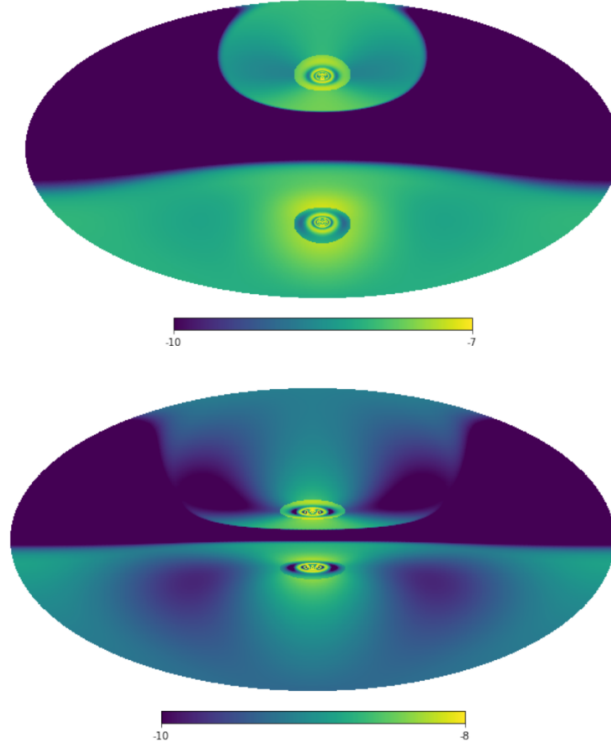
For the resonant conversion in the Galactic halo we can use the model of Galactic magnetic field and electron number density derived from astrophysical observations including synchrotron radiation, Faraday rotation, dispersion of pulsar radiation, angular broadening of extragalactic sources and other effects associated with scattering of radiation by electrons.

Solar system. At $n_e = \{2 \times 10^{-7}, 10^{-5}, 10^{-1}\} \text{cm}^{-3}$, $m_\gamma = \{1.7 \times 10^{-14}, 1.2 \times 10^{-13}, 1.2 \times 10^{-11}\} \text{eV}$ respectively. For $10^{-14} \lesssim m_a \lesssim 10^{-12}$ (the upper limit coming from the density in the local bubble of $n_e \sim 5 \times 10^{-3} \text{cm}^{-3}$), there is only one resonance and we can assume production in vacuum ($\cos \theta_0 = 1$) and detection at high densities ($\cos \theta = -1$) giving the level crossing probability

$$P(\gamma \rightarrow a) = 1 - p \approx \frac{\pi \gamma_{\text{ad}}}{2} \approx \frac{2\pi \Delta_{\gamma a}^2}{|\nabla \Delta_e|} \lesssim 10^{-4}$$

$$\Delta_{\gamma a} \lesssim \left(\frac{10^{-4} |\nabla \Delta_e|}{2\pi} \right)^{1/2}$$

where we assumed that $p \approx 1$ to satisfy COBE constraint that the fractional change in CMB spectrum should be $\lesssim 10^{-4}$. For $10^{-14} \lesssim m_a \lesssim 10^{-13} \text{eV}$, the resonance happens outside the Galactic halo with $|\nabla \Delta_e| \approx 2.5 \times 10^4 \text{Mpc}^{-2}$ at 100 GHz and for $10^{-13} \lesssim m_a \lesssim 10^{-11} \text{eV}$ there will be a resonance inside the Galactic halo with $|\nabla \Delta_e| \approx 2.6 \times 10^{11} \text{Mpc}^{-2}$ at 100 GHz giving



$$\begin{aligned} g_{\gamma a} B_T &< 4 \times 10^{-10} \text{GeV}^{-1} \text{nG} \mid 10^{-14} \lesssim m_a \lesssim 10^{-13} \text{eV} \\ g_{\gamma a} B_T &< 13 \times 10^{-10} \text{GeV}^{-1} \mu\text{G} \mid 10^{-13} \lesssim m_a \lesssim 10^{-11} \text{eV} \end{aligned}$$

3.15 Non-Resonant Conversion:

For axion mass $m_a \leq 10^{-14}$ the only resonances are the ones during the dark ages when the effective mass of the photons becomes imaginary for $\nu \gtrsim 150 \text{GHz}$ (observed frequency today). In this case we can also expect competitive constraints from non-resonant photon-axion oscillations in the stochastic primordial magnetic fields in voids. Previous studies of non-resonant conversion have relied on the toy model of randomly oriented magnetic field domains. This is however an unrealistic oversimplification. More realistically we should expect the primordial magnetic fields to be Gaussian random vector fields with a power spectrum that depends on the production mechanism. In this case we cannot separate the voids into homogeneous regions across which the magnetic fields change abruptly. The variation in magnetic fields across the void would be smooth and the adiabaticity parameter, plays an important role in this case. For the adiabatic evolution the conversion probability would be determined by the high density regions at the edge of void with small mixing angles rather the low density regions near the center of the void with large mixing angles. For adiabatic evolution therefore we expect the photon-axion conversion to be highly suppressed.

For low axion masses, $\Delta_a \ll \Delta_e$ and small mixing angle $\Delta_{\gamma a} \ll \Delta_e$ we have $\cos 2\theta \approx 1$ and the expression for adiabaticity parameter can be simplified, taking a single Fourier mode k_B, k_e for the magnetic field and electron distribution respectively,

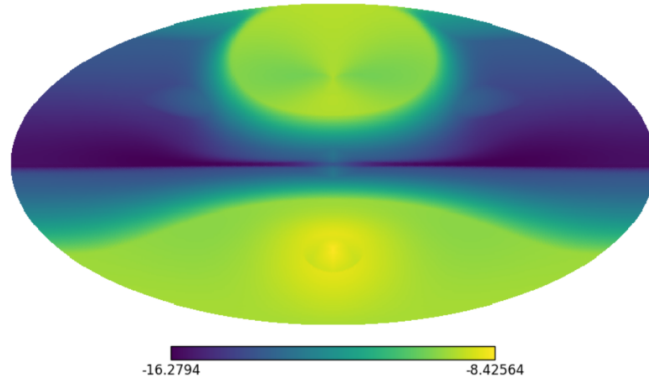
$$\begin{aligned} \gamma_{\text{ad}} &\approx \left| \frac{\Delta_{\text{osc}}}{\sin(2\theta) (k_B + k_e)} \right| \\ &\approx 2 \left(\frac{n_e}{10^{-9} \text{cm}^{-3}} \right)^2 \left(\frac{10^{-10} \text{GeV}^{-1}}{g_{\gamma a}} \right) \left(\frac{1 \text{nG}}{B_T} \right) \left(\frac{0.1 \text{pc}^{-1}}{k_B} \right) \end{aligned}$$

where we have assumed that the magnetic field changes more rapidly than the electron density. We also need the magnetic field to change randomly, therefore the evolution should be non-adiabatic w.r.t the changes in the magnetic field. We see that we have adiabatic evolution on scales $\gg \text{pc}$, in particular for Mpc scale magnetic fields considered by rendering their calculations unrealistic. Most of the contribution to $P(\gamma \rightarrow a)$ would come from magnetic fields on parsec scales or smaller where the contributions from different domains can add incoherently. We are

therefore in the regime where $\Delta_{\text{osc}} s \ll 1$ in each domain of size $s \sim 10\text{pc}$. In this limit we get for a void of radius R_V

$$\begin{aligned} \bar{P}(\gamma \rightarrow a) &\approx \frac{P(\gamma \rightarrow a) R_V}{2s} \\ &\approx \frac{\Delta_{\gamma a}^2 R_V s}{2} = 10^{-4} \left(\frac{g_{\gamma a}}{10^{-10} \text{GeV}^{-1}} \right)^2 \left(\frac{B_T}{1 \text{nG}} \right)^2 \left(\frac{R_V}{1 \text{Gpc}} \right) \left(\frac{s}{10 \text{pc}} \right) \end{aligned}$$

We note that in this limit the conversion probability is independent of frequency. The COBE-FIRAS limit on change in the CMB frequency spectrum at the peak of blackbody of $\lesssim 5 \times 10^{-5}$ translates into $g_{\gamma a} B_T \lesssim 10^{-10} \text{GeV}^{-1} \text{nG}$, which is a factor of ~ 20 weaker than the limits obtained. Our limit is still a very rough limit. To get precise constraints we must evolve the CMB photons through a realistic void profile with a realization of Gaussian random magnetic field which we leave for future work. We can however make the following important observations from above discussion. We see that smaller scales are more non-adiabatic and should therefore contribute the most to the photon-axion



Chapter 4

Searching from Neutron Stars and FRB Magnetars

It is interesting to plot the conversion probabilities in a realistic astrophysical setup. Neutron stars have been the subject of considerable attention as targets for indirect detection of axions . Here, the plasma profile can be approximated by the Goldreich-Julian model . This model consists of an oblique rotating magnetic dipole , whose components, in polar coordinates, read

$$\begin{aligned} B_R &= B_0 \left(\frac{R}{r} \right)^3 (\cos \alpha \cos \theta_{\text{NS}} + \sin \alpha \sin \theta_{\text{NS}} \cos \psi) \\ B_\theta &= \frac{B_0}{2} \left(\frac{R}{r} \right)^3 (\cos \alpha \sin \theta_{\text{NS}} - \sin \alpha \cos \theta_{\text{NS}} \cos \psi) \\ B_\phi &= \frac{B_0}{2} \left(\frac{R}{r} \right)^3 \sin \alpha \sin \psi \end{aligned}$$

where α is the angle between the rotation axis and magnetic dipole moment and $\psi(t) = \varphi_{\text{NS}} - \Omega t$, where $(\theta_{\text{NS}}, \varphi_{\text{NS}})$ are polar coordinates defined with respect to the rotational axis. The neutron star has a surface magnetic field B_0 , radius R and rotational frequency $\Omega = 2\pi/P$, where P is the period. The Goldreich-Julian model gives the density of charge carriers as

$$n_{\text{GJ}}(\mathbf{r}) = \frac{2\Omega \cdot \mathbf{B}}{e} \frac{1}{1 - \Omega^2 r^2 \sin^2 \theta_{\text{NS}}}$$

where $\Omega = \Omega \hat{\mathbf{z}}$ is the constant neutron star rotation vector. Neglecting the relativistic terms in the denominator, we arrive at

$$n_{\text{GJ}} = \frac{B_0 \Omega}{2e} \left(\frac{R}{r} \right)^3 [\cos \alpha + 3 \cos \alpha \cos (2\theta_{\text{NS}}) + 3 \sin \alpha \cos \psi \sin 2\theta_{\text{NS}}]$$

Next, we assume an electrosphere model, in which the magnetosphere is separated into positive and negative charges with mass m_e , so that the plasma frequency is given by $\omega_p = \sqrt{e^2 |n_{\text{GJ}}| / m_e}$. For a fixed frequency ω , the conversion surface on which $k_a^\mu = k_\gamma^\mu$ is then defined by

$$\omega_p^2 = \frac{m_a^2 \omega^2}{m_a^2 + k^2 \sin^2 \theta} \simeq m_a^2$$

where θ is the angle between $\mathbf{B}$ and $\mathbf{k}$, not to be confused with the polar angle θ_{NS} . The final approximation holds for non-relativistic axions with $k \ll m_a$. In this limit, all the kinematic surfaces in the foliation $\Sigma_{\mathbf{k}}$ become independent of $\mathbf{k}$ and collapse to a single surface Σ , given by $\omega_p = m_a$. It is useful to define a coordinate system for the vector $\hat{\mathbf{k}}$ given by

$$\hat{\mathbf{k}} = \sin \theta_k \cos \phi_k \hat{\mathbf{x}} + \sin \theta_k \sin \phi_k \hat{\mathbf{y}} + \cos \theta_k \hat{\mathbf{z}}$$

where $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ are defined by the Cartesian axis related to the polar coordinates $(r, \theta_{\text{NS}}, \varphi_{\text{NS}})$ and (θ_k, ϕ_k) give the direction of propagation of the axion. We can then define an averaged probability over all $\hat{\mathbf{k}}$ directions by

$$\langle P_{a\gamma} \rangle \equiv \int d\Omega_{\mathbf{k}} v_p \cos \theta_{\mathbf{n}} P_{a\gamma}$$

4.1 conversion probability :

$$\partial_k \mathcal{H} \partial_x f_\gamma - \partial_x \mathcal{H} \partial_k f_\gamma = \varepsilon_\mu^* \varepsilon_\nu \Pi_{\text{ax}}^{<\mu\nu}$$

We recall that the object $\mathcal{H} = \mathcal{H}(k, x)$ is the Hamiltonian of the mode, with k_μ and x^μ giving the 4-momentum and coordinates of the photon, respectively. Setting $\mathcal{H}(k, x) = 0$ gives the dispersion relation for that mode. For example, in an unmagnetised cold plasma, one has $\mathcal{H} = k_\mu k^\mu - \omega_p^2$. The object $\Pi_{\text{ax}}^{<\mu\nu}(k, x)$ is the Wigner transform of the photon Wightman self-energy, and corresponds to production due to axions. It is defined by

$$\Pi_{\text{ax}}^{<\mu\nu}(x, x') = \langle j_\phi^\nu(x') j_\phi^\mu(x) \rangle$$

Here, j_ϕ^μ is the effective current arising from the axion-photon interaction

$$\mathcal{L}_{\text{int}} = -\frac{g_{a\gamma\gamma}}{4} \phi F_{\mu\nu} \bar{F}^{\mu\nu} = A_\mu j_\phi^\mu, \quad j_\phi^\mu = g_{a\gamma\gamma} \partial_\nu \phi \bar{F}_{\text{ext}}^{\mu\nu}$$

where $\bar{F}_{\text{ext}}^{\mu\nu}$ is the dual external field strength tensor, originating, for instance, from a background magnetic field. The form of the production term has the form of the standard photon-production term for gauge theories, except that here it applies to more general polarisations relevant for arbitrary inhomogeneous and anisotropic media. It is straightforward to show that, to leading order in $g_{a\gamma\gamma}$, the Wightman self-energy contribution from axions is given by

$$\Pi_{\text{ax}}^{<\mu\nu}(k, X) = g_{a\gamma\gamma}^2 k_\rho k_\sigma \bar{F}_{\text{ext}}^{\mu\rho} \bar{F}_{\text{ext}}^{\nu\sigma} 2\pi \delta(k^2 - m_\phi^2) f_\phi(k, X)$$

where $f_\phi(k, x)$ is the phase-space density of axions.

Furthermore, we should remember that k_0 is on-shell with respect to the photon dispersion relation, i.e., $k_0 = E_\gamma(\mathbf{k}, x)$. Since the axion energy satisfies $E_\phi(\mathbf{k})^2 = |\mathbf{k}|^2 + m_\phi^2$, it follows that the delta function can be written as

$$\delta(k^2 - m_\phi^2) = \delta(E_\gamma(\mathbf{k}, x)^2 - E_\phi(\mathbf{k})^2)$$

then we arrive at the following form of the transport equation for photons:

$$\partial_k \mathcal{H} \partial_x f_\gamma - \partial_x \mathcal{H} \partial_k f_\gamma = g_{a\gamma\gamma}^2 |k \cdot \bar{F}_{\text{ext}} \cdot \varepsilon|^2 2\pi \delta(E_\gamma(\mathbf{k}, x)^2 - E_\phi(\mathbf{k})^2) f_\phi$$

With this in mind, we construct solutions using a method of characteristics, which consists of identifying those integral curves $(x^\mu(\lambda), k_\mu(\lambda))$ in phase space that satisfy

$$\frac{dx^\mu(\lambda)}{d\lambda} = \frac{\partial \mathcal{H}}{\partial k_\mu}, \quad \frac{dk_\mu(\lambda)}{d\lambda} = -\frac{\partial \mathcal{H}}{\partial x^\mu}$$

where λ is a worldline parameter. From the chain rule, it follows that along these curves:

$$\frac{df_\gamma(k(\lambda), x(\lambda))}{d\lambda} = g_{a\gamma\gamma}^2 \left| k \cdot \tilde{F}_{\text{ext}} \cdot \varepsilon \right|^2 2\pi \delta(E_\gamma(\lambda)^2 - E_\phi(\lambda)^2) f_\phi$$

By solving this equation and propagating the solutions out to infinity, one immediately obtains the asymptotic distribution of photons measured by an external observer. Note also that $E_{\gamma,\phi}(\lambda)$ is a shorthand for $E_{\gamma,\phi}(\mathbf{k}(\lambda), x(\lambda))$ and that the axion energy function is evaluated along the photon worldline. Working, as before, in temporal gauge with $\varepsilon^0 = 0$ and integrating gives

$$f_\gamma(\lambda) = \frac{\pi g_{a\gamma\gamma}^2 E_\gamma^2 |\mathbf{B}_{\text{ext}} \cdot \varepsilon|^2}{E_\gamma |E'_\gamma(\lambda_c) - E'_\phi(\lambda_c)|} f_\phi(\mathbf{k}(\lambda_c), x(\lambda_c))$$

We now come to one of the main physical situations in which this conversion can be active: magnetised astrophysical plasmas, as are encountered in the magnetospheres of neutron stars.

4.2 Weakly magnetised plasmas:

In a weakly magnetised, cold plasma, $\omega_c \ll \omega, \omega_p$, where $\omega = E_\gamma$ is the photon-frequency (in natural units), $\omega_c = e|\mathbf{B}|/m_e$ is the cyclotron frequency, and $\omega_p = \sqrt{e^2 n_e / m_e}$ is the plasma frequency with n_e being the electron number density. In that limit, the photon dispersion relation becomes

$$E_\gamma^2 = |\mathbf{k}|^2 + \omega_p^2$$

In addition, $\mathbf{v}_p \cdot \nabla E_\gamma = (\omega_p / E_\gamma) \mathbf{v}_p \cdot \nabla \omega_p$, and the polarisation vectors are transverse to the direction of propagation, so that $(\mathbf{B}_{\text{ext}} \cdot \boldsymbol{\varepsilon})^2 = \sin^2 \theta |\mathbf{B}_{\text{ext}}|^2$, where θ is the angle between $\mathbf{k}$ and $\mathbf{B}_{\text{ext}}$. The ratio of the total and electric energy in a weakly magnetised plasma turns out to be $U/U_E = 2$. All together, this yields

$$P_{a\gamma\gamma}^{\text{iso}} = \frac{\pi}{2} \frac{g_{a\gamma\gamma}^2 \sin^2 \theta |\mathbf{B}_{\text{ext}}|^2 E_\gamma}{|\mathbf{v}_p \cdot \nabla \omega_p| \omega_p}.$$

This reproduces the 1D formula for propagation perpendicular to $\mathbf{B}(\theta = \pi/2)$ in the nonrelativistic limit $E_\gamma / \omega_p \simeq 1$.

4.3 Strongly magnetised plasmas:

For a strongly magnetised plasma, we can use the relation between the permittivity and the polarisation tensor

$$\Pi^{ij} = \omega^2 (\delta^{ij} - \epsilon^{ij})$$

We choose coordinates in which $\mathbf{k} = (0, 0, k)$ points in the z -direction, and $\mathbf{B}_{\text{ext}}$ lies in the $y-z$ plane, with $\mathbf{B}_{\text{ext}} = |\mathbf{B}_{\text{ext}}| (-\sin \theta \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}})$, so that θ is the angle between $\mathbf{k}$ and $\mathbf{B}$. For a strongly magnetised plasma, neglecting relativistic effects,² we find the permittivity 3-tensor is given by [93]

$$\epsilon = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 - \frac{\omega_p^2}{\omega^2} \sin^2 \theta & \frac{\omega_p^2}{\omega^2} \cos \theta \sin \theta \\ 0 & \frac{\omega_p^2}{\omega^2} \cos \theta \sin \theta & 1 - \frac{\omega_p^2}{\omega^2} \cos^2 \theta \end{pmatrix}.$$

From this, we can read off, using the equation for the electric field, the following dispersion relations:

$$k_0^2 - |\mathbf{k}|^2 = 0$$

$$k_0^2 - \frac{1}{2} \left(|\mathbf{k}|^2 + \omega_p^2 + \sqrt{|\mathbf{k}|^4 + \omega_p^4 + 2\omega_p^2 |\mathbf{k}|^2 (1 - 2 \cos^2 \theta)} \right) = 0$$

$$k_0^2 - \frac{1}{2} \left(|\mathbf{k}|^2 + \omega_p^2 - \sqrt{|\mathbf{k}|^4 + \omega_p^4 + 2\omega_p^2 |\mathbf{k}|^2 (1 - 2 \cos^2 \theta)} \right) = 0$$

which follow by setting the $\mathcal{H}_c = 0$. These are the magnetosonic-t, Langmuir-O (LO) and Alfvén modes, respectively. To leading order in $g_{a\gamma\gamma}$, the LO mode is the only one which can both be produced on resonance and propagate out of the magnetosphere as $\omega_p \rightarrow 0$. The LO mode has electric polarisation unit 3-vector

$$\boldsymbol{\varepsilon}_{\text{LO}} = \frac{-1}{\sqrt{1 + \frac{\omega_p^4 \cos^2 \theta \sin^2 \theta}{(\omega^2 - \omega_p^2 \cos^2 \theta)^2}}} \left(\hat{\mathbf{y}} - \frac{\omega_p^2 \cos \theta \sin \theta}{\omega^2 - \omega_p^2 \cos^2 \theta} \hat{\mathbf{z}} \right)$$

It is also straightforward to show that for the LO mode, $U/U_E = 2$. We verified this by explicit calculation. This can also be seen by noting that, since the permittivity takes the form $\epsilon = I - A/\omega^2$ for some matrix A , it follows that $\partial_\omega(\omega\epsilon) = I + A/\omega = 2I - \epsilon$, so that $U = |\mathbf{B}|^2 + \mathbf{E}^* \cdot \partial_\omega(\omega\epsilon) \cdot \mathbf{E} = |\mathbf{B}|^2 + 2|\mathbf{E}|^2 - \mathbf{E}^* \cdot \epsilon \cdot \mathbf{E}$. Again, using Faraday's law to write $|\mathbf{B}|^2 = [|\mathbf{k}|^2 |\mathbf{E}|^2 - (\mathbf{k} \cdot \mathbf{E})^2] / \omega^2$, we have that $U = \omega^2 E_i^* (|\mathbf{k}|^2 \delta_{ij} - k_i k_j - \omega^2 \epsilon_{ij}) E_j + 2E_i^* E_i$, but the first term vanishes due to the equations of motion, leaving $U = 2|\mathbf{E}|^2 = 2U_E$, as required.

, we therefore read off

$$P_{a\gamma} = \frac{\pi}{2} \frac{g_{a\gamma\gamma}^2 |\mathbf{B}_{\text{ext}}|^2 E_\gamma^4 \sin^2 \theta}{\cos^2 \theta \omega_p^2 (\omega_p^2 - 2E_\gamma^2) + E_\gamma^4 |\mathbf{v}_p \cdot \nabla_{\mathbf{x}} E_\gamma|} \frac{1}{\omega_p^2}$$

This result reproduces the isotropic result in the limit $\theta \rightarrow \pi/2$ in which the LO mode becomes ordinary and behaves as though in an isotropic plasma. is the central result of our paper, providing the conversion probability for axions to photons in a strongly magnetised plasma. Note that we have also verified, via an alternative and more explicit calculation. This we did in covariant gauge using the covariant form and deriving an explicit form of the eigenvalue derivative $\partial_{k_0} \mathcal{H}$.

We also state here the explicit form of the energy gradient, which corresponds to taking spatial derivatives while keeping $\mathbf{k}$ constant. This gives

$$\nabla_{\mathbf{x}} E_\gamma(\mathbf{k}, x) = \frac{\omega \omega_p (\cos \theta \omega_p (\omega_p^2 - \omega^2) \nabla_{\mathbf{x}} \cos \theta + \omega^2 \sin^2 \theta \nabla_{\mathbf{x}} \omega_p)}{\cos^2 \theta (\omega_p^4 - 2\omega^2 \omega_p^2) + \omega^4}$$

where

$$\begin{aligned} \nabla_{\mathbf{x}} \cos \theta &= \nabla_{\mathbf{x}} \left(\frac{\mathbf{k} \cdot \mathbf{B}}{|\mathbf{k}| |\mathbf{B}_{\text{ext}}|} \right) \\ &= \frac{k_i \nabla_{\mathbf{x}} B^i}{|\mathbf{k}| |\mathbf{B}_{\text{ext}}|} - \frac{\cos \theta \nabla_{\mathbf{x}} |\mathbf{B}_{\text{ext}}|}{|\mathbf{B}_{\text{ext}}|} \end{aligned}$$

4.4 Dispersion Relations and Ray Tracing :

Firstly, the refractive index of the medium takes on a covariant form and becomes ,

$$n^2 = 1 + \frac{k_a k^a}{(k_b U^b)^2},$$

where k^a is the photons 4 -momentum and U^a is a global unit timelike vector such that $U^a U_a = -1$, recalling that U^a is the 4 -velocity of an observer. When the Minkowski metric is used, reduces to the flat-space case of $n = \omega/k$ where ω is the photon frequency and k is the photon wavenumber.

4.4.1 Vacuum

For the simple case when no plasma is present, only gravity will affect the path the photon travels. The dispersion relationship in a vacuum is just,

$$D(k) = k_a k^a = 0,$$

where the sum over the indices contains the metric, which will account for the curvature of space. With the Minkowski metric, this simply becomes $k^2 - \omega^2$. In this case, photons travel along geodesics around the star.

4.4.2 Isotropic Plasma

With the inclusion of plasma, there is now a scalar function $\omega_p(x^a)$ present. It can be considered a forcing term in the dispersion relationship, which alters the trajectory of the photons from the vacuum case. The function ω_p is a scalar independent of the metric. Hence, it remains the same as the flat-space case, as it is unaffected by the inclusion of GR. In an isotropic plasma, the dispersion relationship is,

$$D(k) = k_a k^a + \omega_p = 0$$

4.4.3 Anisotropic Plasma

The covariant form of the anisotropic dispersion relationship requires some covariant plasma expressions. As in previous sections, let U^a be a global unit timelike vector such that $U^a U_a = -1$. Then, the photon's effective energy measured by an observer with 4-velocity U^a is $W = -k_a U^a$. Also, define the unit vector in the direction of the magnetic field $b^a = B^a / \sqrt{B^t B_t}$ with $b^a b_a = 1$ (where in Schwarzschild coordinates $B^0 = B^t = 0$). Then, the wave vector component parallel to the magnetic field can be represented by the sum $K_{\parallel} = k_a b^a$. In the non-relativistic plasma limit, we have that the GR anisotropic dispersion relation is,

$$D(k) = k_a k^a + \omega_p^2 \left(1 - \frac{K_{\parallel}^2}{W^2} \right) = 0$$

When the appropriate Minkowski limit is taken, where $K_I = k \cos \hat{\theta}$ and $W = \omega$, the above equation simplifies to the flatspace case.

Paths of geodesics in flat space-time are simply straight lines. In a curved spacetime additional terms involving the Christoffel symbols appear in the geodesic equation. In a curved spacetime with the presence of a plasma, the paths will change further. Below we give the equations governing null geodesics in a curved spacetime with plasma.

4.4.4 Geometric Optics

The geometric optics limit is formed by taking the WKB approximation with the eikonal form. The general relativistic equations for ray propagation can then be found by first representing a wave packet in the form,

$$A_b = \int A_b(k) e^{k_s x^n} \sqrt{|g|} d^4 k$$

having used the eikonal form with a Fourier transform. In the exponential, k_b should satisfy $D(k, x) = 0$, and in particular, there should be a sharp maximum when $k = k_0$. So, we can make the substitution $k = q + k_0$. Then, along the ray $x^a = x^a(\lambda)$, where λ is an affine parameter along the light path, the phase $q_a x^a$ should be stationary. So, we will have $\mathbf{q}_a (dx^a/d\lambda) = 0$. One can then expand the dispersion relation about k_0 to obtain $\mathbf{q}_a (\partial D / \partial k_a) = 0$. This yields,

$$\frac{dx^a}{d\lambda} = \frac{\partial D}{\partial k_a}$$

Then using $dD/d\lambda = 0$,

$$\frac{\partial D}{\partial k_a} \frac{dk_a}{d\lambda} + \frac{\partial D}{\partial x_a} \frac{dx^a}{d\lambda} = 0$$

Hence, we will have a system of ODEs that describes the ray path. They are expressed as,

$$\begin{aligned} \frac{dx^\alpha}{d\lambda} &= \frac{\partial D}{\partial k_a} \\ \frac{dk_a}{d\lambda} &= -\frac{\partial D}{\partial x^\alpha} \end{aligned}$$

The solution to the first equation prescribes the spacetime position, whilst the second equation controls the energy and direction of propagation. The convenience of this form is once we have defined the dispersion relation in covariant form, we may then directly use it to find the ray paths. The only caveat is this method requires the dispersion relation to have the 4-wavevector expressed as a covariant vector due to the derivative present in while the position 4-vector remains in contravariant form.

4.5 Detection :

When performing single object observations, one has to compute the power flowing into a particular solid angle $d\mathcal{P}(\theta, \phi)/d\Omega$ pointing away from the star in the direction of the observer

located along the line of sight in the direction (θ, ϕ) . Computing the angular power $d\mathcal{P}/d\Omega$ in a given (θ, ϕ) direction requires detailed plasma ray-tracing to track photons from their emission to detection. However, since we are dealing here with populations, integrating over different orientations of stars within the population is equivalent to averaging over all (θ, ϕ) the total flux, F , from the population, measured on Earth, is given by

$$F = \sum_{\mathbf{x}_i, B_i, P_i} \sum_{\theta_i, \phi_i} \frac{1}{|\mathbf{x}_i|^2} \frac{d\mathcal{P}(\theta_i, \phi_i, \mathbf{x}_i, B_i, P_i)}{d\Omega} n_{\text{ns}}(\mathbf{x}_i, B_i, P_i) \\ \simeq \sum_{\mathbf{x}_i, B_i, P_i} \frac{L(\mathbf{x}_i, B_i, P_i)}{|\mathbf{x}_i|^2} n_{\text{ns}}(\mathbf{x}_i, B_i, P_i)$$

where $L(\mathbf{x}_i, B_i, P_i)$ is the integrated luminosity of a star with magnetic field and period given by B_i and P_i , respectively, and n_{ns} is the corresponding number density of NSs at the point $\mathbf{x}_i$ with those values. Of course, this assumes that the flux is collected from an infinitesimal volume across which the dark matter density can be treated as constant, and where all stars can be treated as being the same distance from Earth. This allows the averaging over (θ_i, ϕ_i) to be performed in a separable way from other population parameters.

The luminosity of a single star from resonant conversion of axions has been computed and is given by

$$L = \int d^3\mathbf{k} \int d\Sigma_{\mathbf{k}} \cdot \mathbf{v}_a P_{a\gamma} \omega f_a$$

where

$$P_{a\gamma} = \frac{\pi}{2} \frac{g_{a\gamma\gamma}^2 |\mathbf{B}|^2 E_\gamma^4 \sin^2 \theta_B}{\cos^2 \theta_B \omega_P^2 (\omega_P^2 - 2E_\gamma^2) + E_\gamma^4 |\mathbf{v}_a \cdot \nabla_{\mathbf{x}} E_\gamma|} \frac{1}{|\mathbf{v}_a \cdot \nabla_{\mathbf{x}} E_\gamma|}$$

is a conversion probability which describes the ratio of axion and photon phase-space densities at the point of conversion. θ_B is the angle between $\mathbf{k}$ and $\mathbf{B}$, and $E_\gamma = \omega$ is the photon energy, which for the modes relevant for transport (the Langmuir Ordinary (LO) mode) is given by

$$E_\gamma^2 = \frac{1}{2} \left[|\mathbf{k}|^2 + \omega_P^2 + \sqrt{|\mathbf{k}|^4 + \omega_P^4 + 2\omega_P^2 |\mathbf{k}|^2 (1 - 2\cos^2 \theta_B)} \right]$$

The gradient in the denominator is understood to act on $\mathbf{B}(x)$ contained implicitly in the angle θ_B , as well as the plasma frequency $\omega_P^2(x)$. The axion phase-space density f_a and we use the fact that the axion density is enhanced due to the gravitational field of the star, so that

$$f_a(\mathbf{x}, \mathbf{k}) = v_a n_{a,c} \frac{\delta(\omega - \omega_c)}{4\pi k^2}, \quad n_a \simeq n_a^\infty \sqrt{\frac{2GM_{\text{NS}}}{|\mathbf{x}|}} \frac{1}{v_0}$$

where M_{NS} is the mass of the star, v_a is the phase velocity at the point $\mathbf{x}$ around the star and n_a^∞ is the number density of axions in the halo where the star is located. We show the signal variation as a function of pulsar parameters and frequency, there was some speculation as to whether refraction of photons relative to incoming axions could lead to suppression of the conversion probability. However, it has now been shown that no such effects occur, and that refraction of the photon is already accounted for by the energy gradients. The validity of this formula has been further confirmed through numerical simulations and independent analytic calculations, giving three cross-checks of this result.

instead of needing to carry out ray tracing for each value (θ_i, ϕ_i) and then re-summing the results, one should formally marginalise over unknown parameters $(\theta_i, \phi_i, \mathbf{B}_i, P_i)$, compute the average signal and then quote the statistical uncertainty on this quantity, which scales as $\sigma/\sqrt{N}$, where σ is the variance due to the distribution of these parameters, and N is the number of stars in the sample. This provides a rigorous basis on which to perform a stochastic analysis for populations of stars. The contribution to the statistical error from (θ, ϕ) (which necessitates a ray-tracing study).

Note: There are two further more subtle caveats to the discussion above which relate to the most strongly magnetised NSs. Namely that photons in strongly magnetised NSs experience reabsorption both from standard model processes (in the form of the Cyclotron resonance) and

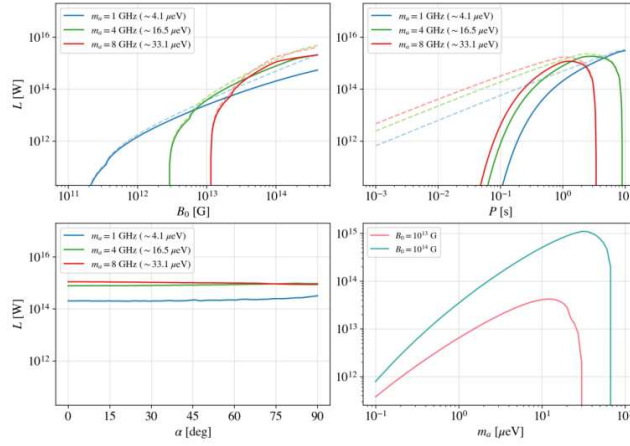


Figure 4.1: *

from re-conversion into axions when the conversion probability becomes so large the conversion is strongly adiabatic.

- overall damping factor $e^{-\tau}$ to the flux, where $\tau \simeq \frac{\pi}{3} \left(\frac{\omega_p^2}{\omega} \right) r$ is the optical depth, and all quantities are understood to be evaluated at the Cyclotron resonance $\omega = \omega_c$, where the Cyclotron frequency is given by $\omega_c = eB/m_e$. Estimating $B \simeq (R/r)^3 B_0$ and using a GJ-like plasma distribution $\omega_p^2 = 4\pi\alpha_{EM}n_e/m_e$ with $n_e \simeq 2\Omega B/e$, we can estimate

$$\tau \simeq 1.4 \left(\frac{R}{10 \text{ km}} \right) \left(\frac{1 \text{ s}}{P} \right) \left(\frac{B_0}{10^{14} \text{ G}} \right)^{1/3} \left(\frac{\mu\text{eV}}{\omega} \right)^{1/3}$$

Resonant conversion between axions and photons occurs at points where $k_\gamma^\mu = k_a^\mu$ where k_γ^μ and k_a^μ are the photon and axion 4-momentum, respectively. An attempt was made to understand the conversion probability for axions to photons $p_{a\gamma}$ leading to

$$p_{a\gamma} = \frac{\pi g_{a\gamma\gamma}^2 \sin^2 \theta_B |\mathbf{B}|^2}{2 |\mathbf{k}_\gamma| |\omega'_p|} \cdot \frac{m_a^5}{(|\mathbf{k}_\gamma|^2 + m_a^2 \sin^2 \theta_B)^2}$$

which attempts to incorporate 3D effects into the conversion probability. This characterises the ratio of the energy density between an axion wavepacket and a photon wavepacket, the latter being subsequently transported out of the magnetosphere along geodesics determined by the photon dispersion relation in the strongly magnetised plasma. Note in the present work we include simultaneously the effects of gravity (by incorporating the curved spacetime metric of the neutron star) and strongly magnetised fields. This results in a covariant dispersion relation for photons in a magnetised plasma which have a dispersion relation .

$$\mathcal{D}(k) = g^{\mu\nu} k_\mu k_\nu - (\omega^2 - k_\parallel^2) \sum_s \frac{4\pi q_s^2 n_s}{\gamma_s^2 \left[\mu_s (\omega - k_\parallel)^2 - c_s^2 (\omega v_s - k_\parallel)^2 \right]}$$

4.6 Search for time dependent signals through matched filter and Time domain analysis works for both FRBs and PSRs ! :

Matched filters are a standard technique in signal processing and they are often used in astronomy to search for signals with a known, or parameterizable profile. This can be done in the spatial, frequency or time domains. In this section, we will derive the standard matched filter before discussing some of its properties. May be use it to quantify the pros and cons of time domain

observations and we will show how it can be used to recover an injected signal in simulated radio data.

There are a number of ways of formulating the matched filter. Here, we will use a discrete matched filter in which the data is represented by a vector of finite length. This can be generalised to continuous functions in which the vector inner products become convolution integrals over functions. The discrete formulation has the advantage of simplifying notation.

Our starting point is the so-called "data vector", $\mathbf{d}$. We will assume the data is the sum of a signal $\mathbf{S} = S_0 \mathbf{F}(\mathbf{p})$ and some additive noise $\mathbf{n}$

$$\mathbf{d} = S_0 \mathbf{F}(\mathbf{p}) + \mathbf{n}$$

Here, we have decomposed the signal according to

$$|\mathbf{F}| \equiv \sqrt{\mathbf{F} \cdot \mathbf{F}} = 1$$

so that S_0 , gives the root mean squared flux density of the signal

$$S_0 = \sqrt{\mathbf{S} \cdot \mathbf{S}}$$

The signal is further characterised by a "parameter vector" $\mathbf{p}$. In section II we will compute a number of templates $\mathbf{F}(\mathbf{p})$ for the signal where $\mathbf{p} = (m_a, \alpha, \theta)$. The noise vector $\mathbf{n}$ is assumed to be Gaussian with $\langle \mathbf{n} \rangle = 0$ and $\langle \mathbf{n} \mathbf{n}^T \rangle = C$ where $\langle \dots \rangle$ denotes an ensemble average of noise realizations and C_{ij} is the covariance matrix with $i, j = 1, \dots, n_d$ where n_d is the total number of data points.

Assuming a Gaussian likelihood, $-2 \log \mathcal{L} = \chi^2_2$, one can calculate the maximum likelihood estimate $\hat{S}_0$ by minimizing

$$\chi^2 = (\mathbf{d} - S_0 \mathbf{F})^T C^{-1} (\mathbf{d} - S_0 \mathbf{F})$$

In the above equation, we assume the data vector $\mathbf{d}$ contains the true signal $\mathbf{S} = \mathbf{S}_{\text{true}}$ so that minimising χ^2 above can be thought of as minimising a generalised leastsquares difference (relative to the noise in each channel hence the factor C^{-1}) between the true signal and possible templates $S_0 \mathbf{F}$. Viewing χ^2 as an unknown function of S_0 , we can find the minimising value, $\hat{S}_0$ given by

$$S_0 = \frac{\mathbf{F}^T C^{-1} \mathbf{d}}{\mathbf{F}^T C^{-1} \mathbf{F}}$$

One can also deduce the "matched filter noise"

$$\sigma_{\text{MF}} = (\mathbf{F}^T C^{-1} \mathbf{F})^{-1/2}$$

and the signal-to-noise estimate $\hat{q}$ may then be written as

$$\hat{q} = \frac{\hat{S}_0}{\sigma_{\text{MF}}} = \frac{\mathbf{F}^T C^{-1} \mathbf{d}}{(\mathbf{F}^T C^{-1} \mathbf{F})^{1/2}}$$

It is important to understand the difference between the noise in the data, characterised by C , and the "matched filter noise", σ_{MF} . They are related, but σ_{MF} also depends on the filter. We return to this issue in the next section.

In order to understand properties of the matched filter we will assume diagonal covariance matrix

$$C_{ij} = \sigma_N^2 \delta_{ij}$$

where δ_{ij} is the Kronecker delta and σ_N is the noise in each channel - all of what said here can be adapted to the case of a general covariance matrix, but it is less simple to see.

In general, the data has a number of dimensions, for example, space, time and/or frequency, then $\mathbf{d}$ has $n_d = n_1 \times \dots \times n_k$ entries where n_i for $i = 1, \dots, k$ are the number of points in each of the dimensions. In our case we will search in the frequency and time directions so the number of entries in the data vector will be $n_d = n_f \times n_t$ where n_f is the number of frequency channels and n_t is the number of time samples. In that case, the data vector could be written as

$$\mathbf{d} = (d_{t_1}^{\mu_1}, \dots, d_{t_{n_t}}^{\mu_1}, \dots, d_{t_1}^{\omega_{n_t}}, \dots, d_{t_{n_t}}^{\omega_{n_t}})$$

where $d_{t_j}^{t_i}$ labels the data vector in the i th frequency bin and j th time-channel. The covariance matrix is then nothing more than the statement that the noise between all possible pairs of time and frequency bins is totally uncorrelated.

Returning to our main discussion, it follows that for a covariance matrix , we have

$$\hat{q} = \frac{\mathbf{F} \cdot \mathbf{d}}{\sigma_N}$$

that is, the dot product of the filter $\mathbf{F}$ with the data vector. When $\mathbf{F}$ matches that in the true signal, we have $\langle \hat{q} \rangle = S_0/\sigma_N$.

Now assume that $\mathbf{p}_{\text{true}}$ has entries which are the true parameters. The ensemble average of $\hat{q}$ for a filter with arbitrary parameter, $\mathbf{p}$ is

$$\langle \hat{q} \rangle = \mathbf{F}(\mathbf{p}) \cdot \mathbf{F}(\mathbf{p}_{\text{true}}) \frac{S_0}{\sigma_N}$$

Next, we note that as a trivial consequence of the Cauchy-Schwarz inequality, we have $\mathbf{F}(\mathbf{p}) \cdot \mathbf{F}(\mathbf{p}_{\text{true}}) \leq |\mathbf{F}(\mathbf{p})| |\mathbf{F}(\mathbf{p}_{\text{true}})|$ with equality if and only if $\mathbf{F}(\mathbf{p}) = \mathbf{F}(\mathbf{p}_{\text{true}})$. Assuming that the template is non-degenerate with respect to the values of $\mathbf{p}$, this occurs only for $\mathbf{p} = \mathbf{p}_{\text{true}}$ for which $\hat{q}$ is then maximal. Thus $\hat{q}$ acts as a likelihood test for the values of $\mathbf{p}$.

In what follows, since the line width of the axion is typically less than the width of our frequency channels, the vector is sparse for a given value of m_a , with nonvanishing entries in only one frequency channel where $\omega = m_a$. This means filters with different values of m_a are orthogonal. More generally, with higher frequency resolution, we would expect to be able to probe both the time and frequency structure of the signal. By contrast, filters with different values of α and θ are not orthogonal and will in general have overlap such that $\mathbf{d}(\theta, \alpha, m_a) \cdot \mathbf{d}(\theta', \alpha', m_a) \neq 0$.

In principle, this means an axion detection would allow us to determine likely values of the pulsar parameters in analogy to the way in which observable gravitational wave signals allow inference of the mass and spin of their associated black holes. However, present approach will be to minimise the expected signal over a conservative subset of values of (α, θ) , thereby obtaining the most conservative constraints on $g_{a\gamma\gamma}$ for allowed values of angles.

In order to turn our continuous axion signals into discrete vectors we must perform some kind of coarsegraining. We therefore define a binning scheme for N time-channels centered on the points $t_i = (i - 1/2)\Delta t$ where $i = 1, \dots, N$ and $\Delta t = 1/N$ so that the discretized signal is given by

$$S_{t_i}^\omega = \frac{1}{\Delta t} \int_{t_i - \Delta t/2}^{t_i + \Delta t/2} dt S(t, \omega)$$

where $S(t, \omega)$ is the flux density of the axion signal as a function of time (and frequency) that we derive using our radio signal models. B. Why do a time domain analysis?

In this section we will discuss the advantages of doing a time domain analysis in terms of increasing the chances of detecting axions. We will also discuss the issue of whether subtraction of the pulse-average from the timedomain data (as is often the case in pulsar observations and as we have done in our data) will have significant impact.

In order to do this we will investigate some properties of the matched filter. Consider applying the matched filter to a given frequency channel whose signal consists of N time-channels is

$$\mathbf{S} = (S_1, \dots, S_N)$$

the matched filter will return

$$\langle \hat{q} \rangle = \frac{\sqrt{\mathbf{S} \cdot \mathbf{S}}}{\sigma_N}$$

where σ_N is the noise on each of the $n_t = N$ timechannels. The noise amplitude σ_N averaged across all times is then given by $\bar{\sigma} = \sigma_N/\sqrt{N}$. We can therefore re-write (18) as

$$\langle \hat{q} \rangle_{\text{time}} = \frac{(\sigma_S^2 + \mu_S^2)^{1/2}}{\sigma}$$

where $\mu_S = \Sigma_i S_i/N$ and $\sigma_S = \sqrt{\Sigma_i S_i^2/N - \mu_S^2}$ are the average and standard deviation of S ,

respectively.

Now let us consider carrying out a measurement with no time resolution. This is the case when one simply uses the telescope to make a total flux measurement over a long observing time. In this case the noise is again $\bar{\sigma}$ given by averaging the noise over all integration time, the signal $\sqrt{\mathbf{S} \cdot \mathbf{S}}$ is simply given by the mean μ_S so that

$$\langle \hat{q} \rangle_{\text{flux-avg.}} = \frac{\mu_S}{\bar{\sigma}}$$

This can be thought of as a trivial matched filter with a single time channel, in which any fine-grained time information has been lost. This is in effect how all previous single pulsar observations for axion dark matter have been carried out.

By comparing the cases of a time-domain analysis with a total-flux measurement it becomes immediately apparent that since $\sigma_S^2 \geq 0$, the time-domain analyses will always equal or outperform the time-averaged measurement. Thus time-domain information increases the potential to detect axions. In particular, when the relative time-variation is large ($\sigma_S/\mu_S \gg 1$), the time-domain search provides a gain in sensitivity by increasing the signal to noise by a factor $\simeq \sqrt{\sigma_S/\mu_S}$, i.e. the squareroot of the relative variance. This is a simple consequence of the fact that $\hat{q}$ is proportional to the root-square of the signal, and so it implicitly encodes information about its variability. The same observation was also made but the matched filter allows this to be justified from first-principles.

Although this is in general not necessary, the observations used later in the paper have had the average of the signal subtracted³. Thus while we retain time-domain information, the baseline μ_S of the signal is essentially renormalised to zero. In this case, the time-domain analysis gives a baseline subtracted (BS) value

$$\langle \hat{q} \rangle_{\text{time-BS}} = \frac{\sigma_S}{\bar{\sigma}}$$

Clearly, when there is only a small time variation, ($\mu_S \gg \sigma_S$), subtracting the baseline leads to a lower value of $\langle \hat{q} \rangle$, relative to retaining it. This is for the simple reason that if the average signal is large, one loses a lot³. Observations of pulsars are often taken in this way since they are probing the time-variable signal from the rotation of the neutron star. Ultimately, there is nothing to prevent the use of the averaged signal as well, but it can require extra work to calibrate it. Here, we see that if the expected time variation is significant, removing the average does not significantly affect sensitivity to axions.

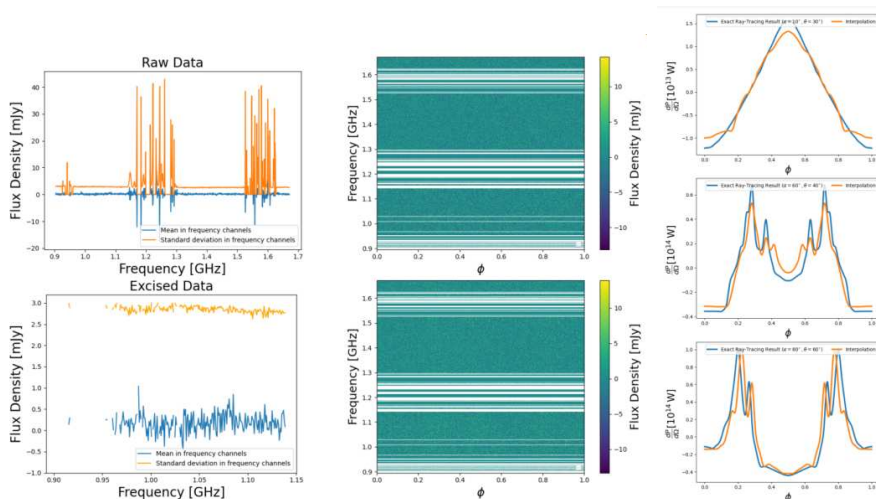


Figure 4.2: *

4.6.1 Coupling constant Bounds :

In the absence of a detection, one can derive constraints on $g_{a\gamma\gamma}$ by comparing the measured value of $q_{\text{meas.}}$ for a particular template, with the expected value $\langle q \rangle$ for that same template. The measured value is defined by

$$q_{\text{meas.}}(\theta, \alpha) = \frac{\mathbf{F}(\theta, \alpha) \cdot \mathbf{d}}{\sigma_N}$$

This must be compared to the expected value $\langle q(g_{a\gamma\gamma}, \theta, \alpha) \rangle$ if a signal were present in the data

$$\langle q(g_{a\gamma\gamma}, \theta, \alpha) \rangle = \frac{\sqrt{\mathbf{S}(g_{a\gamma\gamma}, \theta, \alpha) \cdot \mathbf{S}(g_{a\gamma\gamma}, \theta, \alpha)}}{\sigma_N}$$

In the perturbative limit of the conversion probability, which we have checked is always the case here, this is $\propto g_{a\gamma\gamma}$. Therefore, in order to impose a limit we calculate $\langle q(g_{a\gamma\gamma}^{\text{fid}}, \theta, \alpha) \rangle$ for $g_{a\gamma\gamma}^{\text{fid}} = 10^{-10} \text{GeV}^{-1}$ and exclude any value such that

$$g_{a\gamma\gamma} > g_{a\gamma\gamma}^{\text{fid}} \sqrt{\frac{2q_{\text{meas.}}(\theta, \alpha)}{\langle q(g_{a\gamma\gamma}^{\text{fid}}, \theta, \alpha) \rangle}}$$

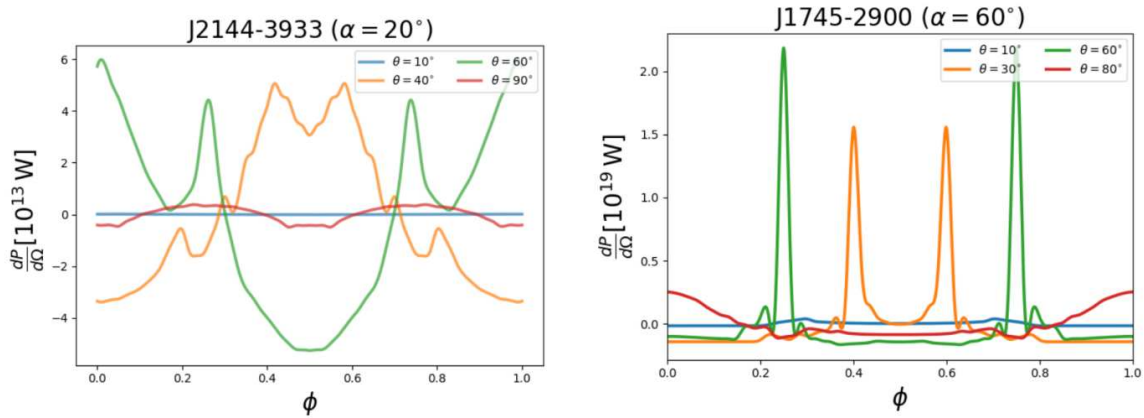


Figure 4.3: *

Pulsar Population Luminosity decreases rapidly (see graph) with respect to magnetic field within a certain bin so its convenient to search for high magnetar source possibly FRBs but still the magnetar origin of FRBs are not well understood , still the first detected FRB Lorimer Burst is well understood and possibly highest Luminosity(30-45 Jy) ever detected.

4.7 Analysis of FRB Data :

We have Analysied burst spectra from FRB010724 was detected in three different beams, 6, 7 and 13 though the detection in beam 6 was so bright . Includes modeling for best flux density , getting Gaussian best fit to separete signal and noise , getting Noise profile and SNR ,plotting the dependance of Noise with respect to Dispersion measure(DM) Archival data for several Parkes FRBs can be found here : <https://data-portal.hpc.swin.edu.au/dataset/parkes-frbs-archival-data>, though lots more FRBs are included in the catalog at <http://frbcat.org/>. I've downloaded the relevant data from the Lorimer burst in ASCII format and put in it the data folder

I've included just the three beams where the FRB was detected, 6, 7 and 13. The data are 2 second samples from around the burst with 1 ms temporal resolution and 96 frequency channels spanning 288 MHz, centered on 1374 MHz. Let's plot those dynamic spectra (frequency vs. time). I've included just the three beams where the FRB was detected, 6, 7 and 13. The data are 2 second samples from around the burst with 1 ms temporal resolution and 96 frequency channels spanning 288 MHz, centered on 1374 MHz. Let's go ahead and plot those dynamic spectra (frequency vs. time). Averaged the amplitudes in 4 ms chunks, producing a new dictionary called amplitudes mapping beam number to numpy arrays. Also average times together to get a new times array.Select the beam(6 or 7 or 13) and separete signal and noise using the gaussian fit .

4.7.1 Modelling:

The most notable feature in the dynamic spectra of the FRB010724 (and of all FRBs), besides its bright, short-duration pulse, is the frequency-dependent arrival time of the burst. This dispersion is a consequence of the fact that the speed of light is not the same for all frequencies of light in many media (like prisms). In particular, the space between galaxies is filled with sparse plasma. The integral of the electron number density along the line of sight at distance d to the source gives us its dispersion measure (DM):

$$DM \equiv \int_0^d n_e(l) dl \dots\dots\dots (1)$$

The dispersion measure is traditionally measured in units of cm^{-3}pc . As Equation 1 of Lorimer et al. (2007) states, the differential arrival time of a dispersed pulse between two frequencies ν_{low} and ν_{high} is given by:

$$\Delta t = 4.148808 \text{ ms} \times \left[\left(\frac{\nu_{\text{low}}}{\text{GHz}} \right)^{-2} - \left(\frac{\nu_{\text{high}}}{\text{GHz}} \right)^{-2} \right] \times \left(\frac{DM}{\text{cm}^{-3}\text{pc}} \right) \dots\dots (2)$$

What is particularly notable is that the measured $DM(> 300)$ is difficult to explain due ionized gas within the Milky Way. Along the direction to the source, only about $25 \text{ cm}^{-3} \text{ pc}$ could be explained by local dispersion of the burst. The rest of the integral must have either come from the region around the source itself or the intergalactic medium (IGM). If the rest of the dispersion were entirely due to the IGM, Lorimer et al. argue that this would put the distance to the source at roughly 1Gpc, roughly 3 billion light years away!

To measure the DM and the other properties of the burst, we will need a model. To do that, I've created a function called *generate_gaussian_frb* below. The idea is to model the FRB as a Gaussian with amplitude amp and frequency-dependent center time. As Lorimer argue, the width of the pulse should scale as ν^{-4} power to scattering in the intervening medium, so the σ of that Gaussian is given by:

$$\sigma(\nu) = \frac{1}{2\sqrt{2 \ln 2}} FWHM(\nu) = \frac{1}{2\sqrt{2 \ln 2}} FWHM(\nu_{\text{high}}) \left[\frac{\nu}{\nu_{\text{high}}} \right]^{-4} \dots\dots\dots (3)$$

So to produce a model, we have a Gaussian profile at ν_{high} centered at $t(\nu_{\text{high}})$, with full-width at half-maximum given by $FWHM(\nu_{\text{high}})$ and amplitude A (for our purposes, let's not worry about the units of this, since the amplitude is multiplied by an unknown beam value) and dispersion measure DM .

Given this model, `scipy.optimize.curve_fit` is used to solve for the arrival time, DM, pulse width, and apparent amplitude in Beam 13 at 4 ms resolution. Then plot the burst, the model of the burst evaluated at best fit values for the four parameters, and the residual between the two.

curve_fit fits 1D functions, use `np. ravel()` to turn 2D functions (like dynamic spectra) into 1D functions. In general, curve fitting functions are iterative algorithms that look for convergence. To prevent the fitter from diverging or finding local minima that aren't the global optimal value, try bounding the parameters within the founding ranges: $-0 \leq t(\nu_{\text{high}}) \leq 2$ - $10 \leq DM \leq 1000$ - $-30 \leq A \leq 30$ - $.004 \leq FWHM(\nu_{\text{high}}) \leq .1$.

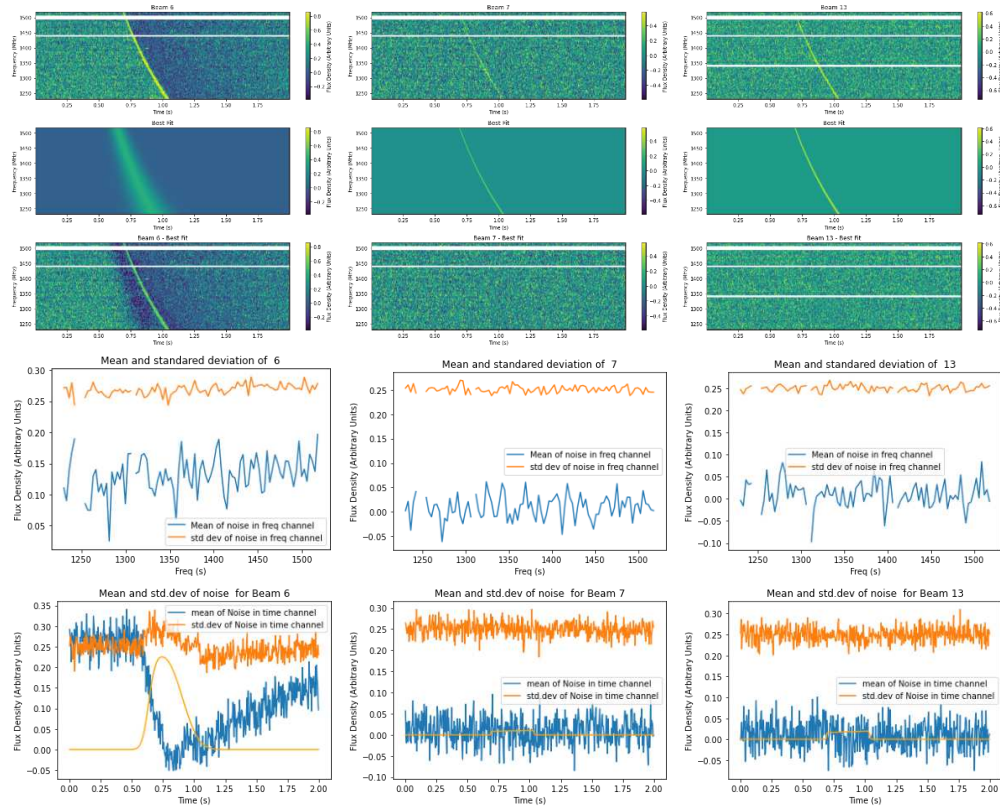


Figure 4.4: *

acc to lorimers paper : minimum detected flux is 300 mJy=0.3 Jy and maximum detectable flux is 30+-10Jy

4.7.2 Measurement of fit:

A good model is one whose differences from the data are consistent with the expected noise in the measurement. We can quantify this by computing χ^2 for a given set of data points d_i and model points m_i computed from a set of model parameters $\vec{\theta}$ with expected noise variance σ_i^2 :

$$\chi^2(\vec{\theta}) = \sum_i \frac{|d_i - m_i(\vec{\theta})|^2}{\sigma_i^2}$$

Setting aside the question of whether the model is consistent with the data up to the noise, one can generally say that, for a given model, the set of parameters that minimizes χ^2 is the "best fit." This is effectively what `scipy.optimize.curve_fit` is doing.

To look at χ^2 closely, let's compute it ourselves for a variety of values of DM around the best fit found above, keeping the other parameters constant. To start, we need to calculate σ_i , the standard deviation of the noise at each time and frequency. For this problem set, we'll assume that it's a constant and so we'll use a very simple estimate the noise by looking at the standard deviation of the unflagged data after subtracting the best-fit model.

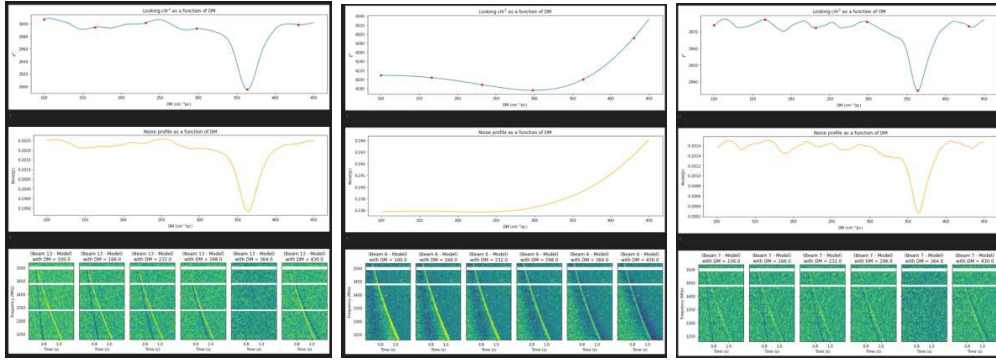


Figure 4.5: *

4.7.3 MCMC

Even if the noise in each individual time and frequency were well-described by a Gaussian (and it's not exactly, though averaging in time helps make it more Gaussian), it's not obvious what the "noise" on the final measurement of, for example, DM should be or even if those errors are Gaussian-distributed. However, there is a very general class of algorithms that helps us estimate the probability distribution. These algorithms, known as Markov Chain Monte Carlo (MCMC), allow us to explore the space of parameters and create a distribution of samples drawn from the underlying probability distribution that we're interested in—the probability distribution that describes our measurement and our uncertainty on it. If we want, we can summarize these distributions with statistics like σ , but that's merely a matter of convenience; ultimately the distribution itself contains the full extent of our information.

In general in statistics, we're interested in the question of how likely it is that our parameters of interest have some value, given the measurement we made and what we know about it. We write this as $P(\vec{\theta} | \vec{d})$ and call it the posterior probability because it reflects our knowledge after the measurement was made (at least in the Bayesian philosophy of probability, but that's a whole other topic). This question can be answered with Bayes' theorem:

$$P(\vec{\theta} | \vec{d}) = \frac{P(\vec{d} | \vec{\theta})P(\vec{\theta})}{P(\vec{d})}$$

where $P(\vec{d} | \vec{\theta})$ is the likelihood of having gotten the data that we did, assuming our model and its parameters are right and $P(\vec{\theta})$ is our prior probability estimate that our model is right. (We often ignore $P(\vec{d})$ the so-called evidence because it does not depend on $\vec{\theta}$, it is often difficult to calculate directly, and can usually be accounted for by normalizing the final posterior distribution to integrate to 1.)

The key idea a MCMC is that it randomly samples the model parameters in a way that's proportional to the posterior. Here's the basic procedure for the simplest MCMC, the Metropolis-Hastings algorithm: 1. Start with some guess set of parameters $\vec{\theta}_0$, perhaps some best fit but it can be anything reasonable. 2. Pick another random set of parameters $\vec{\theta}_1$ from the prior probability distribution $P(\vec{\theta})$. 3. Calculate an "acceptance" ratio of the posteriors using Bayes' theorem:

$$\frac{P(\vec{d} | \vec{\theta}_1) P(\vec{\theta}_1)}{P(\vec{d} | \vec{\theta}_0) P(\vec{\theta}_0)}$$

4. If that ratio is greater than 1 accept the new set of parameters and add it to . list of accepted parameters. If that ratio is less than 1, then pick a random number uniformly distributed between 0 and 1. If . random number is less than . ratio, accept the new set of parameters and add it to . list. Otherwise reject it. 5. If we accepted the new set of parameters, make them . new $\vec{\theta}_0$. 6. Now return to to step 2 and continue into . list of accepted parameter values is long enough to effectively describe underlying probability distribution. This algorithm uses a Markov chain, a random process that only depends on the previous state, to eventually

map out the final posterior probability distribution.

It is often useful to throw away the first several points, since we don't really know if . starting point was representative of the final distribution. Generally speaking, Markov chains tend to lose "memory" of their starting positions, so throwing away some fraction of initial draws is considered best practice. We call this process "burn-in" and it's a bit of an art to figure out how many points to burn. Generally, if we throw out enough points so that the first samples in the chain aren't outliers, we've burned in enough.

For the purposes of this problem set, we're going to make two simplifying assumptions. First we're going to assume uniform priors on the parameters. That means that $P(\vec{\theta})$ is constant inside the bounds on the parameter and 0 outside. So as long as $\vec{\theta}_1$ and $\vec{\theta}_0$ are in bounds, the prior terms cancel the acceptance ratio. Second, we're going to assume that each data point has independent, Gaussian distributed noise. Thus, the likelihood of a given measurement is simply the product of the probabilities of each individual measurement. Up to an overall normalization, this is given by:

$$P(\vec{d} | \vec{\theta}) \propto \prod_i \exp \left[-\frac{|d_i - m_i(\vec{\theta})|^2}{2\sigma_i^2} \right] = e^{-x^2/2}.$$

Thus, the acceptance ratio can be calculated simply as

$$\frac{P(\vec{d} | \vec{\theta}_1)}{P(\vec{d} | \vec{\theta}_0)} = e^{-(\chi_1^2 - \chi_0^2)/2}$$

Implement the Metropolis-Hastings algorithm sample the distribution of values for DM, holding all other parameters fixed at their best-fit values. - Try 10,000 proposed values of DM, drawn uniformly from the range $350 \text{ cm}^{-3}\text{pc} \leq \text{DM} \leq 370 \text{ cm}^{-3}\text{pc}$, but use the first 10% of . accepted samples as burn-in.

- Plot the accepted proposals for both the burn-in and final sample. - Plot a histogram of the posteriori distribution of DM from the accepted, non-burned draws. Perform a 4D MCMC over all 4 parameters of the model, using the above prior and likelihood.

Plot the accepted and burned proposals for DM and the distribution of DMs.

Then use corner to make a corner plot of the 2D covariances (with 68% and 95% confidence regions) and marginalized posteriors of all four parameters.analysis for matched filter is still going on.

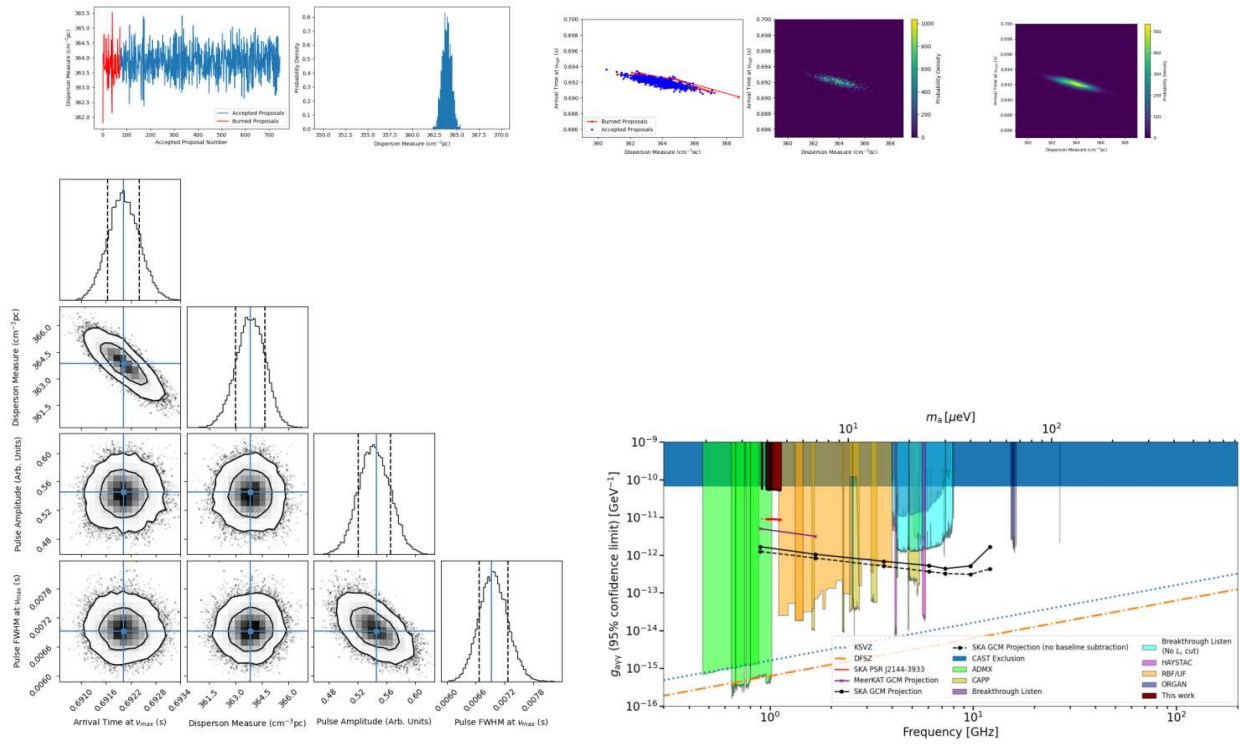


Figure 4.6: *

Chapter 5

Light Shining Through Walls: Experimental Bounds in Astrophysical Situations

Key ingredients	Laboratory	Obscured magnetars	Similarity
Source	Laser	Magnetar	$\checkmark^1$
Magnetic field I	$\sim 10^5$ G	$\sim 10^{14}$ G	$\checkmark\checkmark$
Opaque barrier	Wall (opaque)	Nebula (semi-opaque)	$\checkmark^2$
Magnetic field II	$\sim 10^5$ G	$\sim 10^{-6}$ G	$\checkmark^3$
Detector	Sensitive in-lab	X-ray observations	$\checkmark\checkmark$

¹ Large uncertainty in luminosity estimates of magnetars.

² It is possible for some photons to always reach the detector due to the semi-opaque nature of the nebula.

³ The magnetic field in the ISM is weak ($\sim \mu\text{G}$), although the overall effect would be amplified due to large distances ($\sim \text{kpc}$) traversed.

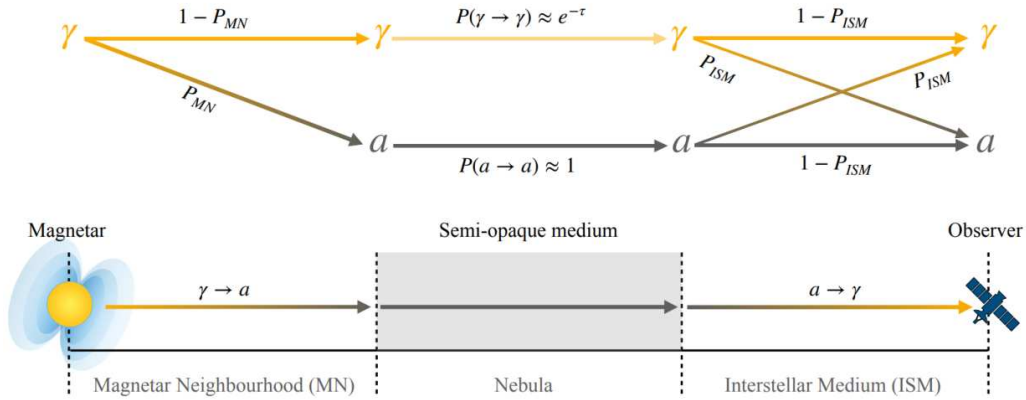


Figure 5.1: *

5.1 Without Resonance

The $\gamma_{\parallel}$ or $A_{\parallel}$ component of an emitted photon couples to ALPs and hence will undergo oscillations in the presence of magnetic field. In the two-state approximation, the evolution of state $(\gamma_{\parallel}, a)$ is represented by the differential equation

$$i \frac{d}{dl} \begin{pmatrix} \gamma_{\parallel} \\ a \end{pmatrix} = \mathcal{H}_{\text{eff}} \begin{pmatrix} \gamma_{\parallel} \\ a \end{pmatrix}$$

The state after traversing a distance l is then given by

$$\begin{pmatrix} \gamma_{\parallel}(l) \\ a(l) \end{pmatrix} = \exp \left[-i \int_0^l \mathcal{H}_{\text{eff}} dl' \right] \begin{pmatrix} \gamma_{\parallel}(0) \\ a(0) \end{pmatrix}$$

where $\mathcal{H}_{\text{eff}}$ is the effective Hamiltonian from previous discussions. The conversion probability $P(\gamma \rightarrow a)$ is given by $P(\gamma \rightarrow a) \equiv |\langle a(l) | \gamma_{\parallel}(0) \rangle|^2$. In the limit of constant electron density and magnetic field, and hence constant $\mathcal{H}_{\text{eff}}$, the photon-ALP conversion probability is

$$P(\gamma \rightarrow a) = 4 \left(\frac{\Delta_{a\gamma}^2}{\Delta_{\text{osc}}^2} \right) \sin^2 \left(\frac{\Delta_{\text{osc}} l}{2} \right)$$

with $P(a \rightarrow \gamma) = P(\gamma \rightarrow a)$. Here the oscillation wavenumber Δ_{osc} is given by

$$\Delta_{\text{osc}}^2 = (\Delta_{\text{pl}} - \Delta_a)^2 + 4\Delta_{a\gamma}^2$$

For propagation in the interstellar medium, we choose the limit of constant electron density and magnetic field, which allows us to easily estimate the behavior of the conversion probability. Note that this probability, regulated by $\Delta_{a\gamma}^2/\Delta_{\text{osc}}^2$, would be small in the limit $\Delta_{a\gamma} \ll |\Delta_{\text{pl}} - \Delta_a|$.

5.2 With Resonance :

As a photon propagates outward from the surface of a magnetar, the effective mass of the photon changes due to its dependence on the magnetic field and electron density. This may lead to fulfillment of the resonance condition $m_{\text{eff}}^2 = m_a^2$. To investigate the effects of resonance on the transition probability, we first calculate the Landau-Zener adiabaticity parameter Γ_{ad} , defined at the point of resonance l_i as

$$\Gamma_i \equiv \left[\frac{4\Delta_{a\gamma}^2}{|d\Delta_{\text{pl}}/dl|} \right]_{l=l_i} = \frac{2g_{a\gamma}^2 \omega_{\gamma}}{m_a^2} B_{T,i}^2 \mathcal{R}_i$$

$$B_{T,i} \equiv B_T(l_i), \quad \mathcal{R}_i \equiv \left| \frac{d \ln m_{\text{eff}}^2}{dl} \right|_{l=l_i}^{-1}$$

For a single resonance, the flip probability $\mathbb{P}_{\text{flip},i}$ between the mass eigenstates can be approximated as

$$\mathbb{P}_{\text{flip},i} \approx e^{-\pi\Gamma_i/2}$$

The flip probability $\mathbb{P}_{\text{flip},i}$ may be interpreted as the probability $\mathbb{P}_i(\gamma \rightarrow \gamma)$ survival probability if far away from resonance, the off-diagonal mixing term in $\mathcal{H}_{\text{eff}}$ is much smaller than the difference between the diagonal terms. That is, for $|l - l_i| \gg 0$, we need to have

$$2|g_{a\gamma} B_T(l) \omega_{\gamma}| \ll |m_{\text{eff}}^2(l) - m_a^2|$$

This condition is fulfilled near the surface of the magnetar, due to the large values of $|m_{\text{eff}}^2|$, and also very far away from the magnetar, due to the very small values of $B_T(l)$. If this condition is satisfied around a resonance, it would imply that the survival probability $\mathbb{P}(\gamma \rightarrow \gamma)$ at that resonance ' i ' would be

$$\mathbb{P}_i(\gamma \rightarrow \gamma) = e^{-\pi\Gamma_i/2}$$

The conversion probability for a single resonance is given by

$$\mathbb{P}_i(\gamma \rightarrow a) = 1 - e^{-\pi\Gamma_i/2}$$

For small Γ_i values, this can be approximated as

$$\mathbb{P}_i(\gamma \rightarrow a) \approx \pi\Gamma_i/2$$

5.3 With a Pair of Resonances

Due to large magnetic fields typically present close to the magnetars, m_{EM}^2 dominates over the ω_{pl}^2 contribution for X-ray photons. This results in $m_{\text{eff}}^2 < 0$, and consequently $m_{\text{eff}}^2 < m_a^2$ in this region. Also, far away from the magnetar $m_{\text{eff}}^2 \rightarrow 0$ due to the decreasing magnetic field and electron density, resulting again in $m_{\text{eff}}^2 < m_a^2$. The flavor conversions will be significant if there are resonances in the intermediate region, where m_{eff}^2 crosses m_a^2 . The above arguments show that the number of such resonances must always be even. Even when there are multiple resonance along the way, the resonances can be split up into pairs where m_{eff}^2 crosses m_a^2 from below at the first resonance and from above in the resonance immediately after.

We therefore first focus on one such pair. The photon survival and conversion probabilities while crossing such a pair can be expressed as

$$\mathbb{P}(\gamma \rightarrow \gamma) = P_1 P_2 + (1 - P_1)(1 - P_2)$$

$$\mathbb{P}(\gamma \rightarrow a) = P_1(1 - P_2) + P_2(1 - P_1)$$

where $P_i \equiv \mathbb{P}_i(\gamma \rightarrow a) = 1 - e^{-\pi \Gamma_i/2}$ is the conversion probability at the resonance 'i'. Note that if P_1 and P_2 are small, the net conversion probability for this pair would become

$$\mathbb{P}(\gamma \rightarrow a) = P_1(1 - P_2) + P_2(1 - P_1) = P_1 + P_2 - 2P_1 P_2 \approx P_1 + P_2$$

As a photon starts propagating from the surface of the magnetar, the resonance condition ($m_{\text{eff}}^2 = m_a^2$) is likely to be fulfilled naturally multiple times in the magnetar neighborhood. This happens because, while B vs. r and n_e vs. r capture the overall behavior, the background electron density and the magnetic field strength typically undergo fluctuations. In the next section, we explore the effects of multiple such pairs of resonances in succession.

5.4 With Multiple Resonances and Different Domains

To calculate the photon to ALP conversion probability in the presence of many fluctuations, we need to know the overall behavior of the environment. We assume that, as the photon propagates outwards from the surface of the magnetar, in addition to fluctuations in the electron density and magnetic field strength, it also encounters many domains with magnetic fields oriented randomly towards an arbitrary direction. Further, due to the randomness of orientation of the magnetic field, we assume that the angles at which each domain are misaligned with the next one are $\sim O(1)$. Let the initial intensities of γ_1, γ_2 and a be $I_{\gamma_1}(0), I_{\gamma_2}(0)$ and $I_a(0)$. In the n -th domain, let the transverse component of the magnetic field be rotated by an angle θ_n compared to the first domain. The photon polarizations, parallel and perpendicular to the transverse magnetic field in this domain can be expressed as

$$|\gamma_{\parallel}(n)\rangle = \cos \theta_n |\gamma_1\rangle + \sin \theta_n |\gamma_2\rangle, \quad |\gamma_{\perp}(n)\rangle = -\sin \theta_n |\gamma_1\rangle + \cos \theta_n |\gamma_2\rangle$$

In terms of the $(|\gamma_1\rangle, |\gamma_2\rangle)$ basis, we then have

$$|\gamma_1\rangle = \cos \theta_n |\gamma_{\parallel}(n)\rangle - \sin \theta_n |\gamma_{\perp}(n)\rangle, \quad |\gamma_2\rangle = \sin \theta_n |\gamma_{\parallel}(n)\rangle + \cos \theta_n |\gamma_{\perp}(n)\rangle$$

The Photon-ALP transition in the n -th domain involves only $|\gamma_{\parallel}(n)\rangle$ and the ALP state $|a\rangle$. Let this transition probability be p_n , then at the end of n -th domain the photon and ALP intensities will be

$$I_{\gamma_1}(n) \sim (1 - \cos^2 \theta_n p_n) I_{\gamma_1}(n-1) + \cos^2 \theta_n p_n I_a(n-1),$$

$$I_{\gamma_2}(n) \sim (1 - \sin^2 \theta_n p_n) I_{\gamma_2}(n-1) + \sin^2 \theta_n p_n I_a(n-1),$$

$$I_a(n) \sim p_n [\cos^2 \theta_n I_{\gamma_1}(n-1) + \sin^2 \theta_n I_{\gamma_2}(n-1)] + (1 - p_n) I_a(n-1).$$

Note that due to the random nature of θ_n , the $\cos^2 \theta_n$ and $\sin^2 \theta_n$ may be replaced by their averaged value of $1/2$ when we are dealing with a large number of domains. Then we get the "averaged" recursion relation

$$\begin{pmatrix} I_{\gamma_1}(n) \\ I_{\gamma_2}(n) \\ I_a(n) \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{2}p_n & 0 & \frac{1}{2}p_n \\ 0 & 1 - \frac{1}{2}p_n & \frac{1}{2}p_n \\ \frac{1}{2}p_n & \frac{1}{2}p_n & 1 - p_n \end{pmatrix} \begin{pmatrix} I_{\gamma_1}(n-1) \\ I_{\gamma_2}(n-1) \\ I_a(n-1) \end{pmatrix}$$

This recursion relation leads to the intensity at the end of n -th domain evolving as

$$\begin{pmatrix} I_{\gamma_1}(n) \\ I_{\gamma_2}(n) \\ I_a(n) \end{pmatrix} = \mathbb{M}_{3 \times 3}^{(n)} \begin{pmatrix} I_{\gamma_1}(0) \\ I_{\gamma_2}(0) \\ I_a(0) \end{pmatrix}$$

Here,

$$\mathbb{M}_{3 \times 3}^{(n)} = \begin{pmatrix} m_{11}^{(n)} & m_{12}^{(n)} & m_{13}^{(n)} \\ m_{12}^{(n)} & m_{11}^{(n)} & m_{13}^{(n)} \\ m_{13}^{(n)} & m_{13}^{(n)} & m_{33}^{(n)} \end{pmatrix}$$

with the individual elements of the matrix given by

$$m_{11}^{(n)} = \frac{1}{3} + \frac{1}{6} \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right) + \frac{1}{2} \prod_{k=1}^n \left(1 - \frac{1}{2}p_k\right)$$

$$m_{12}^{(n)} = \frac{1}{3} + \frac{1}{6} \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right) - \frac{1}{2} \prod_{k=1}^n \left(1 - \frac{1}{2}p_k\right)$$

$$m_{13}^{(n)} = \frac{1}{3} \left[1 - \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right]$$

$$m_{33}^{(n)} = \frac{1}{3} \left[1 + 2 \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right]$$

The method of induction can be used to show that this equation holds for any integer value of n . For $n = 1$, the above equations simplify to the expected expression

$$\mathbb{M}_{3 \times 3}^{(1)} = \begin{pmatrix} 1 - \frac{1}{2}p_1 & 0 & \frac{1}{2}p_1 \\ 0 & 1 - \frac{1}{2}p_1 & \frac{1}{2}p_1 \\ \frac{1}{2}p_1 & \frac{1}{2}p_1 & 1 - p_1 \end{pmatrix}$$

We denote the intensities of the two photon polarizations (defined, for the sake of definiteness, as $\gamma_1 \equiv \gamma_{\perp}(0)$, $\gamma_2 \equiv \gamma_{\parallel}(0)$ at the first domain) and the ALP by I_{γ_1, γ_2} , and I_a , respectively. The random orientation of the magnetic field in each domain ensures that both the photon polarizations mix equally with the ALPs on an average. If p_n denotes the probability of conversion in the n -th domain, then we get the recursion relation

$$\begin{pmatrix} I_{\gamma_1}(n) \\ I_{\gamma_2}(n) \\ I_a(n) \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{2}p_n & 0 & \frac{1}{2}p_n \\ 0 & 1 - \frac{1}{2}p_n & \frac{1}{2}p_n \\ \frac{1}{2}p_n & \frac{1}{2}p_n & 1 - p_n \end{pmatrix} \begin{pmatrix} I_{\gamma_1}(n-1) \\ I_{\gamma_2}(n-1) \\ I_a(n-1) \end{pmatrix}.$$

Here we ignore possible interferences between resonances, i.e., phase effects. Defining $I_{\gamma} \sim I_{\gamma_1} + I_{\gamma_2}$, one may write the above equation as :

$$\begin{pmatrix} I_{\gamma}(n) \\ I_a(n) \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{2}p_n & p_n \\ \frac{1}{2}p_n & 1 - p_n \end{pmatrix} \begin{pmatrix} I_{\gamma}(n-1) \\ I_a(n-1) \end{pmatrix}.$$

$$\begin{pmatrix} I_{\gamma}(n) \\ I_a(n) \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \left[1 + \frac{1}{2} \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right] & \frac{2}{3} \left[1 - \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right] \\ \frac{1}{3} \left[1 - \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right] & \frac{2}{3} \left[\frac{1}{2} + \prod_{k=1}^n \left(1 - \frac{3}{2}p_k\right)\right] \end{pmatrix} \begin{pmatrix} I_{\gamma}(0) \\ I_a(0) \end{pmatrix}.$$

Therefore, the photon to ALP conversion probability in the presence of multiple resonances can be expressed as

$$\mathbb{P}(\gamma \rightarrow a) = \frac{1}{3} \left[1 - \prod_{k=1}^{n_d} \left(1 - \frac{3}{2}p_k\right)\right]$$

where n_d is the number of domains. In the limit of large number n_d of domains, this can be approximated as the limiting value

$$\mathbb{P}(\gamma \rightarrow a) = \frac{1}{3} \left[1 - \exp \left(-\frac{3}{2} \sum_{k=1}^{n_d} p_k \right) \right]$$

Note that, each of the domain may have zero or a finite number of resonances. Therefore, in the approximation that the dominant contribution to flavor conversions are near resonances, we obtain

$$\sum_{k=1}^{n_d} p_k = \sum_{i=1}^{n_r} P_i$$

Here, the index ' k ' counts the number of domains, whereas the index ' i ' counts the number of resonances. The total number of resonances is denoted by n_r . The smallness of the individual P_i 's allows us to approximate

$$\sum_{i=1}^{n_r} P_i \approx \sum_{i=1}^{n_r} \frac{\pi \Gamma_i}{2} = \frac{\pi \Gamma_{\text{tot}}}{2}$$

where Γ_{tot} is given by

$$\Gamma_{\text{tot}} = \frac{2g_{a\gamma}^2 \omega_\gamma}{m_a^2} \sum_i^{n_r} B_{T,i}^2 \mathcal{R}_i$$

Therefore, we can estimate the total conversion probability to be

$$P_{\text{tot}} \approx \frac{1}{3} \left(1 - e^{-\frac{3\pi}{4} \Gamma_{\text{tot}}} \right)$$

Interestingly, in the small Γ_{tot} limit, we obtain $P_{\text{tot}} \approx \frac{1}{2} \frac{\pi \Gamma_{\text{ea}}}{2} = \frac{1}{2} \sum P_i$. This is because, for completely unpolarized light, only half of the photons couple to ALPs at any given point and hence experience resonance. On the other hand, due to the existence of either many domains or a few domains with large Γ_i values, in the scenario where the value of Γ_{tot} becomes large, gives $P_{\text{tot}} \sim 1/3$. This is expected, as the intensities of the photon polarizations and the ALP would be driven towards full depolarization of all the three states in this scenario.

In section 5 we calculate the $\gamma \rightarrow a$ conversion probability in the magnetar neighborhood and $a \rightarrow \gamma$ conversion probability in the ISM.

5.5 Photon-ALP-Photon Conversions During Propagation

For the purpose of this work, we need to estimate the number of photons, emitted from the surface of the magnetar, that may cross the opaque barrier by oscillating into ALPs. To calculate this $\gamma \rightarrow a \rightarrow \gamma$ oscillation probability, we need to calculate the conversion probability $P(\gamma \rightarrow a)$ in the magnetar neighborhood and $P(a \rightarrow \gamma)$ in the ISM. In this section, we discuss in detail the key features of the conversion probabilities and quantify their dependence on different input parameter values.

5.5.1 Locations of the Resonances in the Magnetar Neighborhood

For a photon emitted from a magnetar, the resonance condition ($\Delta_{\text{pl}} = \Delta_a$) is likely to be met several times due to the turbulent nature of the potential caused by the varying electron density and magnetic field. However, to start with, we shall make the assumption that these fluctuations are absent and investigate the effect of resonances in the magnetar neighborhood.

We first estimate the distance at which the resonance happens. The resonance condition is satisfied when $m_{\text{eff}}^2 = m_\alpha^2$. The spatial behavior of the m_{eff}^2 can be obtained as

$$m_{\text{eff}}^2 \equiv \frac{4\pi\alpha}{m_e} n_0 \left(\frac{r}{r_0} \right)^{-3} - \frac{88\alpha^2}{270m_e^4} B_0^2 \omega_\gamma^2 \left(\frac{r}{r_0} \right)^{-6}$$

where we have assumed that the power law dependence $n_e \sim r^{-3}$. We redefine the coefficients as

$$c_n \equiv \frac{4\pi\alpha}{m_e}, \quad c_B \equiv \frac{88\alpha^2}{270m_e^4}$$

This allows us to write the following condition at the point of resonance:

$$m_{\text{eff}}^2 = c_n n_0 \left(\frac{r}{r_0} \right)^{-3} - c_B B_0^2 \omega_\gamma^2 \left(\frac{r}{r_0} \right)^{-6} = m_a^2$$

This condition is satisfied for $r = r_\S$ such that

$$r_\S = r_0 \left[\frac{2c_B \omega_\gamma^2 B_0^2}{c_n n_0 \pm \sqrt{c_n^2 n_0^2 - 4c_B B_0^2 \omega_\gamma^2 m_a^2}} \right]^{1/3}$$

Note that the ' + ' sign leads to smaller values, which corresponds to the solution $r_<$, i.e., the near resonance point, whereas the ' - ' sign (leading to larger values) corresponds to the solution $r_>$, i.e., the far resonance point. For $c_n^2 n_0^2 < 4c_B B_0^2 \omega_\gamma^2 m_a^2$, there is no real solution of $r_\S$. This is because in this case the mass of the ALP is always more than the effective mass of the photon, i.e.,

$$m_a^2 > [m_{\text{eff}}^2]_{\text{max}}$$

and hence the resonance condition can never be fulfilled. The photons emitted from the surface of the magnetar of radius ~ 10 km initially have a negative m_{eff}^2 value due to the contribution from the B_0^2 term. As the photons move outward from the magnetar, this contribution decreases faster than the positive contribution due to the n_0 term, and hence the negative value of m_{eff}^2 moves closer to zero. At sufficiently large r , the value of m_{eff}^2 crosses zero, further attains a maximum and later decreases towards zero. - If the maximum value of m_{eff}^2 is greater than m_a^2 , the resonance condition will be satisfied twice. For example, if $m_a^2 = 10^{-18} \text{eV}^2$ (i.e., $m_a = 10^{-9} \text{eV}$) and $n_0 = 10^{15} \text{cm}^{-3}$, resonances would occur at $r_< \approx 10^4$ km and $r_> \approx 10^5$ km. - On the other hand, if m_{eff}^2 cannot reach m_a^2 , then there will be no resonances. For example, in the range $n_0 \sim (10^{12} - 10^{18}) \text{cm}^{-3}$, the resonance condition can never be fulfilled for $m_a^2 \gtrsim 10^{-10} \text{eV}^2$ (i.e., $m_a \gtrsim 10^{-5} \text{eV}$). - With increasing n_0 , the maximum value of m_{eff}^2 increases and hence the resonance condition can be satisfied for a wider range of m_a . Moreover, increasing n_0 leads to m_{eff}^2 values becoming positive at a lower value of r , thus bringing the near resonance point $r_<$ closer.

The increase in n_0 leading to a smaller value of $r_<$. The position of the near-resonance point does not depend significantly on the value of m_a^2 due to the sharp nature of the zero-crossing, and immediately reaching the maximum m_{eff}^2 value. Although too large m_a^2 values do not meet the resonance criteria, leading to the V-like turnaround behavior.

The far resonance point depends inversely (compared to the near resonance point) to the value of n_0 , as a higher value of n_0 leads to a higher contribution to m_{eff}^2 at further distances, which leads to the far resonance condition being satisfied further away. The far resonance point also depends upon the mass of the ALP, as a higher mass meets the $m_{\text{eff}}^2 = m_a^2$ condition sooner, whereas for lower m_a values, the far resonance point $r_>$ shifts further away from the magnetar.

The dependence of the near and far resonance points on the parameters m_a and n_0 can be approximated. In the presence of resonance, the condition $c_n^2 n_0^2 > 4c_B B_0^2 \omega_\gamma^2 m_a^2$ must be satisfied. Therefore, for small values of m_a , such that the condition $c_n^2 n_0^2 \gg 4c_B B_0^2 \omega_\gamma^2 m_a^2$ is satisfied, we can expand and rewrite :

$$r_\S \approx r_0 \left[\frac{2c_B \omega_\gamma^2 B_0^2}{c_n n_0 \pm c_n n_0 \left(1 - \frac{2c_B B_0^2 \omega_\gamma^2 m_a^2}{c_n^2 n_0^2} \right)} \right]^{1/3}$$

For near resonance $r_<$, which corresponds to the ' + ' sign in the denominator, this gives

$$r_{<} \approx r_0 \left[\frac{c_B \omega_\gamma^2 B_0^2}{c_n n_0} \right]^{1/3}$$

Thus pointing out the lack of dependence of $r_{<}$ on the value of m_a , and the power law dependence of $r_{<} \propto n_0^{-1/3}$. On the other hand, for the far resonance $r_{>}$, which corresponds to the ' - ' sign in the denominator, we get

$$r_{>} \approx r_0 \left[\frac{c_n n_0}{m_a^2} \right]^{1/3}$$

thus, revealing the dependence of $r_{>}$ on both m_a and n_0 . This further emphasizes the point that the location of the near-resonance is independent of the mass of the ALP except near the turnaround point. However, note that the existence of the near-resonance is dependent upon the mass, as not all m_a values may be accessible, depending on the maximum effective mass of the photons in the magnetar neighborhood. The parameter ranges for which the resonance criteria is not satisfied is denoted by the gray hatched region.

5.6 The Conversion Probability P_{MN} in the Magnetar Neighborhood

First, we take the idealized scenario where we do not consider fluctuations in magnetic field and electron density. We assume that the behavior of the transverse magnetic field and the electron density. While passing through a resonance, since $m_{\text{eff}}^2 \approx m_a^2$, the adiabaticity parameter $\Gamma_i \equiv \Gamma(r_i)$ corresponding to a single resonance can be simplified to the form

$$\Gamma(r_i) = 2g_{a\gamma}^2 \omega_\gamma B_{T,i}^2 \left| \frac{dm_{\text{eff}}^2}{dr} \right|_{r_i}^{-1}$$

where the resonance point is denoted as r_i . The inverse relationship of $\Gamma(r_i)$ with the derivative term will be instrumental in estimating the general power law behavior of the conversion probability at the near and far resonance points. At the near-resonance point: The near resonance conditions are satisfied around the region where m_{eff}^2 becomes greater than zero. This means that both the competing contributions - regulated by n_0 and B_0 - are important here. This warrants a careful scrutiny of the power-law behavior of each of the contributions. The derivative of m_{eff}^2 is

$$\begin{aligned} \frac{dm_{\text{eff}}^2}{dr} &= -\frac{3}{r} c_n n_0 \left(\frac{r}{r_0} \right)^{-3} + \frac{6}{r} c_B B_0^2 \omega_\gamma^2 \left(\frac{r}{r_0} \right)^{-6} \\ &= -\frac{3}{r} m_a^2 + \frac{3}{r} c_B B_0^2 \omega_\gamma^2 \left(\frac{r}{r_0} \right)^{-6} \end{aligned}$$

At the near-resonance point $r_{\ll}$, this gives

$$\left. \frac{dm_{\text{eff}}^2}{dr} \right|_{r_{<}} = -\frac{3}{r_{<}} m_a^2 + \frac{3}{r_0} c_B B_0^2 \omega_\gamma^2 \left(\frac{r_{<}}{r_0} \right)^{-7}$$

Therefore, in the limit $m_a \rightarrow 0$, the primary power-law behavior of dm_{eff}^2/dr can be simply expressed as

$$\left. \frac{dm_{\text{eff}}^2}{dr} \right|_{r_{<}} \propto r_{<}^{-7}$$

Since $B_T \propto r^{-3}$, we get the power-law dependence of $\Gamma(r_{<})$ to be

$$\Gamma(r_{<}) \propto B_{T,<}^2 \left| \frac{dm_{\text{eff}}^2}{dr} \right|_{r_{<}}^{-1} \propto r_{<}$$

As discussed previously an increase in n_0 would shift the near-resonance point $r_{<}$ closer to the magnetar surface. This decrease in $r_{<}$ would also decrease Γ_{tot} and hence P_{tot} . On the other

hand, a decrease in n_0 would result in the increase of P_{tot} . As discussed in the previous section, dependence of $r_{<}$ on m_a is negligible, leading to almost no dependence of m_a on P_{tot} .

At the far-resonance point: At the far-resonance point $r_{>}$, the dominant contribution to m_{eff}^2 comes from the n_0 term. This is because the B_0^2 term falls as r^{-6} compared to the n_0 term which falls as $\sim r^{-3}$. Therefore, the derivative term can be approximated as

$$\frac{dm_{\text{eff}}^2}{dr} \approx -\frac{3}{r_0} c_n n_0 \left(\frac{r}{r_0} \right)^{-4}$$

This implies that, at this point the power-law dependence of $\Gamma(r_{>})$ is

$$\Gamma(r_{>}) \propto B_{T,>}^2 \left| \frac{dm_{\text{eff}}^2}{dr} \right|_{r_{>}}^{-1} \propto r_{>}^{-2}$$

As discussed previously, an increase in n_0 would mean that the far-resonance $r_{>}$ occurs further away from the magnetar, as it will take longer for m_{eff}^2 to reach the value of m_a^2 . this would lead to a decrease in Γ_{tot} , and thereby a decrease in P_{tot} . On the other hand, a decrease in n_0 would lead to an increase in P_{tot} . Based on similar arguments, an increase in m_a would lead to the condition $m_{\text{eff}}^2 = m_a^2$ being satisfied sooner at the far resonance, which leads to a decrease in $r_{>}$, and an eventual increase in P_{tot} .

	increase in n_0	decrease in n_0	increase in m_a	decrease in m_a
near-resonance	$r_{<}$ decreases $\Gamma(r_{<})$ decreases	$r_{<}$ increases $\Gamma(r_{<})$ increases	$r_{<}$ unchanged $\Gamma(r_{<})$ unchanged	$r_{<}$ unchanged $\Gamma(r_{<})$ unchanged
far-resonance	$r_{>}$ increases $\Gamma(r_{>})$ decreases	$r_{>}$ decreases $\Gamma(r_{>})$ increases	$r_{>}$ decreases $\Gamma(r_{>})$ increases	$r_{>}$ increases $\Gamma(r_{>})$ decreases

Above table : Qualitative behavior of the location and the adiabaticity parameter at the near and farresonance points with respect to changes in n_0 and m_a .

The dependence of the locations of the near and far resonance points and the adiabaticities and the conversion probabilities on the parameters n_0 and m_a are qualitatively indicated in above table. Quantitatively, the adiabaticities at the near and far resonance points can be expressed as

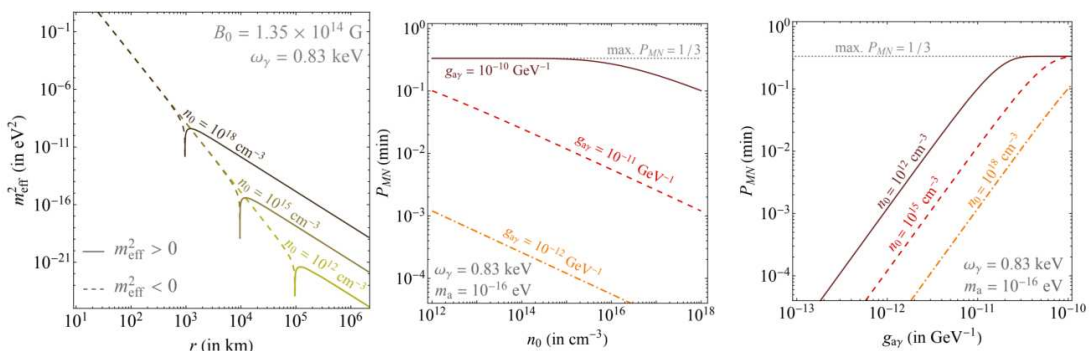
$$\Gamma(r_{<}) \approx 2g_{a\gamma}^2 \frac{1}{\omega_\gamma} \left(\frac{r_{<}}{r_0} \right) \frac{r_0}{3c_B}, \quad \Gamma(r_{>}) \approx 2g_{a\gamma}^2 \omega_\gamma \left(\frac{r_{>}}{r_0} \right)^{-2} \frac{B_0^2}{n_0} \frac{r_0}{3c_n}$$

Note that, $r_{<}$ and $r_{>}$ themselves is a function of B_0, n_0 and m_a . Using the approximate dependence of $r_{<}$ and $r_{>}$ we find that

$$\Gamma(r_{<}) \approx 2g_{a\gamma}^2 \frac{1}{\omega_\gamma} \left(\frac{c_B \omega_\gamma^2 B_0^2}{c_n n_0} \right)^{1/3} \frac{r_0}{3c_B}, \quad \Gamma(r_{>}) \approx 2g_{a\gamma}^2 \omega_\gamma \left(\frac{c_n n_0}{m_a^2} \right)^{-2/3} \frac{B_0^2}{n_0} \frac{r_0}{3c_n}$$

For typical values of parameters for a magnetar, taking $r_0 = 10$ km, $B_0 = 1.35 \times 10^{14}$ G, and $n_0 = 10^{15} \text{ cm}^{-3}$, and further taking $m_a = 10^{-16} \text{ eV}$, for a photon of energy $\omega_\gamma = 0.83 \text{ keV}$, the values of $\Gamma(r_{<})$ and $\Gamma(r_{>})$ are

$$\Gamma(r_{<}) \approx 1.4, \quad \Gamma(r_{>}) \approx 5.1 \times 10^{-12}$$



the adiabaticity parameter at the near-resonance behaves as $\Gamma(r_{<}) \propto n_0^{-1/3}$, whereas the adiabaticity parameter at the far-resonance behaves as $\Gamma(r_{>}) \propto n_0^{-5/3} m_a^{4/3}$. Therefore, the larger ratio of $\Gamma(r_{>})/\Gamma(r_{<})$ can be obtained for the smaller value of n_0 and larger values of m_a . For all ALPs with $m_a \lesssim 10^{-11}$ eV, even taking the smallest possible value of $n_0 = 10^{12} \text{ cm}^{-3}$, we find

$$\Gamma(r_{>})/\Gamma(r_{<})|_{\text{max}} \approx 0.16$$

Therefore, the contribution from the far-resonance can be safely ignored for estimating the transition probability near the magnetar. In the limit of $\Gamma(r_{<}) \gg \Gamma(r_{>})$, we have

$$\mathbb{P}(\gamma_{\parallel} \rightarrow a) \approx P(r_{<}) \approx (1 - e^{-\pi\Gamma(r_{<})/2})$$

The net $\gamma \rightarrow a$ conversion probability will include an extra factor of 1/2 in front to account for $\gamma_{\perp}$ not coupling to ALPs, thus giving us

$$\mathbb{P}(\gamma \rightarrow a) \approx \frac{1}{2} (1 - e^{-\pi\Gamma(r_{<})/2})$$

In reality, we expect that fluctuation and random orientations of the transverse magnetic fields will lead to many resonances being experienced by the photons. Due to the absence of detailed knowledge of the fluctuations involved, we estimate the lower limit of the conversion probability in the magnetar neighborhood to be

$$P_{MN}(\text{min}) = \frac{1}{3} (1 - e^{-3\pi\Gamma(r_{<})/4}) \lesssim \frac{1}{3} (1 - e^{-\frac{3\pi}{4}\Gamma_{\text{tot}}})$$

Here we have used

$$\Gamma(r_{<}) \lesssim \sum \Gamma_i = \Gamma_{\text{tot}}$$

Note that this probability $P_{MN}(\text{min})$ is also less than that obtained by neglecting fluctuations. Therefore, $P_{MN}(\text{min})$ is the conservative lower bound on the conversion probability in the magnetar neighborhood, with or without fluctuations.

5.7 The Conversion Probability P_{ISM} in the Interstellar Medium

The ALPs generated in the magnetar neighborhood will pass through the ISM where they may undergo further oscillations and partially convert back into photons. In this region, the magnetic field and the electron density can be approximated to be

$$B_{\text{ISM}} \sim O(\text{few}) \mu\text{G}, \quad n_{\text{ISM}} \sim O(\text{few} \times 10^{-2}) \text{ cm}^{-3}$$

We take $B_{\text{ISM}} = 2 \mu\text{G}$ and $n_{\text{ISM}} = 2 \times 10^{-2} \text{ cm}^{-3}$ for our ballpark estimates. For soft X-ray ($\sim \text{keV}$) photons, the magnitudes of the elements of the Hamiltonian may be expressed as

$$|\Delta_{a\gamma}| \approx 1.52 \times 10^{-1} \left(\frac{g_{a\gamma}}{10^{-10} \text{ GeV}^{-1}} \right) \left(\frac{B_{\text{ISM}}}{1 \mu\text{G}} \right) \text{ kpc}^{-1}$$

$$|\Delta_{pl}| \approx 1.08 \left(\frac{n_e}{10^{-2} \text{ cm}^{-3}} \right) \left(\frac{1 \text{ keV}}{\omega_{\gamma}} \right) \text{ kpc}^{-1}$$

$$|\Delta_a| \approx 7.82 \times 10^{-2} \left(\frac{m_a}{10^{-12} \text{ eV}} \right)^2 \left(\frac{1 \text{ keV}}{\omega_{\gamma}} \right) \text{ kpc}^{-1}$$

It may be observed that the dominant contribution in the oscillation frequency Δ_{ose} would be from Δ_{pl} , when the ALP mass $m_a \lesssim 10^{-12}$ eV. The oscillation length scale associated with this term is

$$\lambda_{pl} \equiv \frac{2\pi}{|\Delta_{pl}|} \approx 5.82 \times \left(\frac{10^{-2} \text{ cm}^{-3}}{n_e} \right) \left(\frac{\omega_{\gamma}}{1 \text{ keV}} \right) \text{ kpc}$$

This gives, for $n_{\text{ISM}} = 2 \times 10^{-2} \text{ cc}^{-1}$ and for $\omega_{\gamma} = (0.71 - 0.98) \text{ keV}$, an oscillation length of

$$\lambda_{pl} \approx (2.06 - 2.84) \text{ kpc}$$

Note that the distance of the Magnetar candidate under consideration is $L = 9 \text{ kpc}$. This means that the distance traveled alone doesn't allow an averaging out of the probability, as the dominant oscillation length scale λ_{pl} is comparable to distance of $L = 9 \text{ kpc}$. However, due to the large energy band $\omega_\gamma = (0.71 - 0.98)$, the range of corresponding λ_{pl} values is quite large and averaging over oscillation probabilities needs to be performed. The average oscillation probability is given by

$$P_{\text{ISM}}(a \rightarrow \gamma) = \left(\int_{E_{\min}}^{E_{\max}} \left[\frac{4\Delta_{a\gamma}^2}{\Delta_{\text{osc}}^2} \sin^2 \left(\frac{\Delta_{\text{osc}} L}{2} \right) \frac{d\phi}{dE} \right] dE \right) / \left(\int_{E_{\min}}^{E_{\max}} \frac{d\phi}{dE} dE \right)$$

where $d\phi/dE$ is the differential flux in the energy band. Approximating this differential flux to be uniform over the energy band, and averaging over the oscillating term $\sin^2(\Delta_{\text{osc}} L/2)$ we get

$$P_{\text{ISM}}(a \rightarrow \gamma) \approx 2 \frac{\Delta_{a\gamma}^2}{\langle \Delta_{\text{osc}}^2 \rangle}$$

where we take $\langle \Delta_{\text{osc}}^2 \rangle$ to be the value of Δ_{osc}^2 at the average energy in the band. Note that we have taken a constant potential (constant ISM magnetic field and electron density) approximation for the sake of simplification, while in reality we expect a varying electron density.

Note the sharp increase of the conversion probability P_{ISM} at $m_a \sim O(10^{-12} - 10^{-11}) \text{ eV}$. This is caused by the resonance due to the ALP mass matching the photon effective mass in the ISM. However, this sharp feature is only due to the use of the average electron density in the ISM. In reality, due to varying electron density along the line of sight of the photon, this feature will be more spread out and less prominent. Due to the uncertainties involved with the location and the contribution of the ISM resonance peak, results in the range $m_a \sim (10^{-11} - 10^{-12}) \text{ eV}$ have to be interpreted with caution. We therefore focus on the low-mass regime ($m_a \lesssim 10^{-12} \text{ eV}$) and the high-mass regime ($m_a \gtrsim 10^{-11} \text{ eV}$).

In the low-mass ($m_a \lesssim 10^{-12} \text{ eV}$) regime, the average $a \rightarrow \gamma$ conversion probability in the ISM can be approximated as

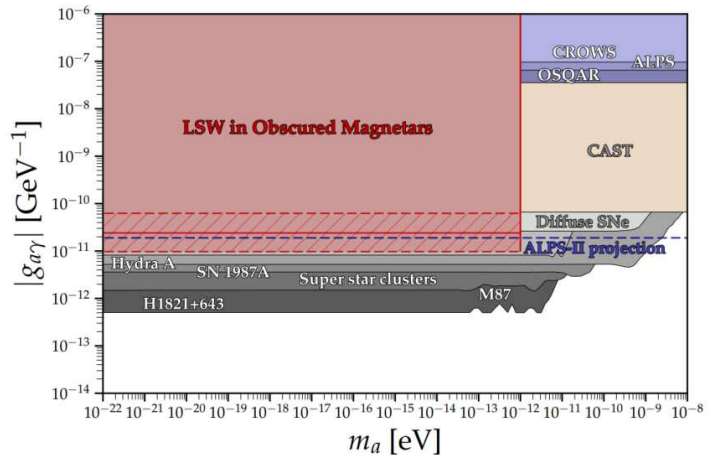
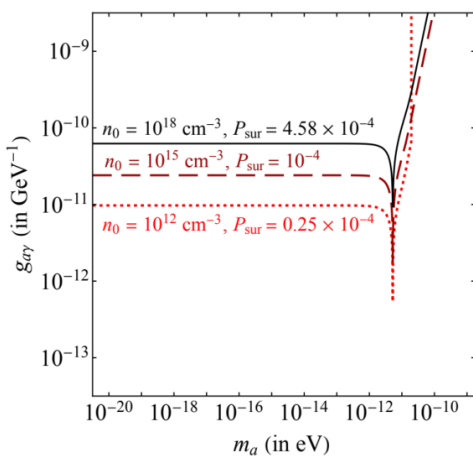
$$P_{\text{ISM}}(a \rightarrow \gamma) \approx 2\Delta_{a\gamma}^2/\Delta_{pl}^2 \propto g_{a\gamma}^2$$

This explains the horizontal contours on the left side of the plot, as the probability in this region only depends on the ALP-photon coupling $g_{a\gamma}$. On the other hand, in the high-mass ($m_a \gtrsim 10^{-11}$) regime, the conversion probability behaves as

$$P_{\text{ISM}}(a \rightarrow \gamma) \approx 2\Delta_{a\gamma}^2/\Delta_a^2 \propto g_{a\gamma}^2/m_a^2$$

leading to a dependence on both $g_{a\gamma}$ and m_a . We observe this dependence on both the parameters on the right side of the plot, where, with increasing m_a the conversion probability drops sharply.

we can find more details of Axion-photon coupling limit in [this link](#).



Chapter 6

The Axion Misalignment Problem

....still incomplete

We start with the dynamics. The action for this system is the following,

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2} (\partial_\mu \phi \partial^\mu \phi) - \frac{1}{2} m^2 \phi^2 \right]$$

where the quantity in the square brackets is the Lagrangian $\mathcal{L}$, and g is the determinant of the metric, for which we will use the FRW one from earlier: $g_{\mu\nu} = \text{diag}(-1, a^2, a^2, a^2)$.

The equation of motion for the field we find by varying the action

$$\frac{\partial \sqrt{-g} \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \sqrt{-g} \mathcal{L}}{\partial (\partial_\mu \phi)} = 0$$

resulting in the Klein-Gordon equation,

$$\square \phi - m_\phi^2 \phi = 0.$$

For the time being, we are just going to be thinking big-picture: we first want to check that the behaviour of a scalar field can be thought of as dark matter with a large-scale density that scales like $\bar{\rho} \sim a^{-3}$. So let us assume that we are following only the homogeneous zero-momentum mode, and bring in any higher modes/perturbations later if it looks like we need to. The upshot of that is simply that when we write out the d'Alembertian ($\square$) we will ignore the spatial derivative,

$$\square = -\partial_t^2 - 3H\partial_t. \quad (31)$$

Plugging this in we get the equation of motion we need to solve,

$$\ddot{\phi} + 3H(t)\dot{\phi} + m^2\phi = 0. \quad (32)$$

Since all of this physics will have to happen in advance of matter-radiation equality for any of this to make sense for dark matter, we assume we will be in a radiation-dominated background where $H = 1/2t$. The equation of motion then has a very familiar form: just a harmonic oscillator with a (decaying) damping term.

Now to solve it, we plug in our initial conditions, for which we will assume the field starts from some arbitrary value misaligned away from zero ($\phi(0) = \phi_i \neq 0$) and with no derivative ($\dot{\phi}(0) = 0$). The solution is then,

$$\phi = \phi_i \left(\frac{2}{m_\phi t} \right)^{\frac{1}{4}} \Gamma\left(\frac{5}{4}\right) J_{\frac{1}{4}}(m_\phi t), \quad (33)$$

where $J_n(x)$ is Bessel's function and $\Gamma(x)$ the gamma function. Notice there are two regimes for this evolution, which depend on how important the damping term ($3H(t)\dot{\phi}$) is compared with the mass term ($m_\phi^2\phi$). When the Hubble parameter is large, the system is overdamped, but once the damping term has decayed away, the field's mass drives its dynamics and so it starts oscillating.

We're interested in the late time behaviour, so it's instructive now to take the limit where mt is large. we can find that $J_{1/4}(x) \approx (\pi x/2)^{-1/2} \cos(x - \pi/3)$ for large x , so the late-time solution

essentially involves an envelope that decays slowly multiplied by quickly oscillating part,

$$\phi(t) \approx \phi_{\text{env}}(t) \cos m_\phi t \quad (34)$$

Now that we know how the field is behaving at late times, we can go and look at the energy density stored in it. The energy density of our scalar field can be read off from its energy-momentum tensor,

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}m^2\phi^2. \quad (35)$$

Working this out during radiation domination ($a \propto t^{1/2}$) and putting everything in terms of a , we see that,

$$\rho_\phi \propto a^{-3} \quad (36)$$

on timescales much longer than the timescale of the oscillations. This dilution with the volume of space is exactly how the energy density of matter scales.

So we have shown that the average density of an oscillating scalar field scales like dark matter—that was our first task. Now let us try to make sure we get the right amount of dark matter.

As mentioned in the previous section, we have measured the large-scale average dark matter density in our Universe to percent-level precision, and in a simple world, one candidate would explain all of it. So let us try to engineer our free parameters to align such that the following equation is true:

$$\Omega_\phi h^2 \equiv \frac{\rho_\phi(\text{today})}{3H_0^2 M_{\text{Pl}}^2} h^2 = 0.12 \quad (37)$$

Since our setup only has two unknowns, m_ϕ and ϕ_i , enforcing the expression above will tell us how those parameters have to be related to each other if our scalar field is to add up to all of the dark matter.

So now what we need is the energy density in the field today, $\rho_\phi(\text{today})$, which we can find by rolling the clock forwards from the density in the scalar field at the point when it started scaling like matter. We already defined this time to be t_{osc} , so the density today will just be the density at that time, diluted by how much space has expanded from then until now:

$$\rho_\phi(\text{today}) = \rho_\phi(t_{\text{osc}}) \left(\frac{a_{\text{today}}}{a_{\text{osc}}} \right)^{-3} \quad (38)$$

where $a_{\text{osc}} = a(t_{\text{osc}})$ is the scale factor when the field started behaving like matter. As a simplifying approximation, let us also assume the field value at this time was $\phi(t_{\text{osc}}) \approx \phi_i$, which shouldn't be too far off since ϕ doesn't evolve much in the overdamped period. Because $\dot{\phi}(t_{\text{osc}}) \approx 0$, this means that we can write $\rho_\phi(t_{\text{osc}}) = \frac{1}{2}m_\phi^2\phi_i^2$.

Now we can bring back Eq.(16) and write this in terms of the temperatures of the Universe now ($T_0 = 2.35 \times 10^{-4}\text{eV}$) and at t_{osc} :

$$\rho_\phi(\text{today}) = \rho_\phi(T_0) = \rho(T_{\text{osc}}) \frac{g_{*s}(T_0)}{g_{*s}(T_{\text{osc}})} \left(\frac{T_0}{T_{\text{osc}}} \right)^3 \quad (39)$$

The reason for doing this is so that we can find the temperature at t_{osc} using the Friedmann equation,

$$3H(T_{\text{osc}})^2 M_{\text{Pl}}^2 = \frac{\pi^2}{30} g_*(T_{\text{osc}}) T_{\text{osc}}^4 \quad (40)$$

where we already know the Hubble parameter when the field began oscillating because that was how we defined it in the first place: $3H(T_{\text{osc}}) = m_\phi$. Plugging these expressions back into Eq.(37), and using the fact that $3H_0^2 M_{\text{Pl}}^2 = 8.07 \times 10^{-11} h^2 \text{eV}^4$, we have numerical values for everything other than m_ϕ and ϕ_i , leaving us with,

$$\Omega_\phi h^2 = 0.12 \left(\frac{\phi_i}{4.7 \times 10^{16} \text{GeV}} \right)^2 \left(\frac{m_\phi}{10^{-21} \text{eV}} \right)^{\frac{1}{2}} \quad (41)$$

To get this I have assumed $g_\star = 3.4$ and $g_{\star,s}(T_0) \approx g_{\star,s}(T_{\text{osc}})$, which will only hold if the number of degrees of freedom are not changing much between t_{osc} and now. The ramification of this choice is that it only applies when t_{osc} is relatively late, or equivalently when the scalar mass is small. To see this, inspect Eq.(40) and enforce the condition $3H(T_{\text{osc}}) = m_\phi$. This reveals the relationship $T_{\text{osc}} \propto m_\phi^{1/2}$, i.e. the heavier the scalar's mass, the higher the temperature when it starts oscillating and hence the earlier it starts oscillating. At the same, if we try to make the scalar mass too light we will run into problems. We need our field to be behaving like dark matter by matter-radiation equality at the latest. Enforcing $T_{\text{osc}} > T_{\text{eq}}$ implies there will be a bound on the scalar mass $m_\phi > T_{\text{eq}}^2/M_{\text{Pl}}\sqrt{\pi^2 g_\star/10} \sim 10^{-28}\text{eV}$. However, this turns out to be far from the most competitive lower limit it is possible to draw on the mass of dark matter,

6.1 Cosmic strings

The name we give to these complicated things is 'topological defects', and they arise as follows. First, notice that the axion field angle exists on the domain $(-\pi, \pi)$. If we consider some continuous field of $\theta(\mathbf{x})$, with every point adopting a randomly chosen angle, we can imagine that there will be certain locations where those angles will just happen to loop around 2π . If this occurs, then somewhere inside that loop there will have to be a point where the angle is undefined. The axion is the phase of a complex scalar field, and so what is occurring at this singular point is that the field is forced into the centre of the complex plane at $|\Phi| = 0$, which is effectively where the PQsymmetry is restored. This point is a potential maximum, so it ought to be unstable, but because of the winding of the field around it, the configuration turns out to be stable. In three dimensions, these singular points all connect up along a one-dimensional line, containing energy associated with the potential for the field's radial mode.

This kind of field configuration that is conjectured to appear after a cosmological phase transition is known as a topological defect [106].¹⁷ In this case, we are breaking a global $U(1)$ and so the topological defects that emerge are called global cosmic strings. The string can either stretch off beyond the horizon or connect back up to itself to form a loop. The strings are thin, but not infinitesimally so—they have a finite high-energy core set by the shape of the potential in the radial direction Φ . It is useful to think in terms of the radial degree of freedom called the "saxion", which will have a mass,

$$m_s = \sqrt{\lambda} f_a \quad (63)$$

To understand the properties of cosmic strings, we can construct a simple static model by imagining we have an infinite one aligned along the z direction in cylindrical coordinates (r, φ, z) . The full field will have some solution: $\Phi = |\Phi(r)|e^{i\theta(\varphi)}$ where the axion field winds around the z axis so we can assume $\theta(\varphi) = \varphi$. The full radial dependence can be obtained numerically by solving

the equation of motion, Eq.(44), but here I will just state its behaviour at the two extremes [108]:

$$|\Phi(r)| \approx \begin{cases} 0.53 f_a m_s r & \text{for } r \rightarrow 0 \\ f_a \left(1 - \frac{1}{2m_s^2 r^2}\right) & \text{for } r \rightarrow \infty \end{cases} \quad (64)$$

We can then use this to look at the energy stored in the cosmic string. The solution is static so the Hamiltonian is just the sum of the gradient and potential energies,

$$\mathcal{H} = \frac{1}{2} ((\nabla|\Phi|)^2 + |\Phi|^2(\nabla\theta)^2) + V_{\text{PQ}}(\Phi) \quad (65)$$

For the time being, we are ignoring $V_{\text{QCD}}(\theta)$, which will allow the string to unwind itself, so we are working at temperatures between PQ breaking and QCD. The useful quantity to calculate is the energy per unit length, also called the string tension μ :

$$\mu \equiv 2\pi \int_0^\infty r \, dr \left(\frac{1}{2} \left(\frac{d|\Phi(r)|}{dr} \right)^2 + \frac{1}{2} \left(\frac{|\Phi(r)|}{r} \right)^2 + V_{\text{PQ}}(|\Phi|) \right) \quad (66)$$

The first and third terms integrate just fine, however looking at the large r behaviour of $\Phi(r)$ we see we need to do $\int^\infty r^{-1} dr$, and so the final answer is divergent. To evaluate the string

tension we therefore need to impose some large-scale cutoff to the integral $r_{\max}$, in which case we get,

$$\mu = f_a^2 \left[4.5 + \pi \ln \left(\frac{m_s r_{\max}}{4} \right) \right] \quad (67)$$

Cosmic strings may well be infinite in extent, but there is actually a physically motivated choice for this cutoff here because we have a cosmological horizon: $r_{\max} = H(t)^{-1}$.

So $\mu \approx f_a^2$, meaning that if cosmic strings exist in the universe they can potentially store a considerable amount of energy. As well as complicating the evolutionary history of the field, their motion through space as they straighten and intersect each other will act to stir up waves in the axion field [109-111]. Although the axion at the moment in our discussion is still massless, it won't stay massless, and so the axions radiated by the cosmic strings will eventually bestow the dark matter with some distribution of momenta over a wide range of scales.

The waves through the axion field stirred up by the motion of cosmic strings will come in a range of wavelengths. There will be long-wavelength modes coming from the large-scale motion of the long horizon-sized strings, as well as short-wavelength modes generated by the collapse and intersection of smaller closed loops of strings. This period of evolution is described by a so-called "scaling solution" in terms of the number of strings inside some comoving volume, V :

$$\xi(t) = \frac{\ell_{\text{tot}}(t)t^2}{V} \quad (68)$$

Given that there could be more than $\xi > 1$ string per horizon, we should slightly shrink the longdistance cutoff in our integral from H^{-1} to the typical distance to the nearest string, $(H\sqrt{\xi})^{-1}$.

$$\mu_{\text{eff}} = \pi f_a^2 \ln \left(\frac{m_s \eta}{H\sqrt{\xi}} \right) \quad (69)$$

$$\xi(t) \approx c_0 + c_1 \ln \left(\frac{m_r}{H(t)} \right) \quad (70)$$

$$\rho_{\text{strings}} = \frac{\xi(t)\mu}{t^2} \approx \xi \times f_a^2 H^2 \log \frac{m_r}{H} \quad (71)$$

Incomplete...

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